

The Earliest Arithmetics in English

Early English Text Society.

Extra Series, No. CXVIII.

1922 (for 1916).



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*** START OF THE PROJECT GUTENBERG EBOOK THE EARLIEST
ARITHMETICS IN ENGLISH ***

This text includes characters that will only display in UTF-8 (Unicode) file encoding:

ȝ, ȝ (yogh, long s)

ȝ, ȝ (n with curl, crossed l: see below)

φ (Greek phi, sometimes used in printed text for 0)

If any of these characters do not display properly, or if the apostrophes and quotation marks in this paragraph appear as garbage, you may have an incompatible browser or unavailable fonts. First, make sure that the browser's "character set" or "file encoding" is set to Unicode (UTF-8). You may also need to change your browser's default font.

In *The Crafte of Nombrynge*, final **n** was sometimes written with an extra curl as **n̄**. It has been rendered as **ȝ** for visual effect; the character is not intended to convey phonetic information. In the same selection, the numeral "0" was sometimes printed as Greek φ (phi); this has been retained for the e-text.

Double **I** with a line **H** is shown as **ȝ**. The first few occurrences of **d** (for "pence") were printed with a curl as **d̄**. The letter is shown with the same **d̄** used in the remainder of the text.

The word "withdraw" or "*with*draw" was inconsistently hyphenated; it was left as printed, and line-end hyphens were retained. All brackets [] are in the original.

The diagrams in "Accomptynge by Counters" may not line up perfectly in all browsers, but the contents should still be intelligible.

The original text contained at least five types of marginal note. Details are given at the [end of the e-text](#).

Typographical errors are shown in the text with mouse-hover popups. Other underlined words are cross-references to the [Index of Technical Terms](#) and the [Glossary](#).

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The Earliest Arithmetics in English

EDITED WITH INTRODUCTION

BY

ROBERT STEELE

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INTRODUCTION

THE number of English arithmetics before the sixteenth century is very small. This is hardly to be wondered at, as no one requiring to use even the simplest operations of the art up to the middle of the fifteenth century was likely to be ignorant of Latin, in which language there were several treatises in a considerable number of manuscripts, as shown by the quantity of them still in existence. Until modern commerce was fairly well established, few persons required more arithmetic than addition and subtraction, and even in the thirteenth century, scientific treatises addressed to advanced students contemplated the likelihood of their not being able to do simple division. On the other hand, the study of astronomy necessitated, from its earliest days as a science, considerable skill and accuracy in computation, not only in the calculation of astronomical tables but in their use, a knowledge of which latter was fairly common from the thirteenth to the sixteenth centuries.

The arithmetics in English known to me are:—

- (1) Bodl. 790 G. VII. (2653) f. 146-154 (15th c.) *inc.* “Of angrym ther be IX figures in numbray . . .” A mere unfinished fragment, only getting as far as Duplation.
- (2) Camb. Univ. LI. IV. 14 (III.) f. 121-142 (15th c.) *inc.* “Al maner of thyngis that prosedeth ffro the frist begynnyng . . .”
- (3) Fragmentary passages or diagrams in Sloane 213 f. 120-3 (a fourteenth-century counting board), Egerton 2852 f. 5-13, Harl. 218 f. 147 and

- (4) The two MSS. here printed; Eg. 2622 f. 136 and Ashmole 396 f. 48.
All of these, as the language shows, are of the fifteenth century.

The CRAFT OF NOMBRYNGE is one of a large number of scientific treatises, mostly in Latin, bound up together as Egerton MS. 2622 in the British Museum Library. It measures 7" × 5", 29-30 lines to the page, in a rough hand. The English is N.E. Midland in dialect. It is a translation and amplification of one of the numerous glosses on the *de algorismo* of Alexander de Villa Dei (c. 1220), such as that of Thomas of Newmarket contained in the British Museum MS. Reg. 12, E. 1. A fragment of another translation of the same gloss was printed by Halliwell in his *Rara Mathematica* (1835) p. 29.¹ It corresponds, as far as p. 71, l. 2, roughly to p. 3 of our version, and from thence to the end p. 2, ll. 16-40.

The ART OF NOMBRYNG is one of the treatises bound up in the Bodleian MS. Ashmole 396. It measures 11½" × 17¾", and is written with thirty-three lines to the page in a fifteenth century hand. It is a translation, rather literal, with amplifications of the *de arte numerandi* attributed to John of Holywood (Sacrobosco) and the translator had obviously a poor MS. before him. The *de arte numerandi* was printed in 1488, 1490 (s.n.), 1501, 1503, 1510, 1517, 1521, 1522, 1523, 1582, and by Halliwell separately and in his two editions of *Rara Mathematica*, 1839 and 1841, and reprinted by Curze in 1897.

Both these tracts are here printed for the first time, but the first having been circulated in proof a number of years ago, in an endeavour to discover other manuscripts or parts of manuscripts of it, Dr. David Eugene Smith, misunderstanding the position, printed some pages in a curious transcript with four facsimiles in the *Archiv für die Geschichte der Naturwissenschaften und der Technik*, 1909, and invited the scientific world to take up the "not unpleasant task" of editing it.

ACCOMPTYNGE BY COUNTERS is reprinted from the 1543 edition of Robert Record's Arithmetic, printed by R. Wolfe. It has been reprinted within the last few years by Mr. F. P. Barnard, in his work on Casting Counters. It is the earliest English treatise we have on this variety of the Abacus (there are Latin ones of the end of the fifteenth century), but there is little doubt in my mind that this method of performing the simple operations of arithmetic is

much older than any of the pen methods. At the end of the treatise there follows a note on merchants' and auditors' ways of setting down sums, and lastly, a system of digital numeration which seems of great antiquity and almost world-wide extension.

After the fragment already referred to, I print as an appendix the 'Carmen de Algorismo' of Alexander de Villa Dei in an enlarged and corrected form. It was printed for the first time by Halliwell in *Rara Mathematica*, but I have added a number of stanzas from various manuscripts, selecting various readings on the principle that the verses were made to scan, aided by the advice of my friend Mr. Vernon Rendall, who is not responsible for the few doubtful lines I have conserved. This poem is at the base of all other treatises on the subject in medieval times, but I am unable to indicate its sources.

THE SUBJECT MATTER.

Ancient and medieval writers observed a distinction between the Science and the Art of Arithmetic. The classical treatises on the subject, those of Euclid among the Greeks and Boethius among the Latins, are devoted to the Science of Arithmetic, but it is obvious that coeval with practical Astronomy the Art of Calculation must have existed and have made considerable progress. If early treatises on this art existed at all they must, almost of necessity, have been in Greek, which was the language of science for the Romans as long as Latin civilisation existed. But in their absence it is safe to say that no involved operations were or could have been carried out by means of the alphabetic notation of the Greeks and Romans. Specimen sums have indeed been constructed by moderns which show its possibility, but it is absurd to think that men of science, acquainted with Egyptian methods and in possession of the abacus,² were unable to devise methods for its use.

THE PRE-MEDIEVAL INSTRUMENTS USED IN CALCULATION.

The following are known:—

- (1) A flat polished surface or tablets, strewn with sand, on which figures were inscribed with a stylus.
- (2) A polished tablet divided longitudinally into nine columns (or more) grouped in threes, with which counters were used, either plain or marked with signs denoting the nine numerals, etc.
- (3) Tablets or boxes containing nine grooves or wires, in or on which ran beads.
- (4) Tablets on which nine (or more) horizontal lines were marked, each third being marked off.

The only Greek counting board we have is of the fourth class and was discovered at Salamis. It was engraved on a block of marble, and measures 5 feet by 2½. Its chief part consists of eleven parallel lines, the 3rd, 6th, and 9th being marked with a cross. Another section consists of five parallel lines, and there are three rows of arithmetical symbols. This board could only have been used with counters (*calculi*), preferably unmarked, as in our treatise of *Accomptynge by Counters*.

CLASSICAL ROMAN METHODS OF CALCULATION.

We have proof of two methods of calculation in ancient Rome, one by the first method, in which the surface of sand was divided into columns by a stylus or the hand. Counters (*calculi*, or *lapilli*), which were kept in boxes (*loculi*), were used in calculation, as we learn from Horace's schoolboys (Sat. 1. vi. 74). For the sand see Persius I. 131, "Nec qui abaco numeros et secto in pulvere metas scit risisse," Apul. Apolog. 16 (pulvisculo), Mart. Capella, lib. vii. 3, 4, etc. Cicero says of an expert calculator "eruditum attigisse pulverem," (de nat. Deorum, ii. 18). Tertullian calls a teacher of arithmetic "primus numerorum arenarius" (de Pallio, *in fine*). The counters were made of various materials, ivory principally, "Adeo nulla uncia nobis est eboris, etc." (Juv. XI. 131), sometimes of precious metals, "Pro calculis albis et nigris aureos argenteosque habebat denarios" (Pet. Arb. Satyricon, 33).

There are, however, still in existence four Roman counting boards of a kind which does not appear to come into literature. A typical one is of the third class. It consists of a number of transverse wires, broken at the middle. On the left hand portion four beads are strung, on the right one (or two). The left hand beads signify units, the right hand one five units. Thus any number up to nine can be represented. This instrument is in all essentials the same as the Swanpan or Abacus in use throughout the Far East. The Russian stchota in use throughout Eastern Europe is simpler still. The method of using this system is exactly the same as that of *Accomptynge by Counters*, the right-hand five bead replacing the counter between the lines.

THE BOETHIAN ABACUS.

Between classical times and the tenth century we have little or no guidance as to the art of calculation. Boethius (fifth century), at the end of lib. II. of his *Geometria* gives us a figure of an abacus of the second class with a set of counters arranged within it. It has, however, been contended with great probability that the whole passage is a tenth century interpolation. As no rules are given for its use, the chief value of the figure is that it gives the signs of the nine numbers, known as the Boethian “apices” or “notae” (from whence our word “notation”). To these we shall return later on.

THE ABACISTS.

It would seem probable that writers on the calendar like Bede (A.D. 721) and Helpericus (A.D. 903) were able to perform simple calculations; though we are unable to guess their methods, and for the most part they were dependent on tables taken from Greek sources. We have no early medieval treatises on arithmetic, till towards the end of the tenth century we find a revival of the study of science, centring for us round the name of Gerbert, who became Pope as Sylvester II. in 999. His treatise on the use of the Abacus was written (c. 980) to a friend Constantine, and was first printed among the works of Bede in the Basle (1563) edition of his works, I. 159, in a somewhat enlarged form. Another tenth century treatise is that of Abbo of Fleury (c. 988), preserved in several manuscripts. Very few treatises on the

use of the Abacus can be certainly ascribed to the eleventh century, but from the beginning of the twelfth century their numbers increase rapidly, to judge by those that have been preserved.

The Abacists used a permanent board usually divided into twelve columns; the columns were grouped in threes, each column being called an “arcus,” and the value of a figure in it represented a tenth of what it would have in the column to the left, as in our arithmetic of position. With this board counters or jetons were used, either plain or, more probably, marked with numerical signs, which with the early Abacists were the “apices,” though counters from classical times were sometimes marked on one side with the digital signs, on the other with Roman numerals. Two ivory discs of this kind from the Hamilton collection may be seen at the British Museum. Gerbert is said by Richer to have made for the purpose of computation a thousand counters of horn; the usual number of a set of counters in the sixteenth and seventeenth centuries was a hundred.

Treatises on the Abacus usually consist of chapters on Numeration explaining the notation, and on the rules for Multiplication and Division. Addition, as far as it required any rules, came naturally under Multiplication, while Subtraction was involved in the process of Division. These rules were all that were needed in Western Europe in centuries when commerce hardly existed, and astronomy was unpractised, and even they were only required in the preparation of the calendar and the assignments of the royal exchequer. In England, for example, when the hide developed from the normal holding of a household into the unit of taxation, the calculation of the geldage in each shire required a sum in division; as we know from the fact that one of the Abacists proposes the sum: “If 200 marks are levied on the county of Essex, which contains according to Hugh of Bocland 2500 hides, how much does each hide pay?”³ Exchequer methods up to the sixteenth century were founded on the abacus, though when we have details later on, a different and simpler form was used.

The great difficulty of the early Abacists, owing to the absence of a figure representing zero, was to place their results and operations in the proper columns of the abacus, especially when doing a division sum. The chief differences noticeable in their works are in the methods for this rule.

Division was either done directly or by means of differences between the divisor and the next higher multiple of ten to the divisor. Later Abacists made a distinction between “iron” and “golden” methods of division. The following are examples taken from a twelfth century treatise. In following the operations it must be remembered that a figure asterisked represents a counter taken from the board. A zero is obviously not needed, and the result may be written down in words.

(a) MULTIPLICATION. 4600×23 .

Thousands					
H u n d r e d s	T e n s	U n i t s	H u n d r e d s	T e n s	U n i t s
		4	6		
		1	8		
	1	2			
	1	2			
	8				
1		5	8		
				2	3

Multiplicand.

600×3 .
 4000×3 .
 600×20 .
 4000×20 .
Total product.

Multiplier.

(b) DIVISION: DIRECT. $100,000 \div 20,023$. Here each counter in turn is a separate divisor.

H.	T.	U.	H.	T.	U.
	2			2	3
1	2				
	2				
	1	9	1		
			9		

Divisors.

Place greatest divisor to right of dividend.

Dividend.

Remainder.

Another form of same.

				8	
	1	9	9	2	
				1	2
	1	9	9		8
					4

Product of 1st Quotient and 20.

Remainder.

Product of 1st Quotient and 3.

Final remainder.

Quotient.

(c) DIVISION BY DIFFERENCES. $900 \div 8$. Here we divide by (10-2).

			H.	T.	U.
					2
					8
			⁴ 9		
			⁴ 1	8	
				2	
			⁴ 1		
				2	
					4
					2
				1	
				1	
				9	
			1	1	2

Difference.

Divisor.

Dividend.

Product of difference by 1st Quotient (9).

Product of difference by 2nd Quotient (1).

Sum of 8 and 2.

Product of difference by 3rd Quotient (1).

Product of difference by 4th Quot. (2). **Remainder.**

4th Quotient.

3rd Quotient.

2nd Quotient.

1st Quotient.

Quotient. (Total of all four.)

DIVISION. $7800 \div 166$.

Thousands					
H.	T.	U.	H.	T.	U.
				3	4
			1	6	6
		⁴ 7	8		
		1			
			1	2	
			9		
		⁴ 2	8	2	

Differences (making 200 trial divisor).

Divisors.

Dividends.

Remainder of greatest dividend.

Product of 1st difference (4) by 1st Quotient (3).

Product of 2nd difference (3) by 1st Quotient (3).

New dividends.

			3	4		Product of 1st and 2nd difference by 2nd Quotient (1).
		⁴ 1	1	6		New dividends.
				2		Product of 1st difference by 3rd Quotient (5).
			1	5		Product of 2nd difference by 3rd Quotient (5).
			⁴ 3	3		New dividends.
			1			Remainder of greatest dividend.
				3	4	Product of 1st and 2nd difference by 4th Quotient (1).
			1	6	4	Remainder (less than divisor).
					1	4th Quotient.
					5	3rd Quotient.
				1		2nd Quotient.
				3		1st Quotient.
				4	6	Quotient.

DIVISION. $8000 \div 606$.

Thousands						
H.	T.	U.	H.	T.	U.	
				9		Difference (making 700 trial divisor).
					4	Difference.
			6		6	Divisors.
		⁴ 8				Dividend.
		1				Remainder of dividend.
			9	4		Product of difference 1 and 2 with 1st Quotient (1).
		⁴ 1	9	4		New dividends.
			3			Remainder of greatest dividend.
				9	4	Product of difference 1 and 2 with 2nd Quotient (1).
		⁴ 1	3	3	4	New dividends.
			3			Remainder of greatest dividend.
				9	4	Product of difference 1 and 2 with 3rd Quotient (1).
			7	2	8	New dividends.
			6		6	Product of divisors by 4th Quotient (1).
			1	2	2	Remainder.
					1	4th Quotient.

					1	3rd Quotient.
					1	2nd Quotient.
				1		1st Quotient.
				1	3	Quotient.

The chief Abacists are Gerbert (tenth century), Abbo, and Hermannus Contractus (1054), who are credited with the revival of the art, Bernelinus, Gerland, and Radulphus of Laon (twelfth century). We know as English Abacists, Robert, bishop of Hereford, 1095, “abacum et lunarem compotum et celestium cursum astrorum rimatus,” Turchillus Compotista (Thurkil), and through him of Guilielmus R. . . . “the best of living computers,” Gislebert, and Simonus de Rotellis (Simon of the Rolls). They flourished most probably in the first quarter of the twelfth century, as Thurkil’s treatise deals also with fractions. Walcher of Durham, Thomas of York, and Samson of Worcester are also known as Abacists.

Finally, the term Abacists came to be applied to computers by manual arithmetic. A MS. Algorithm of the thirteenth century (Sl. 3281, f. 6, b), contains the following passage: “Est et alius modus secundum operadores sive practicos, quorum unus appellatur Abacus; et modus ejus est in computando per digitos et junctura manuum, et iste utitur ultra Alpes.”

In a composite treatise containing tracts written A.D. 1157 and 1208, on the calendar, the abacus, the manual calendar and the manual abacus, we have a number of the methods preserved. As an example we give the rule for multiplication (Claud. A. IV., f. 54 vo). “Si numerus multiplicat alium numerum auferatur differentia majoris a minore, et per residuum multiplicetur articulus, et una differentia per aliam, et summa proveniet.” Example, 8×7 . The difference of 8 is 2, of 7 is 3, the next article being 10; $7 - 2$ is 5. $5 \times 10 = 50$; $2 \times 3 = 6$. $50 + 6 = 56$ answer. The rule will hold in such cases as 17×15 where the article next higher is the same for both, *i.e.*, 20; but in such a case as 17×9 the difference for each number must be taken from the higher article, *i.e.*, the difference of 9 will be 11.

THE ALGORISTS.

Algorism (augrim, augrym, algram, agram, algorithm), owes its name to the accident that the first arithmetical treatise translated from the Arabic happened to be one written by Al-Khowarazmi in the early ninth century, “de numeris Indorum,” beginning in its Latin form “Dixit Algorismi. . . .” The translation, of which only one MS. is known, was made about 1120 by Adelard of Bath, who also wrote on the Abacus and translated with a commentary Euclid from the Arabic. It is probable that another version was made by Gerard of Cremona (1114-1187); the number of important works that were not translated more than once from the Arabic decreases every year with our knowledge of medieval texts. A few lines of this translation, as copied by Halliwell, are given on p. 72, note 2. Another translation still seems to have been made by Johannes Hispalensis.

Algorism is distinguished from Abacist computation by recognising seven rules, Addition, Subtraction, Duplation, Mediation, Multiplication, Division, and Extraction of Roots, to which were afterwards added Numeration and Progression. It is further distinguished by the use of the zero, which enabled the computer to dispense with the columns of the Abacus. It obviously employs a board with fine sand or wax, and later, as a substitute, paper or parchment; slate and pencil were also used in the fourteenth century, how much earlier is unknown.⁵ Algorism quickly ousted the Abacus methods for all intricate calculations, being simpler and more easily checked: in fact, the astronomical revival of the twelfth and thirteenth centuries would have been impossible without its aid.

The number of Latin Algorisms still in manuscript is comparatively large, but we are here only concerned with two—an Algorism in prose attributed to Sacrobosco (John of Holywood) in the colophon of a Paris manuscript, though this attribution is no longer regarded as conclusive, and another in verse, most probably by Alexander de Villedieu (Villa Dei). Alexander, who died in 1240, was teaching in Paris in 1209. His verse treatise on the Calendar is dated 1200, and it is to that period that his Algorism may be attributed; Sacrobosco died in 1256 and quotes the verse Algorism. Several commentaries on Alexander’s verse treatise were composed, from one of which our first tractate was translated, and the text itself was from time to time enlarged, sections on proofs and on mental arithmetic being added. We have no indication of the source on which Alexander drew; it was most

likely one of the translations of Al-Khowarasmī, but he has also the Abacists in mind, as shewn by preserving the use of differences in multiplication. His treatise, first printed by Halliwell-Phillipps in his *Rara Mathematica*, is adapted for use on a board covered with sand, a method almost universal in the thirteenth century, as some passages in the algorism of that period already quoted show: “Est et alius modus qui utitur apud Indos, et doctor hujusmodi ipsos erat quidem nomine Albus. Et modus suus erat in computando per quasdam figuras scribendo in pulvere. . . .” “Si voluerimus depingere in pulvere predictos digitos secundum consuetudinem algorismi . . .” “et sciendum est quod in nullo loco minorum sive secundorum . . . in pulvere debent scribi plusquam sexaginta.”

MODERN ARITHMETIC.

Modern Arithmetic begins with Leonardi Fibonacci's treatise “de Abaco,” written in 1202 and re-written in 1228. It is modern rather in the range of its problems and the methods of attack than in mere methods of calculation, which are of its period. Its sole interest as regards the present work is that Leonardi makes use of the digital signs described in Record's treatise on *The arte of nombrynge by the hand* in mental arithmetic, calling it “modus Indorum.” Leonardo also introduces the method of proof by “casting out the nines.”

DIGITAL ARITHMETIC.

The method of indicating numbers by means of the fingers is of considerable age. The British Museum possesses two ivory counters marked on one side by carelessly scratched Roman numerals IIIV and VIIII, and on the other by carefully engraved digital signs for 8 and 9. Sixteen seems to have been the number of a complete set. These counters were either used in games or for the counting board, and the Museum ones, coming from the Hamilton collection, are undoubtedly not later than the first century. Frohner has published in the *Zeitschrift des Münchener Alterthumsvereins* a set, almost complete, of them with a Byzantine treatise; a Latin treatise is printed among Bede's works. The use of this method is universal through

the East, and a variety of it is found among many of the native races in Africa. In medieval Europe it was almost restricted to Italy and the Mediterranean basin, and in the treatise already quoted (Sloane 3281) it is even called the Abacus, perhaps a memory of Fibonacci's work.

Methods of calculation by means of these signs undoubtedly have existed, but they were too involved and liable to error to be much used.

THE USE OF "ARABIC" FIGURES.

It may now be regarded as proved by Bubnov that our present numerals are derived from Greek sources through the so-called Boethian "apices," which are first found in late tenth century manuscripts. That they were not derived directly from the Arabic seems certain from the different shapes of some of the numerals, especially the 0, which stands for 5 in Arabic. Another Greek form existed, which was introduced into Europe by John of Basingstoke in the thirteenth century, and is figured by Matthew Paris (V. 285); but this form had no success. The date of the introduction of the zero has been hotly debated, but it seems obvious that the twelfth century Latin translators from the Arabic were perfectly well acquainted with the system they met in their Arabic text, while the earliest astronomical tables of the thirteenth century I have seen use numbers of European and not Arabic origin. The fact that Latin writers had a convenient way of writing hundreds and thousands without any cyphers probably delayed the general use of the Arabic notation. Dr. Hill has published a very complete survey of the various forms of numerals in Europe. They began to be common at the middle of the thirteenth century and a very interesting set of family notes concerning births in a British Museum manuscript, Harl. 4350 shows their extension.

The first is dated Mij^c. lviii., the second Mij^c. lxi., the third Mij^c. 63, the fourth 1264, and the fifth 1266. Another example is given in a set of astronomical tables for 1269 in a manuscript of Roger Bacon's works, where the scribe began to write MCC6. and crossed out the figures, substituting the "Arabic" form.

THE COUNTING BOARD.

The treatise on pp. 52-65 is the only one in English known on the subject. It describes a method of calculation which, with slight modifications, is current in Russia, China, and Japan, to-day, though it went out of use in Western Europe by the seventeenth century. In Germany the method is called “Algorithmus Linealis,” and there are several editions of a tract under this name (with a diagram of the counting board), printed at Leipsic at the end of the fifteenth century and the beginning of the sixteenth. They give the nine rules, but “Capitulum de radicum extractione ad algoritum integrorum reservato, cujus species per ciffrales figuras ostenduntur ubi ad plenum de hac tractabitur.” The invention of the art is there attributed to Appulegius the philosopher.

The advantage of the counting board, whether permanent or constructed by chalking parallel lines on a table, as shown in some sixteenth-century woodcuts, is that only five counters are needed to indicate the number nine, counters on the lines representing units, and those in the spaces above representing five times those on the line below. The Russian abacus, the “tchatui” or “stchota” has ten beads on the line; the Chinese and Japanese “Swanpan” economises by dividing the line into two parts, the beads on one side representing five times the value of those on the other. The “Swanpan” has usually many more lines than the “stchota,” allowing for more extended calculations, see Tylor, *Anthropology* (1892), p. 314.

Record’s treatise also mentions another method of counter notation (p. 64) “merchants’ casting” and “auditors’ casting.” These were adapted for the usual English method of reckoning numbers up to 200 by scores. This method seems to have been used in the Exchequer. A counting board for merchants’ use is printed by Halliwell in *Rara Mathematica* (p. 72) from Sloane MS. 213, and two others are figured in Egerton 2622 f. 82 and f. 83. The latter is said to be “novus modus computandi secundum inventionem Magistri Thome Thorleby,” and is in principle, the same as the “Swanpan.”

The Exchequer table is described in the *Dialogus de Scaccario* (Oxford, 1902), p. 38.

1. Halliwell printed the two sides of his leaf in the wrong order. This and some obvious errors of transcription—‘ferye’ for ‘ferthe,’ ‘lest’ for

‘left,’ etc., have not been corrected in the reprint on pp. 70-71.

[2.](#) For Egyptian use see Herodotus, ii. 36, Plato, *de Legibus*, VII.

[3.](#) See on this Dr. Poole, *The Exchequer in the Twelfth Century*, Chap. III., and Haskins, *Eng. Hist. Review*, 27, 101. The hidage of Essex in 1130 was 2364 hides.

[4.](#) These figures are removed at the next step.

[5.](#) Slates are mentioned by Chaucer, and soon after (1410) Prosdocimo de Beldamandi speaks of the use of a “lapis” for making notes on by calculators.

The Earliest Arithmetics in English.

The Crafte of Nombrynge.

Egerton 2622.

leaf 136 a.

HEc algorismus ars *presens dicitur*; in qua
Talibus indorum fruimur bis quinque figuris.

This boke is called þe boke of algorym, or Augrym after lewder vse. And þis boke tretys þe Craft of Nombryng, þe quych crafte is called also Algorym. Ther was a kyng of Inde, þe quich heyth Algor, & he made þis craft. And after his name he called hit algorym; or els anoþer cause is quy it is called Algorym, for þe latyn word of hit s. Algorismus comes of Algos, grece, *quid est* ars, latine, craft oñ englis, and rides, *quid est numerus*, latine, A nombur oñ englys, inde *dicitur Algorismus per addicionem huius sillabe mus & subtraccionem d & e, quasi ars numerandi.*

¶ fforthermore 3e most vndirstonde þat in þis craft ben vsid teen figurys, as here bene writen for ensampul, ϕ 9 8 7 6 5 4 3 2 1.

¶ Expone þe too versus afore: this present craft ys called Algorismus, in þe quych we vse teen signys of Inde. Questio.

¶ Why ten figuris of Inde? Solucio. for as I haue sayd afore þai were fonde fyrst in Inde of a kynge of þat Cuntre, þat was called Algor.

A derivati
Algorism.

Another
derivati
word.

on and Numeration.

versus
[in margin]

¶ **Prima significat unum; duo vero secunda:**

¶ **Tercia significat tria; sic procede sinistre.**

¶ **Donec ad extremam venias, que cifra vocatur.**

¶ **Capitulum primum de significacione figurarum.**

In þis verse is notifide þe significacion of þese figuris. And þus expone the verse. Þe first signifiyth one, þe secunde leaf 136 b. signi*fiyth tweyne, þe thryd signifiyth thre, & the fourte signifiyth 4. ¶ And so forthe towarde þe lyft syde of þe tabul or of þe boke þat þe figures bene writene in, til þat þou come to the last figure, þat is called a cifre. ¶ Questio. In quych syde sittes þe first figure? Solucio, forsothe loke quich figure is first in þe ryzt side of þe bok or of þe tabul, & þat same is þe first figure, for þou schal write bakeward, as here, 3. 2. 6. 4. 1. 2. 5. The figure of 5. was first write, & he is þe first, for he sittes on þe riht syde. And the figure of 3 is last. ¶ Neuer-þe-les wen he says ¶ *Prima significat vnum &c.*, þat is to say, þe first betokenes one, þe secunde. 2. & fore-þer-more, he vndirstondes nozt of þe first figure of euery rew. ¶ But he vndirstondes þe first figure þat is in þe nombur of þe forsayd teen figuris, þe quych is one of þese. 1. And þe secunde 2. & so forth.

Expositi
versus.

The mean
place of tl
figures.

Which fig
read first.

versus
[in margin]

¶ **Quelibet illarum si primo limite ponas,
¶ Simpliciter se significat: si vero secundo,
Se decies: sursum procedas multiplicando.
¶ Namque figura sequens quamuis signat decies plus.
¶ Ipsa locata loco quam significat pertinente.**

¶ Expone þis verse þus. Euery of þese figuris bitokens hym selfe & no more, yf he stonde in þe first place of þe rewele / this worde *Simpliciter* in þat verse it is no more to say but þat, & no more. ¶ If it stonde in the secunde place of þe rewle, he betokens tene tymes hym selfe, as þis figure 2 here 20 tokens ten tyme hym selfe, leaf 137 a. *þat is twenty, for he hym selfe betokenes tweyne, & ten tymes twene is twenty. And for he stondis on þe lyft side & in þe secunde place, he betokens ten tyme hym selfe. And so go forth. ¶ ffor euery figure, & he stonde aftur a-noþer toward the lyft side, he schal betokene ten tymes as mich more as he schul betoken & he stode in þe place þere þat þe figure afore hym stondes. loo an ensampulle. 9. 6. 3. 4. Þe figure of 4.

Expositio
margin].

An explar
of the prii
of notatio

An exampl

þat hase þis schape . betokens bot hymselfe, for he stondes

in þe first place. The figure of 3. þat hase þis schape ³.
betokens ten tymes more þen he schuld & he stde þere þat þe
figure of 4. stondes, þat is thretty. The figure of 6, þat hase þis

units,

tens,

schape ⁶, betokens ten tymes more þan he schuld & he stode

þere as þe figure of ³. stondes, for þere he schuld tokyne bot
sixty, & now he betokens ten tymes more, þat is sex hundryth.

hundreds,

The figure of 9. þat hase þis schape ⁹. betokens ten tymes
more þane he schuld & he stode in þe place þere þe figure of sex
stondes, for þen he schuld betoken to 9. hundryth, and in þe
place þere he stondes now he betokens 9. þousande. Al þe hole
nombur is 9 thousande sex hundryth & foure & thretty.

thousands

¶ fforthermore, when þou schalt rede a nombur of figure, þou
schalt begyne at þe last figure in the lyft side, & rede so forth to
þe riȝt side as here 9. 6. 3. 4. Thou schal begyn to rede at þe
figure of 9. & rede forth þus. 9. leaf 137 b. *thousand sex hundryth
thritty & foure. But when þou schalle write, þou schalt be-gynne
to write at þe ryȝt side.

How to re
number.

¶ Nil cifra significat sed dat signare sequenti.

Expone þis verse. A cifre tokens noȝt, bot he makes þe figure to
betoken þat comes aftur hym more þan he schuld & he were
away, as þus 10. here þe figure of one tokens ten, & yf þe cifre
were away ¹ & no figure by-fore hym he schuld token bot one,
for þan he schuld stonde in þe first place. ¶ And þe cifre tokens
nothyng hym selfe. for al þe nombur of þe ylke too figures is bot
ten. ¶ Questio. Why says he þat a cifre makys a figure to signifye
(tyf) more &c. ¶ I speke for þis worde significatyf, ffor sothe it
may happe aftur a cifre schuld come a-nopur cifre, as þus 200.
And ȝet þe secunde cifre shuld token neuer þe more excep he
schuld kepe þe order of þe place. and a cifre is no figure
significatyf.

The mean
use of the

¶ Quam precedentes plus ultima significabit /

Expone þis verse þus. Þe last figure schal token more þan alle þe oþer afore, thouȝt þere were a hundryth thousand figures afore, as þus, 16798. Þe last figure þat is 1. betokens ten thousand. And alle þe oþer figures ben bot betokene bot sex thousand seuyne hundryth nynty & 8. ¶ And ten thousand is more þen alle þat nombur, ergo þe last figure tokens more þan all þe nombur afore.

The last fi
means mc
all the oth
since it is
highest v

tree Kinds of Numbers

leaf 138 a.

* ¶ **Post predicta scias breuiter quod tres numerorum
Distincte species sunt; nam quidam digiti sunt;
Articuli quidam; quidam quoque compositi sunt.**

¶ Capitulum 2^m de triplice divisione numerorum.

¶ The auctor of þis tretis departys þis worde a nombur into 3 partes. Some nombur is called digitus latine, a digit in englys. Somme nombur is called articulus latine. An Articul in englys. Some nombur is called a composyt in englys. ¶ Expone þis verse. know þou aftur þe forsayd rewles þat I sayd afore, þat þere ben thre spices of nombur. Oone is a digit, Anoþer is an Articul, & þe toþer a Composyt. versus.

Digits.
Articles.
Composit

, Articles, and Composites.

¶ **Sunt digiti numeri qui citra denarium sunt.**

¶ Here he telles qwat is a digit, Expone versus sic. Nomburs digitus bene alle nomburs þat ben with-inne ten, as nyne, 8. 7. 6. 5. 4. 3. 2. 1.

What are

¶ **Articupli decupli degitorum; compositi sunt
Illi qui constant ex articulis degitisque.**

¶ Here he telles what is a composyt and what is ane articul. Expone sic versus. ¶ Articulis ben² alle þat may be deuidyt into nomburs of ten & nothyng leue ouer, as twenty, thretty, fourty, a hundryth, a thousand, & such oþer, ffor twenty may be

What are
articles.

departyt in-to 2 nomburs of ten, fforty in to foure nomburs of ten, & so forth.

leaf 138 b. *Compositys beſ nomburs þat bene componyt of a digyt & of an articulle as fouretene, fyftene, sextene, & ſuch oþer. ffortene is componyd of foure þat is a digit & of ten þat is an articulle. ffiftene is componyd of 5 & ten, & ſo of all oþer, what þat þai ben. Short-lych euery nombur þat be-gynnes with a digit & endyth in a articulle is a composyt, as fortene bygennyng by foure þat is a digit, & endes in ten.

What nun
are compo

**¶ Ergo, proposito numero tibi scribere, primo
Respicias quid sit numerus; si digitus sit
Primo scribe loco digitum, si compositus sit
Primo scribe loco digitum post articulum; sic.**

¶ here he telles how þou ſchalt wyrch whan þou ſchalt write a nombur. Expone *versum* sic, & fac iuxta exponentis sentenciam; whan þou haſt a nombur to write, loke fyrſt what maner nombur it ys þat þou ſchalt write, whether it be a digit or a composyt or an Articul. ¶ If he be a digit, write a digit, as yf it be ſeuē, write ſeuē & write þat digit in þe firſt place toward þe ryght ſide. If it be a composyt, write þe digit of þe composyt in þe firſt place & write þe articul of þat digit in þe ſecunde place next toward þe lyft ſide. As yf þou ſchal write ſex & twenty. write þe digit of þe nombur in þe firſt place þat is ſex, and write þe articul next aftur þat is twenty, as þus 26. But whan þou ſchalt sowne or ſpeke leaf 139 a. *or rede an Composyt þou ſchalt firſt ſowne þe articul & aftur þe digit, as þou ſeyſt by þe comyne ſpeche, Sex & twenty & nouȝt twenty & ſex. *versus*.

How to w
number,

if it is a d
if it is a
composite

How to re

**¶ Articulus si sit, in primo limite cifram,
Articulum vero reliquis inscribe figuris.**

¶ Here he tells how þou ſchal write when þe nombre þat þou haſe to write is an Articul. Expone *versus* sic & fac *secundum* sentenciam. Ife þe nombur þat þou haſt write be an Articul, write firſt a cifre & aftur þe cifer write an Articulle þus. 20. fforthermore þou ſchalt vndirſtonde yf þou haue an Articul, loke how mych he is, yf he be with-ynne an hundryth, þou ſchalt write bot one cifre, afore, as here .90. If þe articulle be by hym-

How to w
Articles:

tens,

hundreds,

silfe & be an hundrid euene, þen schal þou write .1. & 2 cifers afore, þat he may stonde in þe thryd place, for euery figure in þe thryd place schal token a hundrid tymes hym selfe. If þe articul be a thousant or thousandes³ and he stonde by hym selfe, write afore 3 cifers & so forþ of al oþer.

thousands

**¶ Quolibet in numero, si par sit prima figura,
Par erit & totum, quicquid sibi continuatur;
Impar si fuerit, totum tunc fiet et impar.**

¶ Here he teches a generalle rewle þat yf þe first figure in þe rewle of figures token a nombur þat is euene al þat nombur of figurys in þat rewle schal be euene, as here þou may see 6. 7. 3. 5. 4. Computa & proba. ¶ If þe first leaf 139 b. *figure token an nombur þat is ode, alle þat nombur in þat rewle schalle be ode, as here 5 6 7 8 6 7. Computa & proba. versus.

To tell an number

or an odd.

**¶ Septem sunt partes, non plures, istius artis;
¶ Addere, subtrahere, duplare, dimidiare,
Sextaque diuidere, sed quinta multiplicare;
Radicem extrahere pars septima dicitur esse.**

Even Rules of Arithmetic.

¶ Here telles þat þer beŋ .7. spices or partes of þis craft. The first is called addicioŋ, þe secunde is called subtraccioŋ. The thryd is called duplacioŋ. The 4. is called dimydicioŋ. The 5. is called multiplicacioŋ. The 6 is called diuisioŋ. The 7. is called extraccioŋ of þe Rote. What all þese spices bene hit schalle be tolde singillatim in here caputule.

The sever

¶ Subtrahis aut addis a dextris vel mediabis:

Thou schal be-gynne in þe ryght side of þe boke or of a tabul. loke were þou wul be-gynne to write latyn or englys in a boke, & þat schalle be called þe lyft side of the boke, þat þou writest toward þat side schal be called þe ryght side of þe boke. Versus.

Add, subt halve, fro to left.

A leua dupla, diuide, multiplica.

Here he telles þe in quych side of þe boke or of þe tabul þou schalle be-gyne to wyrch duplacioŋ, diuisioŋ, and multiplicacioŋ. Thou schal begyne to worch in þe lyft side of þe

boke or of þe tabul, but yn what wyse þou schal wyrch in hym
**dicetur singillatim in sequentibus capitulis et de vtilitate
 cuiuslibet artis & sic Completur** leaf 140. ***prohemium &
 sequitur tractatus & primo de arte addicionis que prima ars
 est in ordine.**

Multiply &
 divide fro
 to right.

raft of Addition.

**Addere si numero numerum vis, ordine tali
 Incipe; scribe duas primo series numerorum
 Primam sub prima recte ponendo figuram,
 Et sic de reliquis facias, si sint tibi plures.**

¶ Here by-gynnes þe craft of Addicioñ. In þis craft þou most
 knowe foure thynges. ¶ Fyrst þou most know what is addicioñ.
 Next þou most know how mony rewles of figurys þou most
 haue. ¶ Next þou most know how mony diuers casys happes in
 þis craft of addicioñ. ¶ And next qwat is þe profet of þis craft.
 ¶ As for þe first þou most know þat addicioñ is a castyng to-
 gedur of twoo nomburys in-to one nombre. As yf I aske qwat is
twene & thre. Pou wyl cast þese twene nombres to-gedur & say
 þat it is fyue. ¶ As for þe secunde þou most know þat þou schalle
 haue tweyne rewes of figures, one vndur a-nother, as here þou
 mayst se. ¶ As for þe thryd þou most know þat there ben 1234
 foure diuerse cases. As for þe forthe þou most know þat þe 2168.
 profet of þis craft is to telle what is þe hole nombur þat comes of
 diuerse nomburis. Now as to þe texte of oure verse, he teches
 there how þou schal worch in þis craft. ¶ He says yf þou wilt cast
 one nombur to anoþer nombur, þou most by-gynne on þis wyse.
 ¶ ffyrst write leaf 140 b. *two rewes of figuris & nombris so þat
 þou write þe first figure of þe hyer nombur euene vndir the first
 figure of þe nether nombur, And þe secunde of þe nether 123
 nombur euene vndir þe secunde of þe hyer, & so forthe of 234.
 euery figure of both þe rewes as þou mayst se.

Four thing
 be known

what it is;

how many
 of figures

how many
 what is its

How to se
 the sum.

ases of the Craft of Addition.

¶ **Inde duas adde primas hac condicione:
 Si digitus crescat ex addicione priorum;**

Primo scribe loco digitum, quicunque sit ille.

¶ Here he teches what þou schalt do when þou hast write too rewe of figuris on vnder an-oþer, as I sayd be-fore. ¶ He says þou schalt take þe first figure of þe heyer nombre & þe fyrst figure of þe neþer nombre, & cast hem to-geder vp-on þis condicion. Thou schal loke qweþer þe number þat comys þere-of be a digit or no. ¶ If he be a digit þou schalt do away þe first figure of þe hyer nombre, and write þere in his stede þat he stode Inne þe digit, þat comes of þe ylke 2 figures, & so wrich forth on oþer figures yf þere be ony moo, til þou come to þe ende toward þe lyft side. And lede þe nether figure stonde still euer-more til þou haue ydo. ffor þere-by þou schal wyte wheþer þou hast done wel or no, as I schal tell þe afterward in þe ende of þis Chapter.

Add the figures;

rub out the figure;

write the 1 in its place

¶ And loke allgate leaf 141 a. þat þou be-gynne to worch in þis Craft of Addi*cion in þe ryȝt side, here is an ensampul of þis case. Caste 2 to foure & þat wel be sex, do away 4. & write 1234 in þe same place þe figure of sex. ¶ And lete þe figure of 2 2142. in þe nether rewe stonde stil. When þou hast do so, cast 3 & 4 to-gedur and þat wel be seuen þat is a digit. Do away þe 3, & set þere seuen, and lete þe neþer figure stonde stille, & so worch forth bakward til þou hast ydo all to-geder.

Here is an example.

Et si compositus, in limite scribe sequente

Articulum, primo digitum; quia sic iubet ordo.

¶ Here is þe secunde case þat may happe in þis craft. And þe case is þis, yf of þe casting of 2 nomburis to-geder, as of þe figure of þe hyer rewe & of þe figure of þe neþer rewe come a Composyt, how schalt þou worch. Þus þou schalt worch. Thou shalt do away þe figure of þe hyer number þat was cast to þe figure of þe neþer number. ¶ And write þere þe digit of þe Composyt. And set þe articul of þe composit next after þe digit in þe same rewe, yf þere be no mo figures after. But yf þere be mo figuris after þat digit. And þere he schall be rekend for hym selfe. And when þou schalt adde þat ylke figure þat berys þe articulle ouer his hed to þe figure vnder hym, þou schalt cast þat articul to þe figure þat hase hym ouer his hed, & þere þat Articul schal token hym selfe. lo an Ensampull leaf 141 b. *of all. Cast 326 to 6, & þere-of wil arise twelue. do away þe hyer 6 & write 216.

Suppose i Composit down the and carry tens.

Here is an example.

þere 2, þat is þe digit of þis composit. And þen write þe articulle þat is ten ouer þe figuris hed of twene as þus. Now cast þe
 articulle þat standus vpon þe figuris of twene hed to þe same
 figure, & reken þat articul bot for one, and þan þere wil arise
 thre. Ðan cast þat thre to þe neþer figure, þat is one, & þat wul be
 foure. do away þe figure of 3, and write þere a figure of foure.
 and lete þe neþer figure stonde stil, & þan worch forth. vnde
 versus.

¶ **Articulus si sit, in primo limite cifram,**
 ¶ **Articulum vero reliquis inscribe figuris,**
Vel per se scribas si nulla figura sequatur.

¶ Here he puttes þe thryde case of þe craft of Addicion. & þe
 case is þis. yf of Addicioun of 2 figuris a-ryse an Articulle, how
 schal þou do. thou most do away þe heer figure þat was addid to
 þe neþer, & write þere a cifre, and sett þe articuls on þe figuris
 hede, yf þat þere come ony after. And wyrch þan as I haue tolde
 þe in þe secunde case. An ensampull. Cast 5 to 5, þat wylle be
 ten. now do away þe hyer 5, & write þere a cifer. And sette
 ten vpon þe figuris hed of 2. And reken it but for on þus. lo an
 Ensampulle And leaf 142 a. *þan worch forth. But yf þere
 come no figure after þe cifre, write þe articul next hym in þe
 same rewe as here cast 5 to 5, and it wel be ten. do away
 5. þat is þe hier 5. and write þere a cifre, & write after
 hym þe articul as þus And þan þou hast done.

Suppose i
 Article, se
 a cipher a
 carry the i

Here is ar
 example.

1
2φ
15

5
5

1φ
5

¶ **Si tibi cifra superueniens occurrerit, illam**
Dele superpositam; fac illic scribe figuram,
Postea procedas reliquas addendo figuras.

¶ Here he puttes þe fourt case, & it is þis, þat yf þere come a
 cifer in þe hier rewe, how þou schal do. þus þou schalt do. do
 away þe cifer, & sett þere þe digit þat comes of þe addicioun as
 þus In þis ensampul ben alle þe foure cases. Cast 3 to
 foure, þat wol be seuen. do away 4. & write þere seuen;
 þan cast 4 to þe figure of 8. þat wel be 12. do away 8, & sett þere
 2. þat is a digit, and sette þe articul of þe composit, þat is ten,
 vpon þe cifers hed, & reken it for hym selfe þat is on. þan cast
 one to a cifer, & hit wulle be but on, for noȝt & on makes but

What to d
 you have
 cipher in i
 row.

An exampl
 all the
 difficultie

1φφ84.
 17743

one. þan cast 7. þat stondes vnder þat on to hym, & þat wel be 8. do away þe cifer & þat 1. & sette þere 8. þan go forthermore. cast þe oþer 7 to þe cifer þat stondes ouer hym. þat wul be bot seuen, for þe cifer betokens noȝt. do away þe cifer & sette þere seuen, leaf 142 b. * & þen go forþermore & cast 1 to 1, & þat wel be 2. do away þe hier 1, & sette þere 2. þan hast þou do. And yf þou haue wel ydo þis number þat is sett here-after wel be þe number þat schalle aryse of alle þe addicion as here 27827. ¶ Sequitur alia species.

raft of Subtraction.

**A numero numerum si sit tibi demere cura
Scribe figurarum series, vt in addicione.**

¶ This is þe Chapter of subtraccion, in the quych þou most know foure nessessary thynges. the first what is subtraccion. þe secunde is how mony numbers þou most haue to subtraccion, the thryd is how mony maners of cases þere may happe in þis craft of subtraccion. The fourte is qwat is þe profet of þis craft. ¶ As for þe first, þou most know þat subtraccion is drawynge of one nowmber oute of anoþer number. As for þe secunde, þou most knowe þat þou most haue two rewes of figuris one vnder anoþer, as þou addyst in addicion. As for þe thryd, þou moyst know þat foure maner of diuerse casis mai happe in þis craft. ¶ As for þe fourt, þou most know þat þe profet of þis craft is whenne þou hasse taken þe lasse number out of þe more to telle what þere leues ouer þat. & þou most be-gynne to wyrch in þis craft in þe ryght side of þe boke, as þou diddyst in addicion. Versus.

Four thing
know abo
subtraccio

the first;
the secon

the third;
the fourth

**¶ Maiori numero numerum suppone minorem,
¶ Siue pari numero supponatur numerus par.**

leaf 143 a. * ¶ Here he telles þat þe hier number most be more þen þe neþer, or els euen as mych. but he may not be lasse. And þe case is þis, þou schalt drawe þe neþer number out of þe hyer, & þou mayst not do þat yf þe hier number were lasse þan þat. ffor þou mayst not draw sex out of 2. But þou mast draw 2 out of sex. And þou maiste draw twene out of twene, for þou schal leue noȝt of þe hier twene vnde versus.

Put the gr
number al
the less.

¶ **Postea si possis a prima subtrahe primam**
Scribens quod remanet.

Here is þe first case put of subtraccion, & he says þou schalt begynne in þe ryght side, & draw þe first figure of þe neþer rewe out of þe first figure of þe hier rewe. qwether þe hier figure be more þen þe neþer, or euen as mych. And þat is notified in þe vers when he says “Si possis.” Whan þou has þus ydo, do away þe hiest figure & sett þere þat leues of þe subtraccion, lo an Ensampulle draw 2 out of 4. þan leues 2. do away 4 & write þere 2, & latte þe neþer figure stonde stille, & so go for-by oþer figuris till þou come to þe ende, þan hast þou do.

The first c
subtraccio

Here is ar
example.

234
122

¶ **Cifram si nil remanebit.**

¶ Here he puttes þe secunde case, & hit is þis. yf it happe þat qwen þou hast draw on neþer figure out of a hier, & þere leue no3t after þe subtraccion, þus leaf 143 b. *þou schalt do. þou schalle do away þe hier figure & write þere a cifer, as lo an Ensampull Take foure out of foure þan leus no3t. þerefore do away þe hier 4 & set þere a cifer, þan take 2 out of 2, þan leues no3t. do away þe hier 2, & set þere a cifer, and so worch whare so euer þis happe.

Put a ciph
nothing re

Here is ar
example.

24
24

Sed si non possis a prima demere primam
Precedens vnum de limite deme sequente,
Quod demptum pro denario reputabis ab illo
Subtrahe totalem numerum quem proposuisti
Quo facto scribe super quicquid remanebit.

Here he puttes þe thryd case, þe quych is þis. yf it happe þat þe neþer figure be more þen þe hier figure þat he schalle be draw out of. how schalle þou do. þus þou schalle do. þou schalle borro .1. oute of þe next figure þat comes after in þe same rewe, for þis case may neuer happ but yf þere come figures after. þan þou schalt sett þat on ouer þe hier figures hed, of the quych þou woldist y-draw oute þe neyþer figure yf þou haddyst y-my3t. Whane þou hase þus ydo þou schalle rekene þat .1. for ten. ¶ And out of þat ten þou schal draw þe neyþermost figure, And

Suppose y
cannot tal
lower figi
from the t
borrow te

take the lc
number fr
ten;

alle þat leues þou schalle adde to þe figure on whos hed þat .1. stode. And þen þou schalle do away alle þat, & sett þere alle that arisys of the addicion of þe ylke 2 figuris. And yf yt leaf 144 a. *happe þat þe figure of þe quych þou schalt borro on be hym self but 1. If þou schalt þat one & sett it vpon þe oþer figuris hed, and sett in þat 1. place a cifer, yf þere come mony figures after. lo an Ensampul. take 4 out of 2. it wyl not be, þerfore borro one of þe next figure, þat is 2. and sett þat ouer þe hed of þe fyrst 2. & rekene it for ten. and þere þe secunde stondes write 1. for þou tokest on out of hym. þan take þe neþer figure, þat is 4, out of ten. And þen leues 6. cast to 6 þe figure of þat 2 þat stode vnder þe hedde of 1. þat was borwed & rekened for ten, and þat wylle be 8. do away þat 6 & þat 2, & sette þere 8, & lette þe neþer figure stonde stille. Whanne þou hast do þus, go to þe next figure þat is now bot 1. but first yt was 2, & þere-of was borred 1. þan take out of þat þe figure vnder hym, þat is 3. hit wel not be. þer-fore borowe of the next figure, þe quych is bot 1. Also take & sett hym ouer þe hede of þe figure þat þou woldest haue y-draw oute of þe nether figure, þe quych was 3. & þou myzt not, & rekene þat borwed 1 for ten & sett in þe same place, of þe quych place þou tokest hym of, a cifer, for he was bot 1. Whanne þou hast þus ydo, take out of þat 1. þat is reKent for ten, þe neþer figure of 3. And þere leues 7. leaf 144 b. *cast þe ylke 7 to þe figure þat had þe ylke ten vpon his hed, þe quych figure was 1, & þat wol be 8. þan do away þat 1 and þat 7, & write þere 8. & þan wyrch forth in oþer figuris til þou come to þe ende, & þan þou hast þe do. Versus.

add the ar
to the top
number.

Example.

2122
1134

How to 'F
back' the
borrowed

¶ **Facque nonenarios de cifris, cum remeabis**

¶ **Occurrant si forte cifre; dum dempseris vnum**

¶ **Postea procedas reliquas demendo figuras.**

¶ Here he puttes þe fourte case, þe quych is þis, yf it happe þat þe neþer figure, þe quych þou schalt draw out of þe hier figure be more þan þe hier figur ouer hym, & þe next figure of two or of thre or of foure, or how mony þere be by cifers, how wold þou do. Þou wost wel þou most nede borow, & þou mayst not borow of þe cifers, for þai haue nozt þat þai may lene or spare. Ergo⁴ how woldest þou do. Certayn þus most þou do, þou most borow

A very ha
is put.

on of þe next figure significatyf in þat rewe, for þis case may not happe, but yf þere come figures significatyf after the cifers. Whan þou hast borowede þat 1 of the next figure significatyf, sett þat on ouer þe hede of þat figure of þe quych þou wold haue draw þe neþer figure out yf þou hadest myzt, & reken it for ten as þou diddest in þe oþer case here-a-fore. Whan þou hast þus y-do loke how mony cifers þere were bye-twene þat figure significatyf, & þe figure of þe quych þou woldest haue y-draw the leaf 145 a. *neþer figure, and of euery of þe ylke cifers make a figure of 9. lo an Ensampulle after. Take 4 out of 2. it wel not be. borow 1 out of þe next figure significatyf, þe quych is 4, & þen leues 3. do away þat figure of 4 & write þere 3. & sett þat 1 vpon þe figure of 2 hede, & þan take 4 out of ten, & þan þere leues 6. Cast 6 to the figure of 2, þat wol be 8. do away þat 6 & write þere 8. Whan þou hast þus y-do make of euery 0 betweyn 3 & 8 a figure of 9, & þan worch forth in goddes name. & yf þou hast wel y-do þou⁵ schalt haue þis number

40002
10004

Here is an example.

39998
10004

Sic.

o prove the Subtraction.

**¶ Si subtraccio sit bene facta probare valebis
Quas subtraxisti primas addendo figuras.**

¶ Here he teches þe Craft how þou schalt know, whan þou hast subtrayd, wheþer þou hast wel ydo or no. And þe Craft is þis, ryght as þou subtrayd þe neþer figures fro þe hier figures, ryzt so adde þe same neþer figures to þe hier figures. And yf þou haue well y-wroth a-fore þou schalt haue þe hier nombre þe same þou haddest or þou be-gan to worch. as for þis I bade þou schulde kepe þe neþer figures styll. lo an leaf 145 b. *Ensampulle of alle þe 4 cases togedre. worche welle þis case And yf þou worch welle whan þou hast alle subtrayd þe þat hier nombre here, þis schalle be þe nombre here foloyng whan þou hast subtrayd. And þou schalt know þus. adde þe neþer rowe of þe same nombre to þe hier rewe as þus, cast 4 to 4. þat wol be 8. do away þe 4 & write þere 8. by þe first case of addicion. þan cast 6 to 0 þat wol be 6. do away þe 0, & write þere 6. þan cast 6 to 8, þat wel be 14. do

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How to prove subtraction

39998804
20004664

Here is another example.

Our author makes a 3 for 1.

away 8 & write þere a figure of 4, þat is þe digit, and write a figure of 1. þat schall be-token ten. þat is þe articul vpon þe hed of 8 next after, þan reken þat 1. for 1. & cast it to 8. þat schal be 9. cast to þat 9 þe neþer figure vnder þat þe quych is 4, & þat schalle be 13. do away þat 9 & sett þere 3, & sett a figure of 1. þat schall be 10 vpon þe next figuris hede þe quych is 9. by þe secunde case þat þou hadest in addicion. þan cast 1 to 9. & þat wol be 10. do away þe 9. & þat 1. And write þere a cifer. and write þe articulle þat is 1. betokenynge 10. vpon þe hede of þe next figure toward þe lyft side, þe quych ^{leaf 146 a.} *is 9, & so do forth tyl þou come to þe last 9. take þe figure of þat 1. þe quych þou schalt fynde ouer þe hed of 9. & sett it ouer þe next figures hede þat schal be 3. ¶ Also do away þe 9. & set þere a cifer, & þen cast þat 1 þat stondes vpon þe hede of 3 to þe same 3, & þat schalle make 4, þen caste to þe ylke 4 the figure in þe neyþer rewe, þe quych is 2, and þat schalle be 6. And þen schal þou haue an Ensampulle aʒeyn, loke & se, & but þou haue þis same þou hase myse-wroʒt.

He works
proof thrc

and bring
result.

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raft of Duplation.

Sequitur de duplacione

Si vis duplare numerum, sic incipe primo

Scribe figurarum seriem quamcunque velis tu.

¶ This is the Chapture of duplacion, in þe quych craft þou most haue & know 4 thinges. ¶ Þe first þat þou most know is what is duplacion. þe secunde is how many rewes of figures þou most haue to þis craft. ¶ Þe thryde is how many cases may ⁶ happe in þis craft. ¶ Þe fourte is what is þe profet of þe craft. ¶ As for þe first. duplacion is a doublynge of a nombre. ¶ As for þe secunde þou most ^{leaf 146 b.} *haue on nombre or on rewe of figures, the quych called *numerus duplandus*. As for þe thrid þou most know þat 3 diuerse cases may hap in þis craft. As for þe fourte. qwat is þe profet of þis craft, & þat is to know what a-risyʒt of a nombre I-doublyde. ¶ fforþer-more, þou most know & take gode hede in quych side þou schalle be-gyn in þis craft, or ellis þou mayst

Four thing
be known
Duplation

Here they

Mind whe
begin.

spyl alle þi laber þere aboute. certeyn þou schalt begyn in the lyft side in þis Craft. thenke wel ouer þis verse. ¶ ⁷A leua dupla, diuide, multiplica. ⁷

The sentens of þes verses afore, as þou may see if þou take hede. As þe text of þis verse, þat is to say, ¶ Si vis duplare. þis is þe sentence. ¶ If þou wel double a nombre þus þou most be-gynn. Write a rewe of figures of what nombre þou welt. *versus*.

Remember rules.

**Postea procedas primam duplando figuram
Inde quod excrescit scribas vbi iusserit ordo
Iuxta precepta tibi que dantur in addicione.**

¶ Here he telles how þou schalt worch in þis Craft. he says, fyrst, whan þou hast writen þe nombre þou schalt be-gyn at þe first figure in the lyft side, & doubulle þat figure, & þe nombre þat comes þere-of þou schalt write as þou diddyst in addicion, as ¶ I schal telle þe in þe case. *versus*.

How to write sum.

ases of the Craft of Duplation.

leaf 147 a.

*** ¶ Nam si sit digitus in primo limite scribas.**

¶ Here is þe first case of þis craft, þe quych is þis. yf of duplacion of a figure arise a digit. what schal þou do. þus þou schal do. do away þe figure þat was doublede, & sett þere þe diget þat comes of þe duplacion, as þus. 23. double 2, & þat wel be 4. do away þe figure of 2 & sett þere a figure of 4, & so worch forth till þou come to þe ende. *versus*.

If the answer digit, write it in place of the figure.

**¶ Articulus si sit, in primo limite cifram,
¶ Articulum vero reliquis inscribe figuris;
¶ Vel per se scribas, si nulla figura sequatur.**

¶ Here is þe secunde case, þe quych is þis yf þere come an articulle of þe duplacion of a figure þou schalt do ryzt as þou diddyst in addicion, þat is to wete þat þou schalt do away þe figure þat is doublet & sett þere a cifer, & write þe articulle ouer þe next figuris hede, yf þere be any after-warde toward þe lyft side as þus. 25. begyn at the lyft side, and doubulle 2. þat wel be 4. do away þat 2 & sett þere 4. þan doubul 5. þat wel be 10. do

If it is an

put a cipher the place, 'carry' the

away 5, & sett þere a 0, & sett 1 vpon þe next figuris hede þe quych is 4. & þen draw downe 1 to 4 & þat wolle be 5, & þen do away þat 4 & þat 1, & sett þere 5. for þat 1 schal be rekened in þe drawynge togedre for 1. wen leaf 147 b. *þou hast ydon þou schalt haue þis nombre 50. yf þere come no figure after þe figure þat is addit, of þe quych addicion comes an articulle, þou schalt do away þe figure þat is dowblet & sett þere a 0. & write þe articul next by in þe same rewe toward þe lyft syde as þus, 523. double 5 þat woll be ten. do away þe figure 5 & set þere a cifer, & sett þe articul next after in þe same rewe toward þe lyft side, & þou schalt haue þis nombre 1023. þen go forth & double þe oper numbers þe quych is ly3t y-now3t to do. *versus.*

If there is figure to 'them to, v them dow

**¶ Compositus si sit, in limite scribe sequente
Articulum, primo digitum; quia sic iubet ordo:
Et sic de reliquis faciens, si sint tibi plures.**

¶ Here he puttes þe Thryd case, þe quych is þis, yf of duplacion of a figure come a Composit. þou schalt do away þe figure þat is doublet & set þere a digit of þe Composit, & sett þe articulle ouer þe next figures hede, & after draw hym downe with þe figure ouer whos hede he stondes, & make þere-of an nombre as þou hast done afore, & yf þere come no figure after þat digit þat þou hast y-write, þan set þe articulle next after hym in þe same rewe as þus, 67: double 6 þat wel be 12, do away 6 & write þere þe digit leaf 148 a. *of 12, þe quych is 2, and set þe articulle next after toward þe lyft side in þe same rewe, for þere comes no figure after. þan dowble þat oper figure, þe quych is 7, þat wel be 14. the quych is a Composit. þen do away 7 þat þou doublet & sett þe þe diget of hym, the quych is 4, sett þe articulle ouer þe next figures hed, þe quych is 2, & þen draw to hym þat on, & make on nombre þe quych schalle be 3. And þen yf þou haue wel y-do þou schalle haue þis nombre of þe duplacion, 134. *versus.*

If it is a Composit

write dow digit, and the tens.

Here is an example.

**¶ Si super extremam nota sit monadem dat eidem
Quod tibi contingat si primo dimidiabis.**

¶ Here he says, yf ouer þe fyrst figure in þe ry3t side be such a merke as is here made, —^w, þou schalle fyrst doubulle þe figure, the quych stondes vnder þat merke, & þen þou schalt doubul þat

How to do the mark : half.

merke þe quych stondes for haluendel on. for too haluedels makes on, & so þat wol be on. cast þat on to þat duplacion of þe figure ouer whos hed stode þat merke, & write it in þe same place þere þat þe figure þe quych was doublet stode, as þus 23^w. double 3, þat wol be 6; doubul þat halue on, & þat wol be on. cast on to 6, þat wel be 7. do away 6 & þat 1, & sett þere 7. þan hase þou do. as for þat figure, þan go leaf 148 b. *to þe oþer figure & worch forth. & þou schall neuer haue such a merk but ouer þe hed of þe furst figure in þe ryght side. And zet it schal not happe but yf it were y-halued a-fore, þus þou schalt vnderstonde þe verse. ¶ Si super extremam &c. Et nota, talis figura^w significans medietatem, unitatis veniat, i.e. contingat uel fiat super extremam, i.e. super primam figuram in extremo sic versus dextram ars dat: i.e. reddit monadem. i.e. vnitatem eidem. i.e. eidem note & declinatur hec monos, dis, di, dem, &c. ¶ Quod ergo totum hoc dabis monadem note continget. i.e. eveniet tibi si dimidiasti, i.e. accipisti uel subtulisti medietatem alicuius unius, in cuius principio sint figura numerum denotans imparem primo i.e. principiis.

This can c
stand ove
first figur

raft of Mediation.

¶ Sequitur de mediacione.

Incipe sic, si vis aliquem numerum mediare:

Scribe figurarum seriem solam, velut ante.

¶ In þis Chapter is tazt þe Craft of mediacioun, in þe quych craft þou most know 4 thynges. ffurst what is mediacion. the secunde how many rewes of figures þou most haue in þe wyrchyng of þis craft. þe thryde how many diuerse cases may happ in þis craft.⁸ ¶ As for þe furst, þou schalt vndurstonde þat mediacion is a takyng out of halfe a number out of a holle number, leaf 149 a. *as yf þou wolde take 3 out of 6. ¶ As for þe secunde, þou schalt know þat þou most haue one rewe of figures, & no moo, as þou hayst in þe craft of duplacion. ¶ As for the thryd, þou most vnderstonde þat 5 cases may happe in þis craft. ¶ As for þe

The four t
to be kno
mediation

the first

the secon

the third;
the fourth

fourte, þou schalle know þat the profet of þis craft is when þou hast take away þe haluendel of a nombre to telle qwat þere schalle leue. ¶ Incipe sic, &c. The sentence of þis verse is þis. yf þou wold medye, þat is to say, take halfe out of þe holle, or halfe out of halfe, þou most begynne þus. Write one rewe of figures of what nombre þou wolte, as þou dyddyst be-fore in þe Craft of duplacion. *versus*.

Begin thu

¶ **Postea procedas medians, si prima figura
Si par aut impar videas.**

¶ Here he says, when þou hast write a rewe of figures, þou schalt take hede wheþer þe first figure be euen or odde in nombre, & vnderstonde þat he spekes of þe first figure in þe ryȝt side. And in the ryȝt side þou schalle begynne in þis Craft.

See if the number is or odd.

¶ **Quia si fuerit par,
Dimidiabis eam, scribens quicquid remanebit:**

¶ Here is the first case of þis craft, þe quych is þis, yf þe first figure be euen. þou schal take away fro þe figure euen halfe, & do away þat figure and set þere þat leues ouer, as þus, 4. take leaf
149 b. *halfe out of 4, & þan þere leues 2. do away 4 & sett þere 2. þis is lyght y-nowȝt. *versus*.

If it is eue halve it, a write the in its plac

mediation of an Odd Number.

¶ **Impar si fuerit vnum demas mediare
Quod non presumas, sed quod superest mediabis
Inde super tractum fac demptum quod notat vnum.**

Here is þe secunde case of þis craft, the quych is þis. yf þe first figure betokene a nombre þat is odde, the quych odde schal not be mediete, þen þou schalt medye þat nombre þat leues, when the odde of þe same nombre is take away, & write þat þat leues as þou diddest in þe first case of þis craft. Whan þou hayst write þat. for þat þat leues, write such a merke as is here ^w vpon his hede, þe quych merke schal betoken halfe of þe odde þat was take away. lo an Ensampull. 245. the first figure here is betokenynge odde nombre, þe quych is 5, for 5 is odde; þerefore do away þat þat is odde, þe quych is 1. þen leues 4. þen

If it is odd the even r less than i

Then writ sign for o over it.
Here is an example.

medye 4 & þen leues 2. do away 4. & sette þere 2, & make such a merke^w upon his hede, þat is to say ouer his hede of 2 as þus. 242.^w And þen worch forth in þe oþer figures tyll þou come to þe ende. by þe furst case as þou schalt vnderstonde þat þou schalt leaf 150 a. *neuer make such a merk but ouer þe first figure hed in þe ryzt side. Wheþer þe other figures þat comyn after hym be euen or odde. versus.

Put the m
only over
first figur

ases of the Craft of Mediation.

¶ **Si monos, dele; sit tibi cifra post nota supra.**

¶ Here is þe thryde case, þe quych yf the first figure be a figure of 1. þou schalt do away þat 1 & set þere a cifer, & a merke ouer þe cifer as þus, 241. do away 1, & sett þere a cifer with a merke ouer his hede, & þen hast þou ydo for þat 0. as þus 0^w þen worch forth in þe oþer figurys till þou come to þe ende, for it is lyght as dyche water. vnde versus.

If the first
is one put
cipher.

¶ **Postea procedas hac condicione secunda:
Impar si fuerit hinc vnum deme priori,
Inscribens quinque, nam denos significabit
Monos predictam.**

¶ Here he puttes þe fourte case, þe quych is þis. yf it happen the secunde figure betoken odde nombre, þou schal do away on of þat odde nombre, þe quych is significatiue by þat figure 1. þe quych 1 schall be rekende for 10. Whan þou hast take away þat 1 out of þe nombre þat is signifiede by þat figure, þou schalt medie þat þat leues ouer, & do away þat figure þat is medied, & sette in his stye halfe of þat nombre. ¶ Whan þou hase so done, þou schalt write leaf 150 b. *a figure of 5 ouer þe next figures hede byfore toward þe ryzt side, for þat 1, þe quych made odd nombre, schall stonde for ten, & 5 is halfe of 10; so þou most write 5 for his haluendelle. lo an Ensampulle, 4678. begyn in þe ryzt side as þou most nedes. medie 8. þen þou schalt leue 4. do away þat 8 & sette þere 4. þen out of 7. take away 1. þe quych makes odde, & sett 5. vpon þe next figures hede afore toward þe ryzt side, þe quych is now 4. but afore it was 8. for þat 1 schal be rekenet for

What to d
any other
is odd.

Write a fi
five over
next lowe
number's

Example.

10, of þe quych 10, 5 is halfe, as þou knowest wel. Whan þou hast þus ydo, medye þat þe quych leues after þe takyng away of þat þat is odde, þe quych leuyng schalle be 3; do away 6⁵ & sette þere 3, & þou schalt haue such a nombre after go⁴⁶³⁴ forth to þe next figure, & medy þat, & worch forth, for it is lyzt ynov3t to þe certayn.

¶ Si vero secunda dat vnum.

Illa deleta, scribatur cifra; priori

¶ Tradendo quinque pro denario mediato;

Nec cifra scribatur, nisi deinde figura sequatur:

Postea procedas reliquas mediando figuras

Vt supra docui, si sint tibi mille figure.

¶ Here he puttes þe 5 case, þe quych is leaf 151 a. *þis: yf þe secunde figure be of 1, as þis is here 12, þou schalt do away þat 1 & sett þere a cifer. & sett 5 ouer þe next figure hede afore toward þe ri3t side, as þou diddyst afore; & þat 5 schal be halde of þat 1, þe quych 1 is re kent for 10. lo an Ensampulle, 214. medye 4. þat schalle be 2. do away 4 & sett þere 2. þen go forth to þe next figure. þe quych is bot 1. do away þat 1. & sett þere a cifer. & set 5 vpon þe figures hed afore, þe quych is nowe 2, & þen þou schalt haue þis nombre þen worch forth to þe nex⁵ figure. And also it is no maystery. yf þere come no figure²⁰² after þat on is medyet, þou schalt write no 0. ne now3t ellis, but set 5 ouer þe next figure afore toward þe ry3t, as þus 14. medie 4 then leues 2, do away 4 & sett þere 2. þen medie 1. þe quich is rekende for ten, þe haluendel þere-of wel be 5. sett þat 5 vpon þe hede of þat figure, þe quych is now 2, & do away þat 1, & þou⁵ schalt haue þis nombre yf þou worch wel, vnde versus.^{2,}

If the sec
figure is c
a cipher, a
write five
the next f

How to h
fourteen.

o prove the Mediation.

¶ Si mediatio sit bene facta probare valebis

¶ Duplando numerum quem primo dimediasti

¶ Here he telles þe how þou schalt know wheþer þou hase wel ydo or no. doubul leaf 151 b. *þe nombre þe quych þou hase mediet, and yf þou haue wel y-medyt after þe dupleacion, þou schalt haue þe same nombre þat þou haddyst in þe tabulle or þou

How to p
your med

began to medye, as þus. ¶ The furst ensampulle was þis. 4. þe
 quych I-mediet was laft 2, þe whych 2 was write in þe place þat
 4 was write afore. Now doubulle þat 2, & þou schal haue 4, as
 þou hadyst afore. þe secunde Ensampulle was þis, 245. When
 þou haddyst mediet alle þis nombre, yf þou haue wel ydo þou
 schalt haue of þat mediacion þis nombre, 122^w . Now doubulle
 þis nombre, & begyn in þe lyft side; doubulle 1, þat schal be 2.
 do away þat 1 & sett þere 2. þen doubulle þat oþer 2 & sett þere
 4, þen doubulle þat oþer 2, & þat wel be 4. þen doubul þat merke
 þat stondes for halue on. & þat schalle be 1. Cast þat on to 4, & it
 schalle be 5. do away þat 2 & þat merke, & sette þere 5, & þen
 þou schal haue þis nombre 245. & þis was þe same nombur þat
 þou haddyst or þou began to medye, as þou mayst se yf þou take
 hede. The nombre þe quych þou haddist for an Ensampul in þe 3
 case of mediacion to be mediet was þis 241. whan þou haddist
 mediet alle þis nombur truly leaf 152 a. *by euery figure, þou
 schall haue be þat mediacion þis nombur 120^w . Now dowbul þis
 nombur, & begyn in þe lyft side, as I tolde þe in þe Craft of
 duplacion. þus doubulle þe figure of 1, þat wel be 2. do away þat
 1 & sett þere 2, þen doubul þe next figure afore, the quych is 2,
 & þat wel be 4; do away 2 & set þere 4. þen doubul þe cifer, &
 þat wel be noȝt, for a 0 is noȝt. And twyes noȝt is but noȝt.
 þerefore doubul the merke aboue þe cifers hede, þe quych
 betokenes þe haluendel of 1, & þat schal be 1. do away þe cifer
 & þe merke, & sett þere 1, & þen þou schalt haue þis nombur
 241. And þis same nombur þou haddyst afore or þou began to
 medy, & yf þou take gode hede. ¶ The next ensampul þat had in
 þe 4 case of mediacion was þis 4678. Whan þou hast truly
ymedit alle þis nombur fro þe begynnynge to þe endynge, þou
 schalt haue of þe mediacion þis nombur Now doubul this
 nombur & begyn in þe lyft side, & doubulle 2 þat schal be ⁵ 2334.
 4. do away 2 and sette þere 4; þen doubule 3, þat wol be 6; do
 away 3 & sett þere 6, þen doubul þat oþer 3, & þat wel be 6; do
 away 3 & set þere leaf 152 b. *6, þen doubul þe 4, þat welle be 8;
 þen doubul 5. þe quych stondes ouer þe hed of 4, & þat wol be
 10; cast 10 to 8, & þat schal be 18; do away 4 & þat 5, & sett
 þere 8, & sett that 1, þe quych is an articul of þe Composit þe

First exan

The secor

The third
example.

The fourtl
example.

quych is 18, ouer þe next figures hed toward þe lyft side, þe quych is 6. drav þat 1 to 6, þe quych 1 in þe dravyng schal be rekente bot for 1, & þat 1 & þat 6 togedur wel be 7. do away þat 6 & þat 1. the quych stondes ouer his hede, & sett ther 7, & þen þou schalt haue þis nombur 4678. And þis same nombur þou hadyst or þou began to medye, as þou mayst see in þe secunde Ensampul þat þou had in þe 4 case of mediacion, þat was þis: when þou had mediet truly alle the nombur, a *principio usque ad finem*. þou schalt haue of þat mediacion þis nombur Now ⁵ doubul 1. þat wel be 2. do away 1 & sett þere 2. þen doubul ^{102.} 0. þat will be noȝt. þefore take þe 5, þe quych stondes ouer þe next figures hed, & doubul it, & þat wol be 10. do away þe 0 þat stondes betwene þe two figuris, & sette þere in his stid 1, for þat 1 now schal stonde in þe secunde place, where he schal betoken 10; þen doubul 2, þat wol be 4. do away 2 & sett þere 4. & ^{leaf 153} a. *þou schal haue þus nombur 214. þis is þe same numbur þat þou hadyst or þou began to medye, as þou may see. And so do euer more, yf þou wil knowe wheþer þou hase wel ymedyt or no. ¶. doubulle þe numbur þat comes after þe mediacion, & þou schal haue þe same nombur þat þou hadyst or þou began to medye, yf þou haue welle ydo. or els doute þe noȝt, but yf þou haue þe same, þou hase faylide in þi Craft.

The fifth example.

raft of Multiplication.

Sequitur de multiplicatione.

te down a Multiplication Sum.

S i tu per numerum numerum vis multiplicare

Scribe duas quascunque velis series numerorum

Ordo servetur vt vltima multiplicandi

Ponatur super anteriorem multiplicantis

A leua relique sint scripte multiplicantes.

¶ Here be-gynnes þe Chaptre of multiplication, in þe quych þou most know 4 thynges. ¶ Ffirst, qwat is multiplicacion. The secunde, how mony cases may hap in multiplicacion. The thryde,

Four thing known of Multiplic:

how many rewe of figures þere most be. ¶ The 4. what is þe profet of þis craft. ¶ As for þe first, þou schal vnderstonde þat multiplicacion is a bryngynge to-geder of 2 thynges in on nombur, þe quych on nombur contynes so many tymes on, howe leaf 153 b. *many tymes þere ben vnytees in þe nowmbre of þat 2, as twyes 4 is 8. now here ben þe 2 numbers, of þe quych too nowmbres on is betokened be an aduerbe, þe quych is þe worde twyes, & þis worde thryes, & þis worde foure sythes,⁹ & so furth of such other lyke wordes. ¶ And tweyn nombres schal be tokenyde be a nowne, as þis worde foure showys þes tweyn nombres y-broth in-to on hole nombur, þat is 8, for twyes 4 is 8, as þou wost wel. ¶ And þes nombre 8 conteynes as oft tymes 4 as þere ben vnites in þat other nombre, þe quych is 2, for in 2 ben 2 vnites, & so oft tymes 4 ben in 8, as þou wottys wel. ¶ ffor þe secunde, þou most know þat þou most haue too rewe of figures. ¶ As for þe thryde, þou most know þat 8 maner of diuerse case may happe in þis craft. The profet of þis Craft is to telle when a nombre is multiplyed be a noþer, qwat commys þere of. ¶ fforthermore, as to þe sentence of oure verse, yf þou wel multiply a nombur be a-noþer nombur, þou schalt write leaf 154 a. *a rewe of figures of what nomburs so euer þou welt, & þat schal be called Numerus multiplicandus, Anglice, þe nombur the quych to be multiplied. þen þou schalt write a-nother rewe of figures, by þe quych þou schalt multiplie the nombre þat is to be multiplied, of þe quych nombur þe furst figure schal be write vnder þe last figure of þe nombur, þe quych is to be multiplied. And so write forthe toward þe lyft side, as here you may se, And þis one nombur schalle be called numerus multiplicans.

67324
1234

Anglice, þe nombur multipliynge, for he schalle multiply þe hyer nounbur, as þus one tyme 6. And so forth, as I schal telle the afterwarde. And þou schal begyn in þe lyft side. ¶ ffor-þere-more þou schalt vndurstonde þat þere is two manurs of multiplicacion; one ys of þe wyrchyng of þe boke only in þe mynde of a mon. fyrst he teches of þe fyrst maner of duplacion, þe quych is be wyrchyng of tabuls. Afterwarde he wol teche on þe secunde maner. vnde versus.

the first:

the second

the third:
the fourth

The
multiplica

How to se
the sum.

Two sorts
Multiplica
mentally,

and on pa

ltiply one Digit by another.

In digitum cures digitum si ducere maior

leaf 154 b.

* **Per quantum distat a denis respice debes**

¶ **Namque suo decuplo totiens delere minorem**

Sitque tibi numerus veniens exinde patebit.

¶ Here he teches a rewle, how þou schalt fynde þe nounbre þat comes by þe multiplicacion of a digit be anoper. loke how mony [vny]tes ben. bytwene þe more digit and 10. And reken ten for on vnite. And so oft do away þe lasse nounbre out of his owne decuple, þat is to say, fro þat nounbre þat is ten tymes so mych is þe nounbre þat comes of þe multiplicacion. As yf þou wol multiply 2 be 4. loke how mony vnitees ben by-twene þe quych is þe more nounbre, & be-twene ten. Certen þere wel be vj vnitees by-twene 4 & ten. yf þou reken þere with þe ten þe vnite, as þou may se. so mony tymes take 2. out of his decuple, þe quych is 20. for 20 is þe decuple of 2, 10 is þe decuple of 1, 30 is þe decuple of 3, 40 is þe decuple of 4, And þe oper digetes til þou come to ten; & whan þou hast y-take so mony tymes 2 out of twenty, þe quych is sex tymes, þou schal leue 8 as þou wost wel, for 6 times 2 is twelue. take [1]2 out of twenty, & þere schal leue 8. bot yf bothe þe digettes leaf 155 a. *ben y-lyech mych as here. 222 or too tymes twenty, þen it is no fors quych of hem tweyn þou take out of here decuple. als mony tymes as þat is fro 10. but neuer-þe-lesse, yf þou haue hast to worch, þou schalt haue here a tabul of figures, where-by þou schalt se a-nonȝ ryght what is þe nounbre þat comes of þe multiplicacion of 2 digittes. þus þou schalt worch in þis figure.

How to m
two digits

Subtract t
greater fro

take the le
many tim
ten times

Example.

Better use
table, thoi

1									
2	4								
3	6	9							
4	8	12	16						
5	10	15	20	25					
6	12	18	24	30	36				
7	14	21	28	35	42	49			
8	16	24	32	40	48	56	64		
9	18	27	36	45	54	63	72	81	

1	2	3	4	5	6	7	8	9	
---	---	---	---	---	---	---	---	---	--

yf þe figure, þe quych schalle be multiplied, be euene as mych as þe diget be, þe quych þat oper figure schal be multiplied, as two tymes twayn, or thre tymes 3. or sych other. loke qwere þat figure sittes in þe lyft side of þe triangle, & loke qwere þe diget sittes in þe neþer most rewe of þe triangle. & go fro hym vpwarde in þe same rewe, þe quych rewe gose vpwarde til þou come agaynes þe oper digette þat sittes in þe lyft side of þe triangle. And þat nounbre, þe quych þou leaf 155 b. fyn*des þere is þe nounbre þat comes of the multiplicacion of þe 2 digittes, as yf þou wold wete qwat is 2 tymes 2. loke quere sittes 2 in þe lyft side in þe first rewe, he sittes next 1 in þe lyft side al on hye, as þou may se; þe[n] loke qwere sittes 2 in þe lowyst rewe of þe triangle, & go fro hym vpwarde in þe same rewe tylle þou come a-3enenes 2 in þe hyer place, & þer þou schalt fynd ywrite 4, & þat is þe nounbre þat comes of þe multiplicacion of two tymes tweyn is 4, as þow wotest welle. yf þe diget. the quych is multiplied, be more þan þe oper, þou schalt loke qwere þe more diget sittes in þe lowest rewe of þe triangle, & go vpwarde in þe same rewe tyl ¹⁰ þou come a-nendes þe lasse diget in the lyft side. And þere þou schalt fynde þe nombre þat comes of þe multiplicacion; but þou schalt vnderstonde þat þis rewle, þe quych is in þis verse. ¶ In digitum cures, &c., noþer þis triangle schalle not serue, bot to fynde þe nounbres þat comes of the multiplicacion þat comes of 2 articuls or composites, þe nedes no craft but yf þou wolt multiply in þi mynde. And leaf 156 a. *þere-to þou schalt haue a craft afterwarde, for þou schall wyrch with digettes in þe tables, as þou schalt know afterwarde. versus.

How to us

The way to
the
Multiplica-
table.

ltiply one Composite by another.

¶ **Postea procedas postremam multiplicando**
[Recte multiplicans per cunctas inferiores]
Condicionem tamen tali quod multiplicantes
Scribas in capite quicquid processerit inde
Sed postquam fuit hec multiplicare figure
Anteriores tunc sereis multiplicantis
Et sic multiplica velut isti multiplicasti

Qui sequitur numerum scriptum quiscunque figuris.

¶ Here he teches how þou schalt wyrch in þis craft. þou schalt multiplye þe last figure of þe nombre, and quen þou hast so ydo þou schalt draw alle þe figures of þe neþer nounbre more taward þe ryzt side, so qwen þou hast multiplied þe last figure of þe heyer nounbre by alle þe neþer figures. And sette þe nounbir þat comes þer-of ouer þe last figure of þe neþer nounbre, & þen þou schalt sette al þe oþer figures of þe neþer nounbre more nere to þe ryzt side. ¶ And whan þou hast multiplied þat figure þat schal be multiplied þe next after hym by al þe neþer figures. And worch as þou dyddyst afore til leaf 156 b. *þou come to þe ende. And þou schalt vnderstonde þat euery figure of þe hier nounbre schal be multiplied be alle þe figures of the neþer nounbre, yf þe hier nounbre be any figure þen one. lo an Ensampul here folowyng. þou schalt begyne to multiplye in þe lyft side. Multiply 2 be 2, and twyes 2 is 4. set 4 ouer þe hed of þat 2, þen multiplie þe same hier 2 by 3 of þe nether nounbre, as thryes 2 þat schal be 6. set 6 ouer þe hed of 3, þan multiplie þe same hier 2 by þat 2 þe quych stondes vnder hym, þat wol be 4; do away þe hier 2 & sette þere 4. ¶ Now þou most antery þe nether nounbre, þat is to say, þou most sett þe neþer nounbre more towarde þe ryzt side, as þus. Take þe neþer 2 toward þe ryzt side, & sette it euen vnder þe 4 of þe hyer nounbre, & antery alle þe figures þat comes after þat 2, as þus; sette 2 vnder þe 4. þen sett þe figure of 3 þere þat þe figure of 2 stode, þe quych is now vndur þat 4 in þe hier nounbre; þen sett þe oþer figure of 2, þe quych is þe last figure toward þe lyft side of þe neþer number þere þe figure of 3 stode. þen þou schalt haue such a nombre. leaf 157 a. * ¶ Now multiply 4, þe quych comes next after 6, by þe last 2 of þe neþer nounbur toward þe lyft side. as 2 tymes 4, þat wel be 8. sette þat 8 ouer þe figure the quych stondes ouer þe hede of þat 2, þe quych is þe last figure of þe neþer nounbre; þan multiplie þat same 4 by 3, þat comes in þe neþer rewe, þat wol be 12. sette þe digit of þe composyt ouer þe figure þe quych stondes ouer þe hed of þat 3, & sette þe articule of þis composit ouer al þe figures þat stondes ouer þe neþer 2 hede. þen multiplie þe same 4 by þe 2 in þe ryzt side in þe neþer nounbur, þat wol be 8. do

How to m
one numb
another.

Multiply i
'last' figu
the higher
'first' of t
lower nur

Set the an
ouer the f
the lower:

then mult
second of
lower, and

Then ante
lower nur

as thus.

Now muli
the last bu

2465.
232

464465
232

away 4. & sette þere 8. Euer more qwen þou multiplies þe hier figure by þat figure þe quych stondes vnder hym, þou schalt do away þat hier figure, & sett þer þat nounbre þe quych comes of multiplicacion of ylke digittes. Whan þou hast done as I haue byde þe, þou schalt haue suych an order of figure as is here, þen take and antery þi neþer figures. And sett þe fyrst figure of þe neþer figures¹¹ vndre be figure of 6. ¶ And draw al þe oþer figures of þe same rewe to hym-warde, leaf 157 b. *as þou diddyst afore. þen multiplie 6 be 2, & sett þat þe quych comes ouer þere-of ouer al þe oþer figures hedes þat stondes ouer þat 2. þen multiply 6 be 3, & sett alle þat comes þere-of vpon alle þe figures hedes þat standes ouer þat 3; þan multiplie 6 be 2, þe quych stondes vnder þat 6, þen do away 6 & write þere þe digitt of þe composit þat schal come þereof, & sette þe articull ouer alle þe figures þat stondes ouer þe hede of þat 3 as here, þen antery þi figures as þou diddyst afore, and multipli 5 be 2, þat wol be 10; sett þe 0 ouer all þe figures þat stonden ouer þat 2, & sett þat 1. ouer the next figures hedes, alle on hye towarde þe lyft side. þen multiplie 5 be 3. þat wol be 15, write 5 ouer þe figures hedes þat stonden ouer þat 3, & sett þat 1 ouer þe next figures hedes toward þe lyft side. þen multiplie 5 be 2, þat wol be 10. do away þat 5 & sett þere a 0, & sett þat 1 ouer þe figures hedes þat stonden ouer 3. And þen leaf 158 a. þou schalt haue such a nounbre as here stondes aftur.* ¶ Now draw alle þese figures downe togeder as þus, 6.8.1. & 1 draw to-gedur; þat wolle be 16, do away alle þese figures saue 6. lat hym stonde, for þow þou take hym away þou most write þer þe same azene. þerefore late hym stonde, & sett 1 ouer þe figure hede of 4 toward þe lyft side; þen draw on to 4, þat wolle be 5. do away þat 4 & þat 1, & sette þere 5. þen draw 4221 & 1 togedur, þat wol be 10. do away alle þat, & write þere þat 4 & þat 0, & sett þat 1 ouer þe next figures hede toward þe lyft side, þe quych is 6. þen draw þat 6 & þat 1 togedur, & þat wolle be 7; do away 6 & sett þere 7, þen draw 8810 & 1, & þat wel be 18; do away alle þe figures þat stondes ouer þe hede of þat 8, & lette 8 stonde stil, & write þat 1 ouer þe next figuris hede, þe quych is a 0. þen do away þat 0, & sett þere 1, þe quych

1
82
4648[65]
232.

11
121
828
464825
232

11
1101
1215
82820
4648
232

of the hig

as thus.

Antery the figures ag and multi five:

Then add figures ab line:

stondes ouer þe 0. hede. þen draw 2, 5, & 1 togedur, þat wolle be 8. þen do away alle þat, & write þere 8. ¶ And þen þou schalt haue þis nounbre, 571880.

and you w
have the a

ases of this Craft.

leaf 158 b.

*** ¶ Sed cum multiplicabis, primo sic est operandum,
Si dabit articulum tibi multiplicacio solum;
Proposita cifra summam transferre memento.**

¶ Here he puttes þe fyrst case of þis craft, þe quych is þis: yf þere come an articulle of þe multiplicacion ysette before the articulle in þe lyft side as þus multiplie 5 by 2, þat wol be 10; sette ouer þe hede of þat 2 a 0, & sett þat on, þat is þe articul, in þe lyft side, þat is next hym, þen þou schalt haue þis nounbre ¶ And þen worch forth as þou diddist afore. And þou schalt vnderstonde þat þou schalt write no 0. but whan þat place where þou schal write þat 0 has no figure afore hym noþer after. *versus.*

51
23.

What to d
first
multiplica
results in
article.

1051.
23

**¶ Si autem digitus excreuerit articulusque.
Articulus¹² supraposito digito salit vltra.**

¶ Here is þe secunde case, þe quych is þis: yf hit happe þat þere come a composyt, þou schalt write þe digitte ouer þe hede of þe neþer figure by þe quych þou multiplieth þe hier figure; and sett þe articulle next hym toward þe lyft side, as þou diddist afore, as þus Multiply 8 by 8, þat wol be 64. Write þe 4 ouer 8, þat is to say, ouer þe hede of þe neþer 8; & set 6, þe quych leaf 159 a. *is an articul, next after. And þen þou schalt haue such a nounbre as is here, And þen worch forth.

83.
83

What to d
result is a
composite
number.

6483¹³,
83

¶ Si digitus tamen ponas ipsum super ipsam.

¶ Here is þe thryde case, þe quych is þis: yf hit happe þat of þi multiplicacioun come a digit, þou schalt write þe digit ouer þe hede of þe neþer figure, by the quych þou multiplieth þe hier figure, for þis nedes no Ensampul.

What if it
digit.

¶ Subdita multiplica non hanc que [incidit] illi

Delet eam penitus scribens quod prouenit inde.

¶ Here is þe 4 case, þe quych is: yf hit be happe þat þe neþer figure schal multiplie þat figure, þe quych stondes ouer þat figures hede, þou schal do away þe hier figure & sett þere þat þat comys of þat multiplicacion. As yf þere come of þat multiplicacion an articuls þou schalt write þere þe hier figure stode a 0. ¶ And write þe articuls in þe lyft side, yf þat hit be a digit write þere a digit. yf þat hit be a composit, write þe digit of þe composit. And þe articul in þe lyft side. al þis is ly3t y-now3t, þere-fore þer nedes no Ensampul.

The fourth
of the craft.

**¶ Sed si multiplicat aliam ponas super ipsam
Adiunges numerum quem prebet ductus earum.**

leaf 159 b.

¶ Here is þe 5 case, þe quych is þis: yf *þe neþer figure schul multiplie þe hier, and þat hier figure is not recte ouer his hede. And þat neþer figure hase oþer figures, or on figure ouer his hede by multiplicacion, þat hase be afore, þou schalt write þat nounbre, þe quych comes of þat, ouer alle þe ylke figures hedes, as þus here: Multiply 2 by 2, þat wol be 4; set 4 ouer þe hede of þat 2. þen ¹⁴ multiplies þe hier 2 by þe neþer 3, þat wol be 6. set ouer his hede 6, multiplie þe hier 2 by þe neþer 4, þat wol be 8. do away þe hier 2, þe quych stondes ouer þe hede of þe figure of 4, and set þere 8. And þou schalt haue þis nounbre here And antery þi figures, þat is to say, set þi neþer 4 vnder þe hier 3, and set þi 2 other figures nere hym, so þat þe neþer 2 stonde vndur þe hier 6, þe quych 6 stondes in þe lyft side. And þat 3 þat stondes vndur 8, as þus aftur 3e may se, Now worch forthermore, And multiplie þat hier 3 by 2, þat wol be 6, set þat 6 þe quych stondes ouer þe hede of þat 2, And þen worch as I tazt þe afore.

236
234

46836
234

46836
234

The fifth
the craft.

leaf 160 a.

*** ¶ Si supraposita cifra debet multiplicare
Prorsus eam deles & ibi scribi cifra debet.**

¶ Here is þe 6 case, þe quych is þis: yf hit happe þat þe figure by þe quych þou schal multiplie þe hier figure, þe quych stondes ryght ouer hym by a 0, þou schalt do away þat figure, þe quych

The sixth
the craft.

ouer þat cifre hede. ¶ And write þere þat nounbre þat comes of þe multiplicacion as þus, 23. do away 2 and sett þere a 0. vnde versus.

¶ **Si cifra multiplicat aliam positam super ipsam
Sitque locus supra vacuus super hanc cifram fiet.**

¶ Here is þe 7 case, þe quych is þis: yf a 0 schal multiply a figure, þe quych stondes not recte ouer hym, And ouer þat 0 stonde no thyng, þou schalt write ouer þat 0 anoþer 0 as þus: multiplye 2 be a 0, it wol be nothyng. write þere a 0 ouer þe hede of þe neþer 0, And þen worch forth til þou come to þe ende.

The sever
of the cra:

24
03

¶ **Si supra ¹⁵ fuerit cifra semper est pretereunda.**

¶ Here is þe 8 case, þe quych is þis: yf þere be a 0 or mony cifers in þe hier rewe, þou schalt not multiplie hem, bot let hem stonde. And antery þe figures beneþe to þe next figure sygnificatyf as þus: Ouer-lepe alle þese cifers & sett þat leaf 160 b. *neþer 2 þat stondes toward þe ryght side, and sett hym vndur þe 3, and sett þe oþer nether 2 nere hym, so þat he stonde vndur þe thrydde 0, þe quych stondes next 3. And þan worch. vnde versus.

The eight
of the cra:

00032.
22

¶ **Si dubites, an sit bene multiplicacio facta,
Diuide totalem numerum per multiplicantem.**

¶ Here he teches how þou schalt know wheþer þou hase wel I-do or no. And he says þat þou schalt deuide alle þe nounbre þat comes of þe multiplicacion by þe neþer figures. And þen þou schalt haue þe same nounbur þat þou hadyst in þe begynnyng. but 3et þou hast not þe craft of dyuision, but þou schalt haue hit afterwarde.

How to pr
multiplica

¶ **Per numerum si vis numerum quoque multiplicare
¶ Tantum per normas subtiles absque figuris
Has normas poteris per versus scire sequentes.**

¶ Here he teches þe to multiplie be þow3t figures in þi mynde. And þe sentence of þis verse is þis: yf þou wel multiplie on nounbre by anoþer in þi mynde, þou schal haue þereto rewles in þe verses þat schal come after.

Mental
multiplica

**¶ Si tu per digitum digitum vis multiplicare
Regula precedens dat qualiter est operandum.**

¶ Here he teches a rewle as þou hast afore to multiplie a digit be anoper, as yf þou wolde wete qwat is sex tymes 6. þou leaf 161 a. *schalt wete by þe rewle þat I taȝt þe before, yf þou haue mynde þerof.

Digit by c
easy.

**¶ Articulum si per reliquum reliquum vis multiplicare
In proprium digitum debet vterque resolui.**

**¶ Articulus digitos post se multiplicantes
Ex digitus quociens retenerit multiplicari
Articuli faciunt tot centum multiplicati.**

¶ Here he teches þe furst rewle, þe quych is þis: yf þou wel multiplie an articul be anoper, so þat both þe articuls bene *with-Inne* an hundreth, þus þou schalt do. take þe digit of bothe the articuls, for euery articul hase a digit, þen multiplie þat on digit by þat oper, and loke how many vnytes ben in þe nounbre þat comes of þe multiplicacion of þe 2 digittes, & so many hundrythes ben in þe nounbre þat schal come of þe multiplicacion of þe ylke 2 articuls as þus. yf þou wold wete qwat is ten tymes ten. take þe digit of ten, þe quych is 1; take þe digit of þat oper ten, þe quych is on. ¶ Also multiplie 1 be 1, as on tyme on þat is but 1. In on is but on vnite as þou wost welle, þerefore ten tymes ten is but a hundryth. ¶ Also yf þou wold wete what is twenty tymes 30. take þe digit of twenty, þat is 2; & take þe digitt of thrytty, þat is 3. multiplie 3 be 2, þat is 6. Now in 6 ben 6 vnites, ¶ And so many hundrythes ben in 20 tymes 30*, leaf 161 b. þerefore 20 tymes 30 is 6 hundryth euen. loke & se. ¶ But yf it be so þat one articul be *with-Inne* an hundryth, or by-twene an hundryth and a thowsande, so þat it be not a þowsande fully. þen loke how many vnytes ben in þe nounbur þat comys of þe multiplicacion ¹⁶And so many tymes ¹⁶ of 2 digittes of ylke articuls, so many thowsant ben in þe nounbre, the quych comes of þe multiplicacion. And so many tymes ten thowsand schal be in þe nounbre þat comes of þe multiplicacion of 2 articuls, as yf þou wold wete qwat is 4 hundryth tymes [two hundryth]. Multiply 4 be 2, ¹⁷ þat wol be 8. in 8 ben 8 vnites. ¶ And so many tymes ten thousand be in 4 hundryth tymes [2] ¹⁷

The first c
the craft.

Article by
article;

an examp

another e

o work subtly without Figures.

hundryth, þat is 80 thousand. Take hede, I schall telle þe a generale rewle whan þou hast 2 articuls, And

Mental multiplicat

Another example.

þou wold wete qwat comes of þe multiplicacion of hem 2. multiplie þe digit of þat on articuls, and kepe þat nounbre, þen loke how many cifers schuld go before þat on articuls, and he were write. Als many cifers schuld go before þat other, & he were write of cifers. And haue alle þe ylke cifers togedur in þi mynde, leaf 162 a. *a-rowe ychon aftur other, and in þe last plase set þe nounbre þat comes of þe multiplicacion of þe 2 digittes. And loke in þi mynde in what place he stondes, where in þe secunde, or in þe thryd, or in þe 4, or where ellis, and loke qwat þe figures by-token in þat place; & so mych is þe nounbre þat comes of þe 2 articuls y-multiplied to-gedur as þus: yf þou wold wete what is 20 thousand tymes 3 þowsande. multiply þe digit of þat articulle þe quych is 2 by þe digitte of þat oþer articul þe quych is 3, þat wol be 6. þen loke how many cifers schal go to 20 thousand as hit schuld be write in a tabul. certainly 4 cifers schuld go to 20 þowsant. ffor þis figure 2 in þe fyrst place betokenes twene. ¶ In þe secunde place hit betokenes twenty. ¶ In þe 3. place hit betokenes 2 hundryth. ¶ In þe 4 place 2 thousand. ¶ In þe 5 place hit betokenes twenty þousant. þefore he most haue 4 cifers a-fore hym þat he may stonde in þe 5 place. kepe þese 4 cifers in thy mynde, þen loke how many cifers go to 3 thousand. Certayn to 3 thousante leaf 162 b. *go 3 cifers afore. Now cast ylke 4 cifers þat schuld go to twenty thousand, And thes 3 cifers þat schuld go afore 3 thousand, & sette hem in rewe ycho after oþer in þi mynde, as þai schuld stonde in a tabulle. And þen schal þou haue 7 cifers; þen sett þat 6 þe quych comes of þe multiplicacion of þe 2 digittes aftur þe ylke cifers in þe 8 place as yf þat hit stode in a tabul. And loke qwat a figure of 6 schuld betoken in þe 8 place. yf hit were in a tabul & so mych it is. & yf þat figure of 6 stonde in þe fyrst place he schuld betoken but 6. ¶ In þe 2 place he schuld betoken sixty. ¶ In the 3 place he schuld betoken sex hundryth. ¶ In þe 4 place sex thousand. ¶ In þe 5 place sixty þowsant. ¶ In þe sext place sex hundryth þowsant. ¶ In þe 7 place sex þowsant thousandes. ¶ In þe 8 place sixty þowsant thousandes. þefore sett

Another example.

Notation.

Notation :

6 in octauo loco, And he schal betoken sixty þowsant
thousantes. And so mych is twenty þowsant tymes 3 thousand,
¶ And þis rewle is generalle for alle maner of articuls, Whethir
þai be hundryth or þowsant; but þou most know well þe craft of
þe wryrchynge in þe tabulle leaf 163 a. *or þou know to do þus in
þi mynde aftur þis rewle. Thou most þat þis rewle holdyþe note
but where þere ben 2 articuls and no mo of þe quych ayther of
hem hase but on figure significatyf. As twenty tymes 3 thousand
or 3 hundryth, and such oþur.

Mental
multiplica

¶ **Articulum digito si multiplicare oportet**
Articuli digit[i] sumi quo multiplicare]
Debemus reliquum quod multiplicatur ab illis
Per reliquo decuplum sic summam latere nequibit.

¶ Here he puttes þe thryde rewle, þe quych is þis. yf þou wel
multiply in þi mynde, And þe Articul be a digitte, þou schalt loke
þat þe digitt be *with-Inne* an hundryth, þen þou schalt multiply
the digitt of þe Articulle by þe oþer digitte. And euery vnite in þe
nounbre þat schalle come þere-of schal betoken ten. As þus: yf
þat þou wold wete qwat is twyes 40. multiplie þe digitte of 40,
þe quych is 4, by þe oþer diget, þe quych is 2. And þat wolle be
8. And in þe nombre of 8 ben 8 vnites, & euery of þe ylke vnites
schuld stonde for 10. þere-fore þere schal be 8 tymes 10, þat wol
be 4 score. And so mony is twyes 40. ¶ If þe articul be a
hundryth or be 2 hundryth And a þowsant, so þat hit be notte a
thousant, leaf 163 b. *worch as þou dyddyst afore, saue þou schalt
rekene euery vnite for a hundryth.

The third
the craft;

an examp

¶ **In numerum mixtum digitum si ducere cures**
Articulus mixti sumatur deinde resoluas
In digitum post fac respectu de digitis
Articulusque docet excrescens in diriuando
In digitum mixti post ducas multiplicantem
¶ **De digitis vt norma** ¹⁸[docet] de [hunc]
Multiplica simul et sic postea summa patebit.

Here he puttes þe 4 rewle, þe quych is þis: yf þou multipli on
composit be a digit as 6 tymes 24, ¹⁹þen take þe diget of þat
composit, & multiply þat digitt by þat oþer diget, and kepe þe

The fourtl
of the cra:

nombur þat comes þere-of. þen take þe digit of þat composit, & multiply þat digit by anoþer diget, by þe quych þou hast multiplyed þe diget of þe articul, and loke qwat comes þere-of. þen take þou þat nounbur, & cast hit to þat other nounbur þat þou secheſte as þus yf þou wel wete qwat comes of 6 tymes 4 & twenty. multiply þat articulle of þe composit by þe digit, þe quych is 6, as yn þe thryd rewle þou was tauȝt, And þat ſchal be 6 ſcore. þen multiply þe diget of þe composit, leaf 164 a. *þe quych is 4, and multiply þat by þat other diget, þe quych is 6, as þou waſt tauȝt in þe firſt rewle, yf þou haue mynde þerof, & þat wol be 4 & twenty. caſt all ylke nounburs to-gedir, & hit ſchal be 144. And ſo mych is 6 tymes 4 & twenty.

Composit
digit.

Mental
multiplica

o multiply without Figures.

¶ **Ductus in articulum numerus si compositus sit
Articulum purum comites articulum quoque
Mixti pro digitis post fiat [et articulus vt]
Norma iubet [retinendo quod extra dicta ab illis]
Articuli digitum post tu mixtum digitum duc
Regula de digitis nec precipit articulusque
Ex quibus excrescens summe tu iunge priori
Sic manifesta cito fiet tibi summa petita.**

¶ Here he puttes þe 5 rewle, þe quych is þis: yf þou wel multiply an Articul be a composit, multiplie þat Articul by þe articul of þe composit, and worch as þou was tauȝt in þe ſecunde rewle, of þe quych rewle þe verſe begynnes þus. ¶ *Articulum si per Relicum vis multiplicare.* þen multiply þe diget of þe composit by þat oþir articul aftir þe doctrine of þe 3 rewle. take þerof gode hede, I pray þe as þus. Yf þou wel wete what is 24 tymes ten. Multiplie ten by 20, þat wel be 2 hundryth. þen multiply þe diget of þe 10, þe quych is 1, by þe diget of þe composit, þe quych is 4, & þat leaf 164 b. *wol be 4. þen reken euery vnite þat is in 4 for 10, & þat ſchal be 40. Caſt 40 to 2 hundryth, & þat wol be 2 hundryth & 40. And ſo mych is 24 tymes ten.

The fifth
the craft:

Article by
Composit

An exampl

o work without Figures.

¶ **Compositum numerum mixto si[c] multiplicabis**

**Vndecies tredecim sic est ex hiis operandum
 In reliquum primum demum duc post in eundem
 Vnum post denum duc in tria deinde per vnum
 Multiplicesque demum intra omnia multiplicata
 In summa decies quam si fuerit tibi doces
 Multiplicandorum de normis sufficiunt hec.**

¶ Here he puttes þe 6 rewle, & þe last of alle multiplicacion, þe quych is þis: yf þou wel multiplie a *composit* by a *noþer* *composit*, þou schalt do þus. multiplie þat on *composit*, qwyth þou welt of the twene, by þe articul of þe toþer *composit*, as þou were tauzt in þe 5 rewle, þen multiplie þat same *composit*, þe quych þou hast multiplied by þe *oper* articul, by þe digit of þe *oper* *composit*, as þou was tauzt in þe 4 rewle. As þus, yf þou wold wete what is 11 tymes 13, as þou was tauzt in þe 5 rewle, & þat schal be an hundryth & ten, afterwarde multiply þat same *composit* þat þou hast multiplied, þe quych is a .11. And multiplie hit be þe digit of þe *oper* *composit*, þe quych is 3, for 3 is þe digit of 13, And þat wel be 30. þen take þe digit of þat *composit*, þe quych *composit* þou multiplied by þe digit of þat *oper* *composit*, leaf 165 *a*. *þe quych is a 11. ¶ Also of the quych 11 on is þe digit. multiplie þat digitt by þe digett of þat other *composit*, þe quych diget is 3, as þou was tauzt in þe first rewle in þe begynnynge of þis craft. þe quych rewle begynnes “In *digitum* cures.” And of alle þe multiplicacion of þe 2 digitt comys thre, for onys 3 is but 3. Now cast alle þese nounbers togedur, the quych is þis, a hundryth & ten & 30 & 3. And al þat wel be 143. Write 3 first in þe ryght side. And cast 10 to 30, þat wol be 40. set 40 next aftur towarde þe lyft side, And set aftur a hundryth as here an Ensampulle, 143.

The sixth
the craft:

Composit
Composit

Mental
multiplica
An exampl

of the sixt
of the cra

(Cetera desunt.)

- [1.](#) In MS, ‘awiy.’
- [2.](#) ‘ben’ repeated in MS.
- [3.](#) In MS. ‘thausandes.’
- [4.](#) Perhaps “So.”

- [5.](#) ‘hali’ marked for erasure in MS.
- [6.](#) ‘moy’ in MS.
- [7.](#) ‘Subtrahas aut addis a dextris vel mediabis’ added on margin of MS.
- [8.](#) After ‘craft’ insert ‘the .4. what is þe profet of þis craft.’
- [9.](#) After ‘sythes’ insert ‘& þis wordes fyue sithe & sex sythes.’
- [10.](#) ‘t’l’ marked for erasure before ‘tyl’ in MS.
- [11.](#) Here ‘of þe same rew’ is marked for erasure in MS.
- [12.](#) ‘sed’ deleted in MS.
- [13.](#) 6883 in MS.
- [14.](#) ‘þen’ overwritten on ‘þat’ marked for erasure.
- [15.](#) ‘Supra’ inserted in MS. in place of ‘cifra’ marked for erasure.
- [16–16.](#) Marked for erasure in MS.
- [17.](#) 4 in MS.
- [18.](#) docet. decet MS.
- [19.](#) ‘4 times 4’ in MS.

The Art of Nombryng.
A TRANSLATION OF
John of Holywood's De Arte Numerandi.

[Ashmole MS. 396, fol. 48.]

Boys seying in the begynnyng of his Arsemetrike:—Alle
Fol. 48. thynges that bene fro the first begynnyng of
thynges have procedede, and come forthe, And by resoun
of nombre ben formede; And in wise as they bene, So

**owethe they to be knowene; wherfor in vniuersalle
knowlechyng of thynges the Art of nombrynge is best, and
most operatyfe.**

Therfore sithen the science of the whiche at this tyme we
intendene to write of standithe alle and about nombre:
ffirst we most se, what is the propre name therofe, and fro
whens the name come: Afterwarde what is nombre, And how
manye spices of nombre ther ben. The name is clepede
Algorisme, hade out of Algore, other of Algos, in grewe, That is
clepide in englisshe art other craft, And of Rithmus that is
calledede nombre. So algorisme is clepede the art of nombryng,
other it is had ofe en or in, and gogos that is introduccioun, and
Rithmus nombre, that is to say Interduccioun of nombre. And
thirdly it is hade of the name of a kyng that is clepede Algo and
Rythmus; So callede Algorismus. Sothely .2. manere of nombres
ben notified; Formalle,¹ as nombre is vnitees gadrede to-gedres;
Materialle,² as nombre is a colleccioun of vnitees. Other nombre
is a multitude hade out of vnitees, vnitee is that thyng wher-by
euery thyng is callede oone, other o thyng. Of nombres, that
one is clepede digitalle, that othere Article, Another a nombre
componede oper myxt. Another digitalle is a nombre with-in
.10.; Article is þat nombre that may be dyvydede in .10. parties
egally, And that there leve no residue; Componede or medlede is
that nombre that is come of a digite and of an article. And
vndrestande wele that alle nombres betwix .2. articles next is a
nombre componede. Of this art bene .9. spices, that is forto sey,
numeracioun, addicioun, Subtraccioun, Mediacioun,
Duplacioun, Multiplacioun, Dyvysioun, Progressioun, And of
Rootes the extraccioun, and that may be hade in .2. maners, that
is to sey in nombres quadrat, and in cubices: Amonge the
whiche, ffirst of Numeracioun, and afterwarde of þe opers by
ordure, y entende to write.

The name
art.

Derivation:
Algorism.

Another.

Another.

Kinds of
numbers.

The 9 rule
the Art.

er I. Numeration.

Fol. 48 b.

***For-sothe numeracioun is of euery nombre by competent figures an artificialle representacioun.**

Sothly figure, difference, places, and lynes supposen o thyng other the same, But they ben sette here for dyuers resons. ffigure is clepede for protraccioun of figuracioun; Difference is callede for therby is shewede euery figure, how it hathe difference fro the figures before them: place by cause of space, where-in me writethe: lynees, for that is ordeynede for the presentacioun of euery figure. And vnderstonde that ther ben .9. lymytes of figures that representen the .9. digites that ben these. 0. 9. 8. 7. 6. 5. 4. 3. 2. 1. The .10. is clepede theta, or a cercle, other a cifre, other a figure of nought for nought it signyfieth. Nathelesse she holdyng that place givethe others for to signyfie; for withe-out cifre or cifres a pure article may not be writte. And sithen that by these .9. figures significatifes Ioynede with cifre or with cifres alle nombres ben and may be representede, It was, nether is, no nede to fynde any more figures. And note wele that euery digite shalle be writte with oo figure allone to it aproprede. And alle articles by a cifre, ffor euery article is namede for oone of the digitis as .10. of 1.. 20. of. 2. and so of the others, &c. And alle nombres digitalle owen to be sette in the first difference: Alle articles in the seconde. Also alle nombres fro .10. til an .100. [which] is excludede, with .2. figures mvst be writte; And yf it be an article, by a cifre first put, and the figure y-writte toward the lift honde, that signifieth the digit of the whiche the article is namede; And yf it be a nombre componede, ffirst write the digit that is a part of that componede, and write to the lift side the article as it is seide be-fore. Alle nombre that is fro an hundrede tille a thousande excludede, owithe to be writ by .3. figures; and alle nombre that is fro a thousande til .x. Ml. mvst be writ by .4. figures; And so forthe. And vnderstonde wele that euery figure sette in the first place signyfieth his digit; In the seconde place .10. tymes his digit; In the .3. place an hundrede so moche; In the .4. place a thousande so moche; In the .5. place .x. thousande so moche; In the .6. place an hundrede thousande so moche; In the .7. place a thousande thousande. And so infynyty multiplying by Fol. 49. *these .3. 10, 100, 1000. And vnderstande wele that competently me may sette vpon figure in the place of a

Figures,
difference
places, an
limits.

The 9 figi

The ciphe

The nume

of digits,
of articles

of compo:

The value
position.

thousande, a prike to shewe how many thousande the last figure shalle represent. We writene in this art to the lift side-warde, as arabiene writene, that weren fynders of this science, othere for this resoun, that for to kepe a custumable ordre in redyng, Sette we alle-wey the more nombre before.

Numbers
written from
right to le

er II. Addition.

Addicioun is of nombre other of nombres vnto nombre or to nombres aggregacioun, that me may see that that is cometh of as excrement. In addicioun, 2. ordres of figures and .2. nombres ben necessary, that is to sey, a nombre to be addede and the nombre wherto the addicioun sholde be made to. The nombre to be addede is that þat sholde be addede therto, and shalle be vnderwriten; the nombre vnto the whiche addicioun shalle be made to is that nombre that resceyueth the addicion of þat other, and shalle be writen above; and it is convenient that the lesse nombre be vnderwrit, and the more addede, than the contrary. But whether it happe one other other, the same comythe of, Therfor, yf þow wilt adde nombre to nombre, write the nombre wherto the addicioun shalle be made in the omest ordre by his differences, so that the first of the lower ordre be vndre the first of the omyst ordre, and so of others. That done, adde the first of the lower ordre to the first of the omyst ordre. And of suche addicioun, other þere growith therof a digit, An article, other a composede. If it be digitus, In the place of the omyst shalt thou write the digit excrescyng, as thus:—

Definition

How the
numbers shal
be writen

The methode
working.

Begin at the
right.

If the article; in the place of the omyst put a-way by a cifre writte, and the digit transferrede, of þe whiche the article toke his name, toward the lift side, and be it addede to the next figure folowyng, yf ther be any figure folowyng; or no, and yf it be not, leve it [in the] voide, as thus:—

The resultant	2
To whom it shal be addede	1
The nombre to be addede	1

The Sum
digit,

or an article

Resultans	2	7	8	2	7
Cui <i>debet</i> addi	1	0	0	8	4
Numerus addendus	1	7	7	4	3

The resultant	10
To whom it shalle be addede	7
The nombre to be addede	3

And yf it happe that the figure folowyng wherto the addicioun shalle be made by [the cifre of] an article, it sette a-side; In his place write the Fol. 49 b. *[digit of the] Article as thus:—

The resultant	17
To whom it shalle be addede	10
The nombre to be addede	7

And yf it happe that a figure of .9. by the figure that me mvst adde [one] to, In the place of that 9. put a cifre *and* write þe article towarde þe lift honde as bifore, and thus:—

The resultant	10
To whom it shalle be addede	9
The nombre to be addede	1

And yf ³ [therefrom grow a] nombre componed, ⁴ [in the place of the nombre] put a-way ⁵ [let] the digit [be] ⁶ writ þat is part of þat composit, and þan put to þe lift side the article as before, and þus:—

The resultant	12
To whom it shalle be addede	8
The nombre to be addede	4

or a comp

This done, adde the seconde to the seconde, and write above oper as before. Note wele þat in addicions and in alle spices folowyng, whan he seithe one the other shalle be writen aboue, and me most vse euer figure, as that euery figure were sette by halfe, and by hym-selfe.

The transl
note.

er III. Subtraction.

Subtraccioun is of .2. proposede nombres, the fyndyng of the excesse of the more to the lasse: Other subtraccioun is ablacioun of o nombre fro a-nother, that me may see a some left. The lasse of the more, or even of even, may be *withdraw*; The more fro the lesse may neuer be. And sothly that nombre is more that hathe more figures, So that the last be

Definitior
Subtractic

How it m:
done.

signyficatifes: And yf ther ben as many in that one as in that other, me most deme it by the last, other by the next last. Moreouer in *with-drawyng* .2. nombres ben necessary; A nombre to be *withdraw*, And a nombre that me shalle *with-draw* of. The nombre to be *with-draw* shalle be writ in the lower ordre by his differences; The nombre fro the whiche me shalle *withe-draw* in the omyst ordre, so that the first be vnder the first, the seconde vnder the seconde, And so of alle others. *Withe-draw* therfor the first of the lowere ordre fro the first of the ordre above his hede, and that wolle be other more or lesse, *oper* egalle.

yf it be egalle or even the figure sette beside, put in his place a cifre. And yf it be more put away *perfro* als many of vnitees the lower figure conteynethe, and writ the residue as thus

Fol. 50.

The remanent	20
Wherof me shalle <i>withdraw</i>	22
The nombre to be <i>withdraw</i>	2

The remanent	2	2
Wherof me shalle <i>with-draw</i>	2	8
pe nombre to be <i>withdraw</i>		6

*Remanens	2	2	1	8	2	9	9	9	8
A quo sit subtraccio	8	7	2	4	3	0	0	0	4
Numerus subtrahendus	6	5	7	[6]	.	.	.	.	6

And yf it be lesse, because the more may not be *with-draw* therfro, borow an vnyte of the next figure that is

worthe 10. Of that .10. and of the figure that ye wolde have *with-draw* fro be-fore to-gedre Ioynede, *with-draw* pe figure be-nethe, and put the residue in the place of the figure put a-side as þus:—

And yf the figure wherof me shal borow the vnyte be one, put it a-side, and write a cifre in the place *perof*, lest the figures folowing faile of thaire nombre, and þan worche as it shewith in this figure here:—

And yf the vnyte wherof me shal borow be a cifre, go ferther to the figure signyficatife, and ther borow one, and retournyng

The remanent	1	8
Wherof me shalle <i>with-draw</i>	2	4
The nombre to be <i>with-draw</i>	0	6

What is re

Write the number al

Subtract t figure if possible.

If it is not possible ‘ten,’

and then subtract.

If the secc figure is c

bake, in the place of euery cifre
 þat ye passide ouer, sette figures
 of .9. as here it is specifiende:—

The remanent	3	0	9 ⁸
Wherof me shal <i>with-draw</i>	3	1	2
The nombre to be <i>with-draw</i>	.	.	3

And whan me comethe to the
 nombre wherof me intendithe,
 there remaynethe alle-wayes
 .10. ffor þe whiche .10. &c.
 The reson why þat for euery
 cifre left behynde me setteth
 figures ther of .9. this it is:—

The remenaunt	2	9	9	9	9
Wherof me shalle <i>with-draw</i>	3	0	0	0	3
The nombre to be <i>with-draw</i>					4

If fro the .3. place me
 borowede an vnyte, that vnyte by respect of the figure that he
 came fro representith an .C., In the place of that cifre [passed
 over] is left .9., [which is worth ninety], and yit it remaynethe as
 .10., And the same resone wolde be yf me hade borowede an
 vnyte fro the .4., .5., .6., place, or any other so vpwarde. This
 done, withdraw the seconde of the lower ordre fro the figure
 above his hede of þe omyst ordre, and wirche as before. And
 note wele that in addicion or in subtraccioun me may wele fro
 the lift side begynne and ryn to the right side, But it wol be more
 profitabler to be do, as it is taught. And yf thow wilt prove yf
 thow have do wele or no, The figures that thow hast withdraw,
 adde them ayene to the omyst figures, and they wolle accorde
 with the first that thow haddest yf thow have labored wele; and
 in like wise in addicioun, whan thow hast addede alle thy
 figures, *withdraw* them that thow first Fol. 50 b. *addest, and the
 same wolle retourne. The subtraccioun is none other but a
 prouffe of the addicioun, and the contrarye in like wise.

If the secc
 figure is a

A justific:
 the rule g

Why it is
 to work fr
 right to le

How to pi
 subtractio

and additi

er IV. Mediation.

Mediacioun is the fyndyng of the halfyng of euery
 nombre, that it may be seyne what and how moche is
 euery halfe. In halfyng ay oo order of figures and oo
 nombre is necessary, that is to sey the nombre to be halfede.
 Therfor yf thow wilt half any nombre, write that nombre by his
 differences, and begynne at the right, that is to sey, fro the first
 figure to the right side, so that it be signyficatife other represent

Definitior
 mediation

Where to

vnyte or eny other digitalle nombre. If it be vnyte write in his place a cifre for the figures folowyng, [lest they signify less], and write that vnyte *without* in the table, other resolue it in .60.

If the first is unity.

mynvtes and sette a-side half of tho minutes so, and reserve the remenaunt *without* in the table, as thus .30.; other sette *without* thus .dī: that kepethe none ordre of place, Nathelesse it hathe signyficacioun. And yf the other figure signyfie any other digital nombre fro vnyte forthe, oper the nombre is ode or evene.

If it be even, write this half in this wise:—

Halfede	2	2
to be halfede	4	4

And if it be odde, Take the next even vndre hym conteynede, and put his half in the place of that odde, and of þe vnyte that remaynethe to be halfede do thus:—

halfede	2	3
To be halfede	4	7

[di]

This done, the seconde is to be halfede, yf it be a cifre put it be-side, and yf it be significatife, other it is even or ode: If it be even, write in the place of þe nombres wipede out the halfe; yf it be ode, take the next even vnder it contenythe, and in the place of the Impar sette a-side put half of the even: The vnyte that remaynethe to be halfede, respect hade to them before, is worthe .10.

What to d is not unit Then halv second fig

Dyvide that .10. in .2., 5. is, and sette a-side that one, and adde that other to the next figure precedent as here:—

Halfede			
to be halfede			

And yf þe addicioun sholde be made to a cifre, sette it a-side, and write in his place .5.

If it is ode to the fig before.

And vnder this *fourme* me shalle write and worche, tille the totalle nombre be halfede.

doubledede	2	6	8	9	0	10	17	4
to be doubledede	1	3	4	4	5	5	8	7

er V. Duplation.

Duplicacioun is agregacion of nombre [to itself] þat me may se the nombre growen. In doublynge ay is but one ordre of figures necessarie. And me most be-gynne *with* the lift side, other of the more figure, And after the nombre of the more figure representithe. Fol. 51. *In the other .3. before we begynne alle way fro the right side and fro the lasse nombre, In

Definitior Duplation

Where to

this spice and in alle other folowyng we wolle begynne fro the lift side, ffor and me bigon the double fro the first, omwhile me myght double oo thyng twyes. And how be it that me myght double fro the right, that wolde be harder in techyng and in workyng. Therfor yf thou wolt double any nombre, write that nombre by his differences, and double the last. And of that doublyng other growithe a nombre digital, article, or componede. [If it be a digit, write it in the place of the first digit.]

Why.

If it be article, write in his place a cifre and transferre the article towarde the lift, as thus:—

double	10
to be doubled	5

And yf the nombre be componede, write a digital that is part of his composicioun, and sette the article to the lift hande, as thus:—

What to d
the result.

That done, me most double the last save one, and what growethe þerof me most worche as before. And yf a cifre be, touche it not. But yf any nombre shalle be addede to the cifre, in þe place of þe figure wiped out me most write the nombre to be addede, as thus:—

doubled	16
to be doubled	8

In the same wise me shalle wirche of alle others. And this probacioun:

How to p
your answ

If thou truly double the halfis, and truly half the doubles, the same nombre and figure shalle mete, suche as thou labourede vpone first, And of the contrarie.

doubled	6	0	6
to be doubled	3	0	3

Doubled	6	1	8
to be doubled	3	0	9

er VI. Multiplication.

Multiplicacioun of nombre by hym-self other by a-nother, with proposide .2. nombres, [is] the fyndyng of the thirde, That so oft conteynethe that other, as ther ben vnytes in the oþer. In multiplicacioun .2. nombres pryncipally ben necessary, that is to sey, the nombre multiplying and the nombre to be multipliede, as here;—twies fyve. [The number multiplying] is designede aduerbially. The nombre to be multipliede resceyvethe a *nominalle* appellacioun, as twies .5. 5.

Definitior
Multiplic:

Multiplie
Multiplic:

is the nombre multipliede, and twies is the nombre to be multipliede.

Resultans	9	1	0	1	3	2	6	6	8	0	0	8
Multiplicandus	.	.	5	.	.	4	.	3	4	0	0	4
Multiplicans	.	2	2	.	3	3	2	2	2	.	.	.

Also me may thervpone to assigne the. 3. nombre, the whiche is Fol. 51 b. *clepede product or *provenient*, of takyng out of one fro another: as twyes .5 is .10., 5. the nombre to be multipliede, and .2. the multipliant, and. 10. as before is come therof. And vnderstonde wele, that of the multipliant may be made the nombre to be multipliede, and of the contrarie, remaynyng euer the same some, and herofe comethe the comen speche, that seithe all nombre is convertede by Multiplying in hym-selfe.

ases of Multiplication.

And ther ben .6 rules of Multiplicacioun; ffirst, yf a digit multiplie a digit, considre how many

of vnytees ben betwix the digit by multiplying and his .10. bethe to-gedre accomptede, and so oft *with-draw* the digit multiplying, vnder the article of his denominacioun. Example of grace. If thow wolt wete how moche is .4. tymes .8., ¹¹se how many vnytees ben betwix .8. ¹² and .10. to-geder rekenede, and it shewith that .2.: withdraw ther-for the

quaternary, of the article of his denominacion twies, of .40., And ther remaynethe .32., that is, to some of alle the multiplicacioun. Wher-vpon for more evidence and declaracion the seide table is made. Whan a digit multipliethe an article, thow most bryng the digit into þe digit, of þe whiche the article [has] ¹³ his name, and euery vnyte shalle stonde for .10., and euery article an .100. Whan the digit multipliethe a nombre componede, þou most bryng the digit into aþer part of the nombre componede, so þat

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	10 ¹⁰	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40
5	10	15	20	25	30	35	40	45	50
6	12	18	24	30	36	42	48	56	60
7	14	21	28	35	42	49	56	63	70
8	16	24	32	40	48	56	64	72	80
9	18	27	36	45	54	63	72	81	90
10	20	30	40	50	60	70	80	90	100

Product.

There are of Multiplic:

- (1) Digit l digit.
See the ta above.
- (2) Digit l article.
- (3) Comp by digit.

digit be had into digit by the first rule, into an article by þe seconde rule; and afterwarde Ioyne the produccioun, and þere wol be the some totalle.

Resultans	1	2	6	7	3	6	1	2	0	1	2	0	8
Multiplicandus			2		3	2			6				4
Multiplicans		6	3	2	3			2	0		3	0	2

Whan an article multipliethe an article, the digit wherof he is namede is to be brought Into the digit wherof the oþer is namede, and euery vnyte wol be worthe Fol. 52. *an .100., and euery article. a .1000. Whan an article multipliethe a nombre componede, thow most bryng the digit of the article into aither part of the nombre componede; and Ioyne the produccioun, and euery article wol be worthe .100., and euery vnyte .10., and so wolle the some be opene. Whan a nombre componede multipliethe a nombre componede, euery part of the nombre multiplying is to be hade into euery part of the nombre to be multipliede, and so shalle the digit be hade twies, onys in the digit, that other in the article. The article also twies, ones in the digit, that other in the article. Therfor yf thow wilt any nombre by hym-self other by any other multiplie, write the nombre to be multipliede in the ouer ordre by his differences, The nombre multiplying in the lower ordre by his differences, so that the first of the lower ordre be vnder the last of the ouer ordre. This done, of the multiplying, the last is to be hade into the last of the nombre to be multipliede. Wherof than wolle grow a digit, an article, other a nombre componede.

If it be a digit, even above the figure multiplying is hede write his digit that come of, as it apperethe here:—

And yf an article had be writ ouer the figure multiplying his hede, put a cifre þer and transferre the article towarde the lift hande, as thus:—

And yf a nombre componede be writ ouer the figure multiplying is hede, write the digit in the nombre

The resultant	6
To be multipliede	3
þe nombre multipli yng	2

The resultant	1	0
to be multipliede		5
þe nombre multipli yng		2

(4) Article
article.

(5) Comp
by article.

(6) Comp
by compo

How to se
your num

If the resu
digit,

an article,

componede is place, and sette the article to the lift hande, as thus:—

Resultant	1	2
to be multipliede		4
the nombre multiplieng		3

This done, me most bryng the last save one of the multiplieng into the last of þe nombre to be multipliede, and se what comythe therof as before, and so do with alle, tille me come to the first

of the nombre multiplying, that must be brought into the last of the nombre to be multipliede, wherof growithe oþer a digit, an article, Fol. 52 b. *other a nombre componede.

If it be a digit, In the place of the ouerer, sette a-side, as here:

Resultant	6	6
to be multipliede		3
the nombre multiplieng	2	2

If an article happe, there put a cifre in his place, and put hym to the lift hande, as here:

The resultant	1	1	0
to be multipliede			5
þe nombre multiplying		2	2

If it be a nombre componede, in the place of the ouerer sette a-side, write a digit that ¹⁴ is a part of the componede, and sette on the left honde the article, as here:

The resultant	1	3 ¹⁵	2
to be multipliede			4
þe nombre multipliant		3	3

That done, sette forwarde the figures of the nombre multiplying by oo difference, so that the first of the multipliant be vnder the last save one of the nombre to be multipliede, the other by o place sette forwarde. Than me shalle brynge the last of the multipliant in hym to be multipliede, vnder the whiche is the first multipliant. And than wolle growe oþer a digit, an article, or a componede nombre. If it be a digit, adde hym even above his hede; If it be an article, transferre hym to the lift side; And if it be a nombre componede, adde a digit to the figure above his hede, and sette to the lift hande the article. And alle-wayes euery figure of the nombre multipliant is to be brought to the last save one nombre to be multipliede, til me come to the first of the multipliant, where me shalle wirche as it is seide before of the first, and afterwarde to put forwarde the figures by o difference and one tille they alle be multipliede. And

or a comp

Multiply
the last by
and so on

Then ante
multiplier
place.

Work as b

How to di
ciphers.

yf it happe that the first figure of þe multipliant be a cifre, and boue it is sette the figure signyficatife, write a cifre in the place of the figure sette a-side, as thus, *etc.*:

The resultant	1	2	0
to be multipliede			6
the multipliant		2	0

And yf a cifre happe in the lower order be-twix the first and the last, and even above be sette the figure signyficatif, leve it vntouchede, as here:—

The resultant	2	2	6	4	4
To be multipliede			2	2	2
The multipliant	1	0	2		

And yf the space above sette be voide, in that place write thow a cifre. And yf the cifre happe betwix þe first and the last to be multipliede, me most sette forwarde the ordre of the figures by thaire differences, for oft of duccioun of figures in cifres nought is the resultant, as here,

Fol. 53. *wherof it is evident and open, yf that the first figure of the nombre be to be multipliede be a cifre, vndir it shalle be none sette as here:—

Resultant	8	0	0	8	
to be multipliede	4	0	0	4	
the multipliant	2	.	.	.	

Resultant	3	2	0 ¹⁶
To be multipliede		8	0
The multipliant		4	

Vnder[stand] also that in multiplicacioun, divisioun, and of rootis the extraccioun, competently me may leve a mydel space betwix .2. ordres of figures, that me may

write there what is come of addyng other withe-drawyng, lest any thyng sholde be ouer-hippede and sette out of mynde.

er VII. Division.

For to dyvyde oo nombre by a-nother, it is of .2. nombres proposede, It is forto depart the moder nombre into as many partis as ben of vnytees in the lasse nombre. And note wele that in makynge of dyvysioun ther ben .3. nombres necessary: that is to sey, the nombre to be dyvydede; the nombre dyvydyng and the nombre exeant, other how oft, or quocient. Ay

How to de
ciphers.

Leave roc
between t
rows of fi

Definitior
division.

Dividend,
Divisor,
Quotient.

shalle the nombre that is to be dyvydede be more, other at the lest evene with the nombre the dyvysere, yf the nombre shalle be made by hole nombres. Therfor yf thow wolt any nombre dyvyde, write the nombre to be dyvydede in þe ouerer bordure by his differences, the dyvisere in the lower ordure by his differences, so that the last of the dyviser be vnder the last of the nombre to be dyvyde, the next last vnder the next last, and so of the others, yf it may competently be done; as here:—

How to se
your Sum

An exampl

The residue		2	7
The quotient			5
To be dyvydede	3	4	2
The dyvyser		6	3

Residuum			8				2	7			2	6
Quociens		2	1	2	2			5				9
Diuidendus	6	8	0	6	6	3	4	2	3	3	2	
Diuiser	3	2		3			6	3		3	4	

Examples

And ther ben .2. causes whan the last figure may not be sette vnder the last, other that the last of the lower nombre may not be *with-draw* of the last of the ouerer nombre for it is lasse than the lower, other how be it, that it myght be *with-draw* as for hymself fro the ouerer the remenaunt may not so oft of them above, other yf þe last of the lower be even to the figure above his hede, and þe next last oþer the figure be-fore þat be more þan the figure above sette. Fol. 53². *These so ordeynede, me most wirche from the last figure of þe nombre of the dyvyser, and se how oft it may be *with-draw* of and fro the figure aboue his hede, namely so that the remenaunt may be take of so oft, and to se the residue as here:—

When the
the diviso
not be set
the last of
dividend.

How to be

The residue		2	6
The quociens			9
To be dyvydede	3	3	2
The dyvyser		3	4

And note wele that me may not withe-draw more than .9. tymes nether lasse than ones. Therfor se how oft þe figures of the lower ordure may be *with-draw* fro the figures of the ouerer, and the nombre that shewith þe quociens most be writ ouer the hede of þat figure, vnder the whiche the first figure is, of the dyviser; And

An exampl

by that figure me most withe-draw alle *oper* figures of the lower ordir and that of the figures aboue thaire hedis. This so done, me most sette forwarde þe figures of the diuiser by o difference towards the right honde and worche as before; and thus:—

Where to
quotiente

Examples

Residuum										.	1	2	
quociens				6	5	4				2	0	0	4
Diuidendus	3	5	5	1	2	2	8	8	6	3	7	0	4
Diuisor		5	4	3			4	4	2	3			

The quocient				6	5	4
To be dyvydede	3	5	5	1	2	2
The dyvyser		5	4	3		

And yf it happe after þe setting forwarde of the figures þat þe last of the divisor may not so oft be *withdraw* of the figure above his hede, above þat figure vnder the whiche the first of the diuiser is writ me most sette a cifre in ordre of the nombre quocient, and sette the figures forwarde as be-fore be o difference alone, and so me shalle do in alle nombres to be dyvydede, for where the dyviser may not be *with-draw* me most sette there a cifre, and sette forwarde the figures; as here:—

A special

The residue						1	2
The quocient				2	0	0	4
To be dyvydede	8	8	6	3	7	0	4
The dyvyser	4	4	2	3			

And me shalle not cesse fro suche setting of figures forwarde, nether of settinge of þe quocient into the dyviser, neþer of subtraccioun of the dyvyser, tille the first of the dyvyser be *with-draw* fro þe first

Another
example.

to be dividede. The whiche done, or ought, ¹⁷ *oper* nought shalle remayne: and yf it be ought, ¹⁷ kepe it in the tables, And euer vny it to þe divisor. And yf þou wilt wete how many vnytees of þe divisioun Fol. 53³. *wol growe to the nombre of the divisere, the nombre quocient wol shewe it: and whan suche divisioun is made, and þou lust prove yf thou have wele done or no, Multiplie the quocient by the divisor, And the same figures wol come ayene that thou haddest bifore and none other. And yf ought be residue, than *with* addicioun therof shalle come the same figures: And so multiplicacioun provithe divisioun, and

What the
quotient s

How to pi
your divis

dyvisioun multiplicacioun: as thus, yf multiplicacioun be made, divide it by the multipliant, and the nombre quocient wol shewe the nombre that was to be multipliede, *etc.*

or multipl

er VIII. Progression.

Progressioun is of nombre after egalle excesse fro oone or twayne take agregacioun. of progressioun one is naturelle or contynuelle, þat oþer broken and discontynuelle. Naturelle it is, whan me begynneth *with* one, and kepeth the ordure ouerlepyng one; as .1. 2. 3. 4. 5. 6., *etc.*, so þat the nombre folowyng passith the other be-fore in one. Broken it is, whan me lepithe fro o nombre tille another, and kepith not the contynuel ordire; as 1. 3. 5. 7. 9, *etc.* Ay me may begynne *with* .2., as þus; .2. 4. 6. 8., *etc.*, and the nombre folowyng passeth the others by-fore by .2. And note wele, that naturelle progressioun ay begynneth *with* one, and Intercise or broken progressioun, omwhile begynnyth *with* one, omwhile *with* twayne. Of progressioun naturell .2. rules ther be yove, of the whiche the first is this; whan the progressioun naturelle endithe in even nombre, by the half therof multiplie þe next totale ouerere nombre; Example of grace: .1. 2. 3. 4. Multiplie .5. by .2. and so .10. cometh of, that is the totale nombre þerof. The seconde rule is suche, whan the progressioun naturelle endithe in nombre ode. Take the more porcioun of the oddes, and multiplie therby the totale nombre. Example of grace 1. 2. 3. 4. 5., multiplie .5. by .3, and thryes .5. shalle be resultant. so the nombre totale is .15. Of progresioun intercise, ther ben also .2. ¹⁸ rules; and þe first is þis: Whan the Intercise progression endithe in even nombre by half therof multiplie the next nombre to þat halfe as .2. ¹⁸ 4. 6. Multiplie .4. by .3. so þat is thryes .4., and .12. the nombre of alle the progressioun, wolle folow. The seconde rule is this: whan the progressioun interscise endithe in ode, take þe more porcioun of alle þe nombre, Fol. 53⁴. *and multiplie by hym-selfe; as .1. 3. 5. Multiplie .3. by hym-selfe, and þe some of alle wolle be .9., *etc.*

Definitior
Progressi

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The first r
Broken
Progressi

The secur

er IX. Extraction of Roots.

Here folowithe the extraccioun of rotis, and first in nombre quadrates. Wherfor me shalle se what is a nombre quadrat, and what is the rote of a nombre quadrat, and what it is to draw out the rote of a nombre. And before other note this divisioun: Of nombres one is lyneal, anoper superficialle, anoper quadrat, anoper cubike or hoole. lyneal is that þat is considrede after the processe, havynge no respect to the direccioun of nombre in nombre, As a lyne hathe but one dymensioun that is to sey after the lengthe. Nombre superficial is þat comethe of ledynge of oo nombre into a-nother, wherfor it is callede superficial, for it hathe .2. nombres notyng or mesuryng hym, as a superficialle thyng hathe .2. dimensions, þat is to sey lengthe and brede. And for bycause a nombre may be hade in a-nother by .2. maners, þat is to sey other in hym-selfe, oþer in anoper, Vnderstonde yf it be had in hym-self, It is a quadrat. ffor dyvisioun write by vnytes, hathe .4. sides even as a quadrangille. and yf the nombre be hade in a-noper, the nombre is superficiel and not quadrat, as .2. hade in .3. makethe .6. that is þe first nombre superficiele; wherfor it is open þat alle nombre quadrat is superficiel, and not conuertide. The rote of a nombre quadrat is þat nombre that is had of hym-self, as twies .2. makithe 4. and .4. is the first nombre quadrat, and 2. is his rote. 9. 8. 7. 6. 5. 4. 3. 2. 1. / The rote of the more quadrat .3. 1. 4. 2. 6. The most nombre quadrat 9. 8. 7. 5. 9. 3. 4. 7. 6. / the remenent ouer the quadrat .6. 0. 8. 4. 5. / The first caas of nombre quadrat .5. 4. 7. 5. 6. The rote .2. 3. 4. The seconde caas .3. 8. 4. 5. The rote .6. 2. The thirde caas .2. 8. 1. 9. The rote .5. 3. The .4. caas .3. 2. 1. The rote .1. 7. / The 5. caas .9. 1. 2. 0. 4. / The rote 3. 0. 2. The solide nombre or cubike is þat þat comythe of double ledynge of nombre in nombre; And it is clepede a solide body that hathe þer-in .3 [dimensions] þat is to sey, lengthe, brede, and thiknesse. so þat nombre hathe .3. nombres to be brought forthe in hym. But nombre may be hade twies in nombre, for other it is hade in hym-selfe, oþer in a-noper. If a nombre be hade twies in hym-self, oþer ones in his quadrat, þat is the same, þat a cubike Fol. 54. *is, And is the same that is solide. And yf a nombre twies be hade in a-noper, the nombre is clepede solide and not cubike, as twies .3. and þat .2. makithe .12. Wherfor it is opyne that alle

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Linear,
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numbers.

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Notes of s
examples
square ro
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Solid nun

Three
dimension
solids.

Cubic nur

cubike nombre is solide, and not conuertide. Cubike is þat nombre þat comythe of ledynge of hym-selfe twyes, or ones in his quadrat. And here-by it is open that o nombre is the roote of a quadrat and of a cubike. Natheles the same nombre is not quadrat and cubike. Opyne it is also that alle nombres may be a rote to a quadrat and cubike, but not alle nombre quadrat or cubike. Therfor sithen þe ledynge of vnyte in hym-self ones or twies nought comethe but vnytes, Seithe Boice in Arsemetrike, that vnyte potencially is al nombre, and none in act. And vndirstonde wele also that betwix eury .2. quadrates ther is a meene proporcionalle, That is openede thus; lede the rote of o quadrat into the rote of the oper quadrat, and þan wolle þe meene shew.

All cubics
solid num

No numbr
be both li
and solid.

Unity is n
number.

Examples
square ro

Residuum			0				4			0					0		
Quadrande	4	3	5	6	3	0	2	9	1	7	4	2	4	1	9	3	6
Duplum	1	2			1	0			2	6				[8]	19		
Subduplum		6		6		5		5	1	3		2		4		4	

Also betwix the next .2. cubikis, me may fynde a double meene, that is to sey a more meene and a lesse. The more meene thus, as to brynge the rote of the lesse into a quadrat of the more. The lesse thus, If the rote of the more be brought Into the quadrat of the lesse.

A note on
proportion

er X. Extraction of Square Root.

To²⁰ draw a rote of the nombre quadrat it is What-euer nombre be proposede to fynde his rote and to se yf it be quadrat. And yf it be not quadrat the rote of the most quadrat fynde out, vnder the nombre proposede. Therfor yf thou wilt the rote of any quadrat nombre draw out, write the nombre by his differences, and compt the nombre of the figures, and wete yf it be ode or even. And yf it be even, than most thou begynne worche vnder the last save one. And yf it be ode with the last; and forto sey it shortly, al-weyes fro the last ode me shalle begynne. Therfor vnder the last in an od place sette, me most fynde a digit, the whiche lade in hym-selfe it puttithe away that, þat is ouer his hede, oper as neighe as me may: suche a

To find a
root.

Begin wit
last odd p

Find the r
square ro
that numbr
subtract,

Multiply 1
double by
second fig
and add a
the square
second fig
and subtra
Examples

The residue													5	4	3	2	
To be quadrede	4	1	2	0	9	1	5	1	3	9	9	0	0	5	4	3	2
The double		4	0				2		4			6		0			0
The vnder double	2		0		3	1		2		3	[3]		[0]		[0]		0

The residu

1	2	3	4	5	6	7	8	9
2	8	12	16	20	24	28	32	36
3	18	27	36	45	54	63	72	81
4	32	48	64	80	96	112 ²⁴	128	144

This table
constructe
use in cut
sums, giv
value of a

5	50	75	100	125	150	175	200	225
6	72	108	144	180	216	252	288	324
7	98	147	196	245	294	343	393	441
8	128	192	256	320	384	448	512	576
9	168	243	324	405	486	567	648	729 ²⁵

The rote of the most quadrat conteynede vndre the nombre proposede. Therfor yf thow wilt *prove* yf thow have wele do or no, Multiplie the digit last founde *with* the vnder-double oper vnder-doublis, and thow shalt fynde the same figures that thow haddest before; And so that nought be the Fol. 55. *residue. And yf thow have any residue, than *with* the addicioun perof that is reseruede *with-out* in thy table, thow shalt fynde thi first figures as thow haddest them before, *etc.*

How to pr
square ro
without o
remainder

er XI. Extraction of Cube Root.

Heere folowithe the extraccioun of rotis in cubike nombres; wher-for me most se what is a nombre cubike, and what is his roote, And what is the extraccioun of a rote. A nombre cubike it is, as it is before declarede, that comethe of ledyng of any nombre twies in hym-selfe, other ones in his quadrat. The rote of a nombre cubike is the nombre that is twies hade in hym-selfe, or ones in his quadrat. Wher-thurghe it is open, that euey nombre quadrat or cubike have the same rote, as it is seide before. And forto draw out the rote of a cubike, It is first to fynde be nombre proposede yf it be a cubike; And yf it be not, than thow most make extraccioun of his rote of the most cubike vndre the nombre proposide his rote founde. Therfor *proposede* some nombre, whos cubical rote pou woldest draw out; First thow most compt the figures by fourthes, that is to sey in the place of thousandes; And vnder the last thousande place, thow most fynde a digit, the whiche lade in hym-self cubikly puttithe a-way that pat is ouer his hede as in respect of hym, other as nyghe as thow maist. That done, thow most trebille the digit, and that triplat is to be put vnder the .3. next figure towarde the right honde, And the vnder-trebille vnder the trebille; Than me most fynde a digit vndre the next figure bifore the triplat, the whiche *with* his vnder-trebille had into a trebille, afterwarde other

Definitior
cubic nun
and a cub

Mark off
places in

Find the f
digit;

treble it a
place it ur
next but o
multiply t
digit.

vnder[trebille]²⁶ had in his produccioun, puttethe a-way alle that is ouer it in regarde of²⁷ [the triplat. Then lade in hymself puttithe away that þat is over his hede as in respect of hym, other as nyghe as thou maist:] That done, thow most trebille the digit ayene, and the triplat is to be sette vnder the next .3. figure as before, And the vnder-trebille vnder the trebille: and than most thow sette forwarde the first triplat with his vndre-trebille by .2. differences. And than most thow fynde a digit vnder the next figure before the triplat, the whiche withe his vnder-triplat had in his triplat afterwarde, other vnder-treblis lad in product Fol. 55 b.

*It sittethe a-way all that is ouer his hede in respect of the triplat than had in hym-self cubikly,²⁸ or as nyghe as ye may.

Then find second di

Multiply i triplate an second di, twice by t digit.

Subtract.

Examples

Residuum						5					4		1	0	1	9		
Cubicandus	8	3	6	5	4	3	2	3	0	0	7	6	7	1	1	6	6	7
Triplum			6	0						1	8				4			
Subtriplum	2			0			[3]			6		7		2			2	

Nother me shalle not cesse of the fyndynge of that digit, neither of his triplacioun, neþer of the triplat-is²⁹ anterioracioun, that is to sey, setting forwarde by .2. differences, Ne therof the vndre-triple to be put vndre the triple, Nether of the multiplicacioun þerof, Neither of the subtraccioun, tille it come to the first figure, vnder the whiche is a digitalle nombre to be founde, the whiche withe his vndre-treblis most be hade in tribles, After-warde without vnder-treblis to be hade into produccioun, setting away alle that is ouer the hede of the triplat nombre, After had into hymselfe cubikly, and sette alle-way that is ouer hym.

Continue process ti first figur reached.

Examples

Also note wele that the produccion comynge of the ledying of a digite founde³⁰ me may adde to, and also with-draw fro of the totalle

To be <u>cubicede</u>	1	7	2	8	3	2	7	6	8
The triple			3	2				9	
The vnder triple			1	2	[3]		3	3	

nombre sette above that digit so founde.³¹ That done ought or nought most be the residue. If it be nought, It is open that the nombre proposede was a cubike nombre, And his rote a digit founde last with the vnder-triples: If the rote therof wex bade in hym-selfe, and afterwarde product they shalle make the first

The resid

Special ca

Examples

of Numbers, &c.

A table of
numbers;
probably :
the Abacu

[Ende.]

- [1.](#) MS. Materialle.
- [2.](#) MS. Formalle.
- [3.](#) 'the' in MS.
- [4.](#) 'be' in MS.
- [5.](#) 'and' in MS.
- [6.](#) 'is' in MS.
- [7.](#) 6 in MS.
- [8.](#) 0 in MS.
- [9.](#) 2 in MS.
- [10.](#) *sic*.
- [11.](#) 'And' inserted in MS.
- [12.](#) '4 the' inserted in MS.
- [13.](#) 'to' in MS.
- [14.](#) 'that' repeated in MS.
- [15.](#) '1' in MS.
- [16.](#) Blank in MS.
- [17.](#) 'nought' in MS.
- [18.](#) 3 written for 2 in MS.
- [19.](#) 7 in MS.
- [20.](#) runs on in MS.
- [21.](#) 'so' in MS.
- [22.](#) 'nought' in MS.
- [23.](#) MS. adds here: 'wher-vpone se the table in the next side of the next leefte.'
- [24.](#) 110 in MS.
- [25.](#) 0 in MS.
- [26.](#) double in MS.
- [27.](#) 'it hym-selfe' in MS.
- [28.](#) MS. adds here: 'it settethe a-way alle his respect.'
- [29.](#) 'aucterioracioun' in MS.

[30.](#) MS. adds here: '*with* an vndre-triple / other of an vndre-triple in a triple or triplat is And after-warde *with* out vndre-triple other vndre-triplis in the product and ayene that product that comethe of the ledynge of a digit founde in hym-selfe cubicalle' /

[31.](#) MS. adds here: 'as ther had be a divisioun made as it is openede before.'

[32.](#) MS. anteriocacioun.

[33.](#) 4 in MS.

Accomptynge by counters.

The original text was printed as a single continuous paragraph, with no break between speakers; all examples were shown inline. It has been broken up for this e-text.

116 *b.*

* ¶ The seconde dialoge of accomptynge by counters.

Mayster.

Nowe that you haue learned the comen kyndes of Arithmetyke with the penne, you shall se the same art in counters: whiche feate doth not only serue for them that can not write and rede, but also for them that can do bothe, but haue not at some tymes theyr penne or tables redye with them. This sorte is in two fourmes *commonly*. The one by lynes, and the other without lynes: in that y^t hath lynes, the lynes do stande for the order of places: and in y^t that hath no lynes, there must be sette in theyr stede so many counters as shall nede, for eche lyne one, and they shall supplye the stede of the lynes.

S. By examples I shuld better *perceaue* your meanyng.

M. For example of the 117 *a ly*nes*:

Lo here you se .vi. lynes whiche stande for syxe places so that the nethermost

standeth for y^e fyrst place, and the next about it, for the second: and so vppward

tyll you come to the hyghest, which is the syxte lyne, and standeth for the syxte place. Now what is the valewe of euery place or lyne, you may *perceaue* by the figures whiche I haue set on them, which is accordynge as you learned before in the Numeration of figures by the penne: for the fyrste place is the

	1	0	0	0	0	0
	1	0	0	0	0	
x	1	0	0	0		
	1	0	0			
	1	0				
	1					
	1					

Numerati

place of vnities or ones, and euery counter set in that lyne betokeneth but one: *and* the seconde lyne is the place of 10, for euery counter there, standeth for 10. The thyrd lyne the place of hundredes: the fourth of thousandes: *and* so forth.

S. Syr I do perceauē that the same order is here of lynes, as was in the other figures 117 *b*. *by places, so that you shall not nede longer to stande about Numeration, excepte there be any other difference.

M. Yf you do vnderstande it, then how wyll you set 1543?

S. Thus, as I suppose.

X	1
	5
	4
	3

M. You haue set y^e places truly, but your figures be not mete for this vse: for the metest figure in this behalfe, is the figure of a counter round, as you se here, where I haue expressed that same summe.

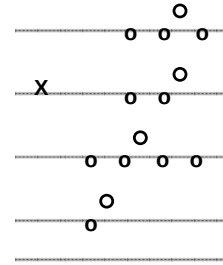
S. So that you haue not one figure for 2, nor 3, nor 4, and so forth, but as many digettes as you haue, you set in the lowest lyne: and for euery 10 you set one in the second line: and so of other. But I know not by what reason you set that one counter for 500 betwene two lynes.

x	o
	o
	o o o o
	o o o

M. you shall remember this, that when so euer you nede to set downe 5, 50, or 500, or 5000, or so forth any other number, whose numerator 118 *a*. *is 5, you shall set one counter for it, in the next space aboue the lyne that it hath his denomination of, as in this example of that 500, bycause the numerator is 5, it must be set in a voyd space: and bycause the denominator is hundred, I knowe that his place is the voyde space next aboue hundredes, that is to say, aboue the thyrd lyne. And farther you shall marke, that in all workynge by this sorte, yf you shall sette downe any summe betwene 4 and 10, for the fyrste parte of that nomber you shall set downe 5, & then so many counters more, as there reste numbers aboue 5. And this is true bothe of digettes and articles. And for example I wyll set downe this summe 287965, which summe yf you marke well, you nede none other examples for to lerne the numeration of 118

x	
	o o

b. *this forme. But this shal you marke, that as you dyd in the other kynde of arithmetike, set a pricke in the places of thousandes, in this worke you shall sette a starre, as you se here.



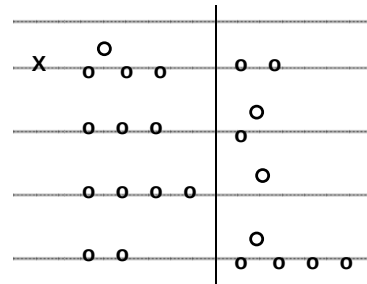
on on the Counting Board.

S. Then I perceave numeration, but I praye you, howe shall I do in this arte to adde two summes or more together?

Addition.

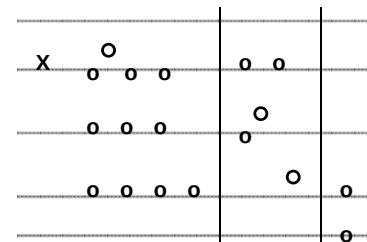
M. The easyest way in this arte is, to adde but 2 summes at ones together: how be it you may adde more, as I wyll tell you anone. Therefore when you wyll adde two summes, you shall fyrst set downe one of them, it forseth not whiche, *and* then by it drawe a lyne crosse the other lynes. And afterward set downe the other summe, so that that lyne may be betwene them, as yf you wolde adde 2659 to 8342, you must set your summes as you se here.

And then yf you lyst, you ¹¹⁹ a. *may adde the one to the other in the same place, or els you may adde them both together in a newe place: which waye, bycause it is moste playnest, I wyll shoue you fyrst. Therefore wyl I begynne at the vnites, whiche in the

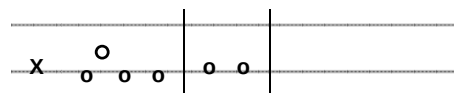


fyrst summe is but 2, *and* in y^e second summe 9, that maketh 11, those do I take vp, and for them I set 11 in the new rounge, thus,

Then do I take vp all y^e articles vnder a hundred, which in the fyrst summe are 40, and in the second summe 50, that maketh 90: or you may saye better, that in the fyrste summe there are 4 articles of 10, and in the seconde summe 5, which make 9, but then take hede that you sette them in theyr ¹¹⁹ b. *ryght lynes as you se here.

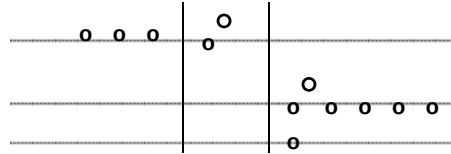


Where I haue taken awaye 40 from the fyrste summe, and 50



from y^e second, and in theyr stede

I haue set 90 in the thyrde,
 whiche I haue set playnely y^t you
 myght well perceauie it: how be it
 seyng that 90 with the 10 that



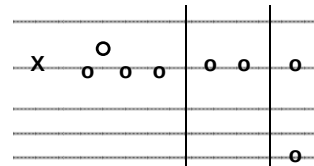
was in y^e thyrde rume all redy, doth make 100, I myghte better
 for those 6 counters set 1 in the thyrde lyne, thus:

For it is all one summe as you may se, but it is beste,
 neuer to set 5 counters in any line, for that may be done
 with 1 counter in a hygher place.

S. I iudge that good reason, for many are vnnefull,
 where one wyll serue.

M. Well, then ^{120 a.} *wyll I adde forth of hundredes: I fynde 3 in
 the fyrste summe, and 6 in the seconde, whiche make 900, them
 do I take vp *and* set in the thyrde rume where is one hundred all
 redy, to whiche I put 900, and it wyll be 1000, therfore I set one
 counter in the fourth lyne for them all, as you se here.

Then adde I y^e thousandes together, whiche
 in the fyrst summe are 8000, *and* in y^e
 second 2000, that maketh 10000: them do I
 take vp from those two places, and for
 them I set one counter in the fyfte lyne, and then appereth as
 you se,



to be 11001, for so many doth amount of the addition of
 8342 to 2659.

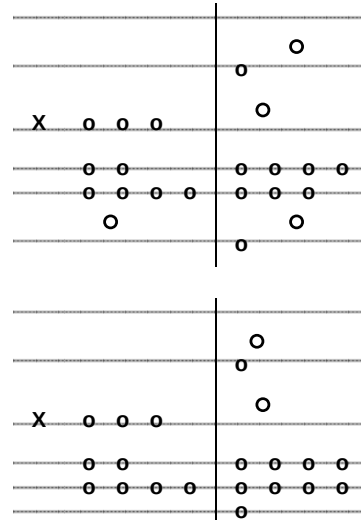
^{120 b.} *S. Syr, this I do perceave: but how shall I set one
 summe to an other, not chaungynge them to a thyrde
 place?

M. Marke well how I do it: I wyll adde together 65436, and
 3245, whiche fyrste I set downe thus.

Then do I begynne with the smalest, which in the fyrst summe is
 5, that do I take vp, and wold put to the other 5 in the seconde
 summe, sauynge that two counters can not be set in a voyd place
 of 5, but for them bothe I must set 1 in the seconde lyne, which is
 the place of 10, therfore I take vp the 5 of the fyrst summe, *and*

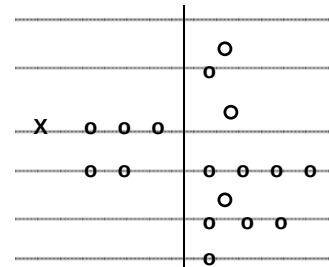
the 5 of the seconde, and for them I set 1 in the second lyne, 121 *a*. *as you se here.

Then do I lyke wayes take vp the 4 counters of the fyrste summe *and* seconde lyne (which make 40) and adde them to the 4 counters of the same lyne, in the second summe, and it maketh 80, But as I sayde I maye not conueniently set aboue 4 counters in one lyne, therefore to those 4 that I toke vp in the fyrst summe, I take one also of the seconde summe, and then haue I taken vp 50, for whiche 5 counters I sette downe one in the space ouer y^e second lyne, as here doth appere.



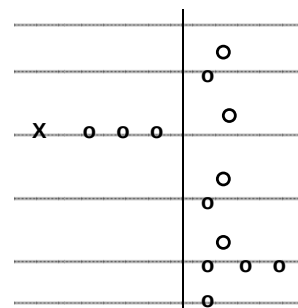
121 *b* *and then is there 80, as well $\underline{w}^t$ those

4 counters, as yf I had set downe y^e other 4 also. Now do I take the 200 in the fyrste summe, and adde them to the 400 in the seconde summe, and it maketh 600,



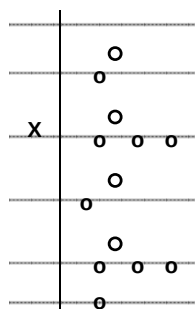
therefore I take vp the 2 counters in the fyrste summe, and 3 of them in the seconde summe, and for them

5 I set 1 in y^e space aboue, thus.

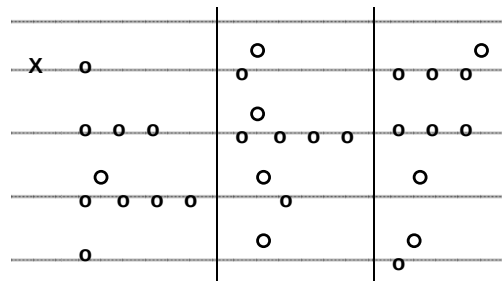
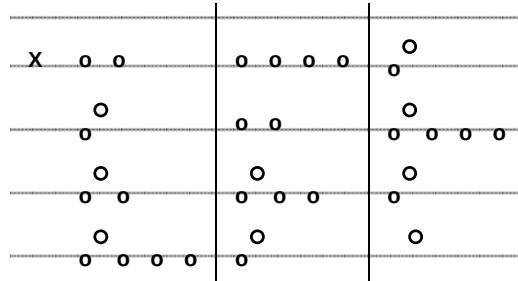


122 *a* *And so you see the hole summe, that amounteth of the addytion of 65436 with 3245

to be 6868[1]. And yf you haue marked these two examples well, you nede no farther enstruction in Addition of 2 only summes: but yf you haue more then two summes to adde, you may adde them thus.



Fyrst adde two of them, and then adde the thyrd^e, and y^e fourth, or more yf there be so many: as yf I wolde adde 2679 with 4286 and 1391. Fyrste I adde the two fyrste summes thus. 122 *b*. *And then I adde the thyrd^e thereto thus. And so of more yf you haue them.



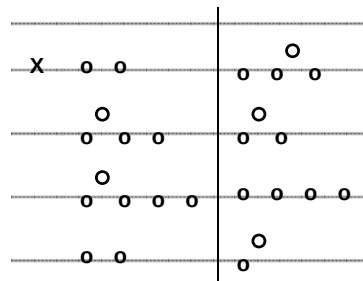
ction on the Counting Board.

S. Nowe I thynke beste that you passe forth to Subtraction, except there be any wayes to examyn this maner of Addition, then I thynke that were good to be knowen nexte.

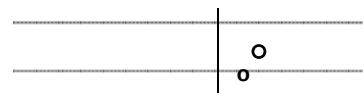
M. There is the same profe here that is in the other Addition by the penne, I meane Subtraction, for that onely is a sure waye: but consyderynge that Subtraction must be fyrste knowen, I wyl fyrste teache you the arte of Subtraction, and that by this example: I wolde subtracte 2892 out of 8746. These summes must I set downe as I dyd in Addition: but here it is best 116 *a* (*sic*). *to set the lesser number fyrste, thus.

Subtractic

Then shall I begynne to subtracte the greatest nombres fyrste (contrary to the vse of the penne) y^t is the thousandes in this example: therfore I fynd amongst the thousandes 2, for which I withdrawe so many from the seconde summe (where are 8) and so remayneth there 6, as this example showeth.



Then do I lyke wayes with the hundredes, of whiche in the fyrste summe 116 *b*. *I fynde 8, and is the



seconde summe but 7, out of whiche I
can not take 8, therefore thus muste I do:

I muste loke how moche my summe
dyffereth from 10, whiche I fynde here
to be 2, then must I bate for my summe

of 800, one thousande, and set downe the excesse of hundredes,
that is to saye 2, for so moche 100[0] is more then I shuld take
vp. Therefore from the fyrste summe I take that 800, and from the
second summe where are 6000, I take vp one thousande, and leue
5000; but then set I downe the 200 unto the 700 y^t are there all
redye, and make them 900 thus.

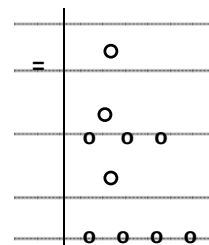
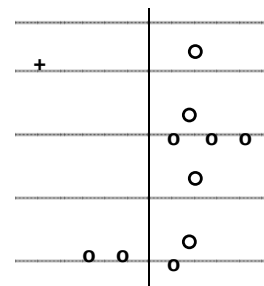
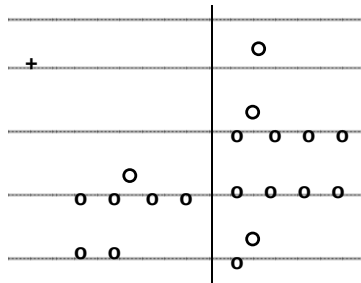
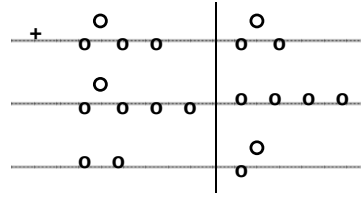
Then come I to the articles of tennes
where in the fyrste summe I fynde 90,
117 a. *and in the seconde summe but
only 40: Now consydering that 90 can
not be bated from 40, I loke how moche
y^t 90 doth dyffer from the next summe
about it, that is 100 (or elles whiche is

all to one effecte, I loke how moch 9 doth dyffer from 10) and I
fynd it to be 1, then in the stede of that 90, I do take from the
second summe 100: but consyderynge that it is 10 to moche, I set
downe 1 in y^e nexte lyne beneth for it, as you se here.

Sauynge that here I haue set one counter in y^e
space in stede of 5 in y^e nexte lyne. And thus
haue I subtracted all saue two, which I must
bate from the 6 in the second summe, and
there wyll remayne 4, thus.

So y^t yf I subtracte 2892 from 8746, the
remayner wyll be 5854, 117 b. *And that this is
truely wrought, you maye proue by Addition: for
yf you adde to this remayner the same summe that
you dyd subtracte, then wyll the formar summe
8746 amount agayne.

S. That wyll I proue: and fyrst I set the summe that
was subtracted, which was 2892, and then the remayner 5854,



thus.

Then do I adde fyrst y^e 2 to 4, whiche maketh 6, so take I vp 5 of those counters, and in theyr stede I sette 1 in the space, as here appereth.

118 a. *Then do I adde the 90 nexte aboue to the 50, and it maketh 140, therefore I take vp those 6 counters, and for them I sette 1 to the hundredes in y^e thyrde lyne, and 4 in y^e second lyne, thus.

Then do I come to the hundredes, of whiche I fynde 8 in the fyrst summe, and 9 in y^e second, that maketh 1700, therefore I take vp those 9 counters, and in theyr stede I sette 1 in the .iiii. lyne, and 1 in the space nexte beneth, and 2 in the thyrde lyne, as you se here.

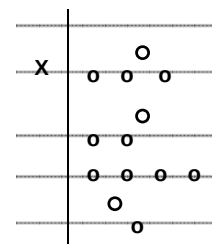
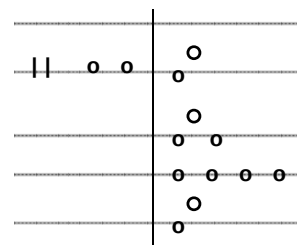
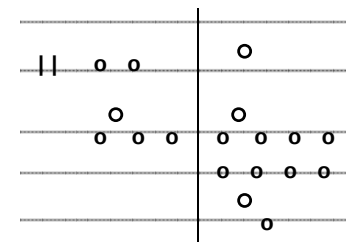
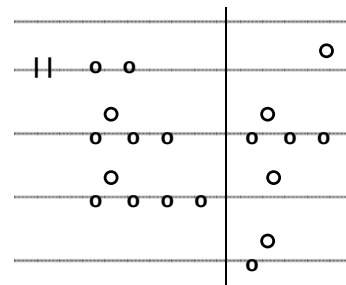
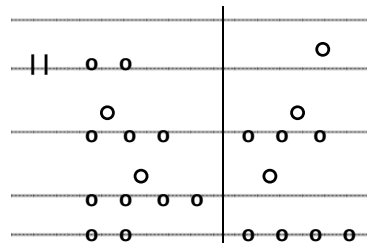
Then is there lefte in the fyrste summe but only 2000, whiche I shall take vp from thence, and set 118 b. *in the same lyne in y^e second summe, to y^e one y^t is there all redy: and then wyll the hole summe appere (as you may wel se) to be 8746, which was y^e fyrst grosse summe, and therefore I do perceauē, that I hadde well subtracted before.

And thus you may se how Subtraction maye be tryed by Addition.

S. I perceauē the same order here w^t counters, y^t I lerned before in figures.

M. Then let me se howe can you trye Addition by Subtraction.

S. Fyrste I wyl set forth this example of Addition where I haue added 2189 to 4988, and the hole summe appereth to be 7177,



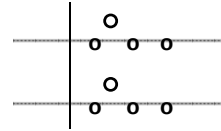
Therefore I take vp 5000,
downe so many in
age there all redy do
thus.

and set down the same in
here.
summe
econde



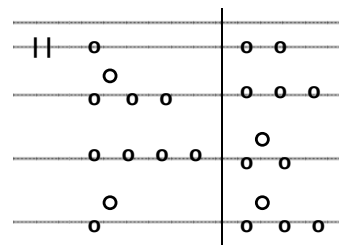
somme || ○ ○ ○ ○
 econde ○
 ○ ○ ○ ○

summe of my addition, and therefore I perceave,
I haue wel done.



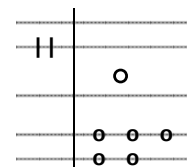
M. To stande longer about this, it is but folye:
excepte that this you maye also vnderstande, that many do
begynne to subtracte with counters, not at the hyghest summe, as
I haue taught you, but at the nethermoste, as they do vse to adde:
and when the summe to be abatyd, in any lyne appeareth greater
then the other, then do they borowe one of the next hygher
roume, as for example: yf they shuld abate 1846 from 2378, they
set y^e summes thus.

120 b. *And fyrste they take 6 whiche is in
the lower lyne, and his space from 8 in
the same rouses, in y^e second summe,
and yet there remayneth 2 counters in the
lowest lyne. Then in the second lyne must
4 be subtracte from 7, and so remayneth



there 3. Then 8 in the thyrde lyne and his space, from 3 of the
second summe can not be, therefore do they bate it from a hygher
roume, that is, from 1000, and bycause that 1000 is to moch by
200, therefore must I sette downe 200 in the thyrde lyne, after I
haue taken vp 1000 from the fourth lyne: then is there yet 1000
in the fourth lyne of the fyrst summe, whiche yf I withdrawe
from the seconde summe, then doth all y^e figures stande in this
order.

So that (as you se) it differeth not greatly whether
you begynne subtraction at the hygher lynes, or at
121 a. *the lower. How be it, as some menne lyke the
one waye beste, so some lyke the other: therefore
you now knowyng bothe, may vse whiche you lyst.



lication by Counters.

But nowe touchynge Multiplication: you shall set your numbers
in two rouses, as you dyd in those two other kyndes, but so that
the multiplier be set in the fyrste rouse. Then shall you begyn
with the hyghest numbers of y^e seconde rouse, and multiply
them fyrst after this sort. Take that ouermost lyne in your fyrst

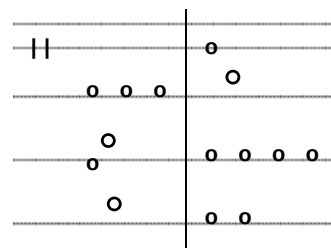
Multiplic:

workynge, as yf it were the lowest lyne, setting on it some mouable marke, as you lyste, and loke how many counters be in hym, take them vp, and for them set downe the hole multiplyer, so many tymes as you toke vp counters, reckenynge, I saye that lyne for the vnites: *and* when you haue so done with the hygheest number then come to the nexte lyne beneth, *and* do euen so with it, and so with y^e next, tyll you haue done all. And yf there be any number in a space, then for it ¹²¹ *b.* *shall you take y^e multiplyer 5 tymes, and then must you reckon that lyne for the vnites whiche is nexte beneth that space: or els after a shorter way, you shall take only halfe the multiplyer, but then shall you take the lyne nexte aboue that space, for the lyne of vnites: but in suche workynge, yf chaunce your multiplyer be an odde number, so that you can not take the halfe of it iustly, then muste you take the greater halfe, and set downe that, as if that it were the iuste halfe, and farther you shall set one counter in the space beneth that line, which you reckon for the lyne of vnities, or els only remoue forward the same that is to be multiplyed.

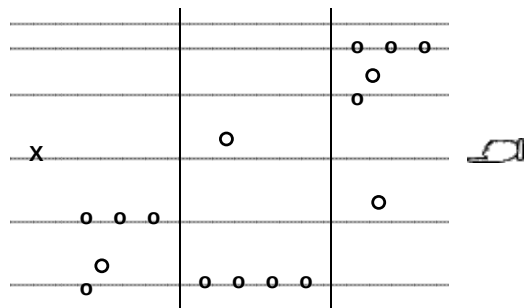
S. Yf you set forth an example hereto I thynke I shal perceauce you.

M. Take this example: I wold multiply 1542 by 365, therefore I set y^e numbers thus.

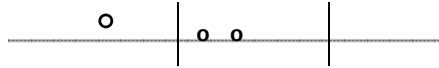
¹²² *a.* *Then fyrste I begynne at the 1000 in y^e hygheest roume, as yf it were y^e fyrst place, & I take it vp, settinge downe for it so often (that is ones) the multiplyer, which is 365, thus, as you se here:



where for the one counter taken vp from the fourth lyne, I haue sette downe other 6, whiche make y^e summe of the multiplyer, reckenynge that fourth lyne, as yf it were the fyrste: whiche thyng I haue marked



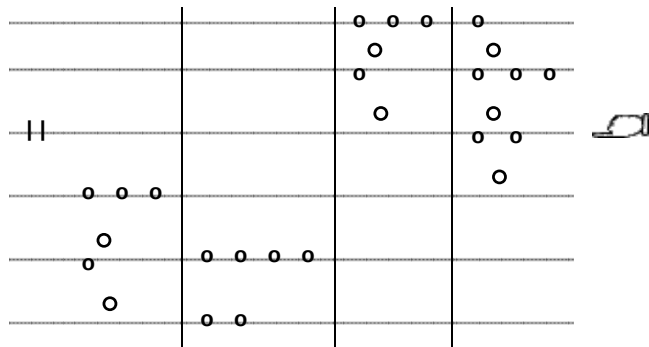
by the hand set at the
begynnyng of y^e same,



S. I perceave this well: for in dede, this summe that you haue set
downe is 365000, for so moche doth amount 122 *b.* *of 1000,
multiplied by 365.

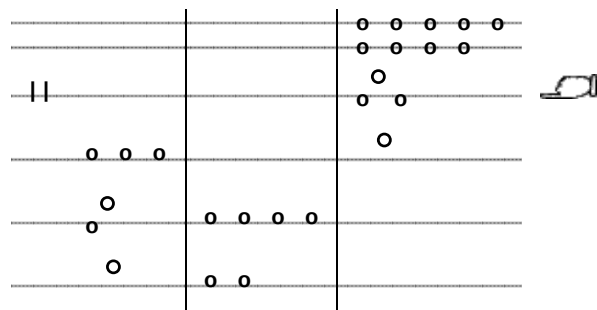
M. Well then to go forth, in the nexte space I fynde one counter
which I remoue forward but take not vp, but do (as in such case I
must) set downe the greater halfe of my multiplier (seyng it is an
odde number) which is 182, *and* here I do styll let that fourth
place stand, as yf it were y^e fyrst:

as in this fourme
you se, where I
haue set this
multiplication with
 y^e other: but for the
ease of your
vnderstandynge,
I haue set a lytell



lyne betwene them: now shulde they both in one summe stand
thus.

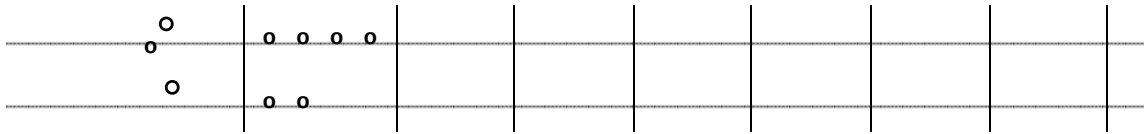
123 *a.* *Howe be it an
other fourme to
multyplye suche
counters in space is
this: Fyrst to remoue
the fynger to the lyne
nexte benethe y^e space,



and then to take vp y^e

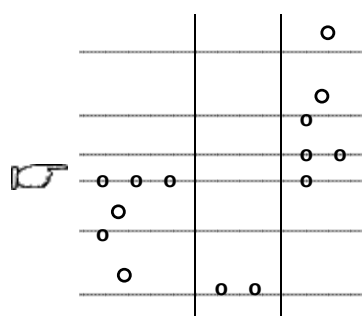
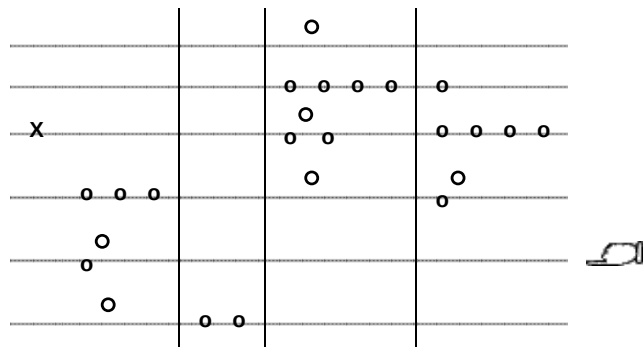
counter, *and* to set downe y^e multiplier .v. tymes, as here you se.





Which summes yf you do adde together into one summe, you
 shal perceauē that it wyll be y^e same y^t appeareth of y^e other
 working before, so that 123 *b*. *bothe sortes are to one entent, but
 as the other is much shorter, so this is playner to reason, for
 suche as haue had small exercyse in this arte. Not withstandynge
 you maye adde them in your mynde before you sette them
 downe, as in this example, you myghte haue sayde 5 tymes 300
 is 1500, and 5 tymes 60 is 300, also 5 tymes 5 is 25, whiche all
 put together do make 1825, which you maye at one tyme set
 downe yf you lyste. But nowe to go forth, I must remoue the
 hand to the nexte counters, whiche are in the second lyne, and
 there must I take vp those 4 counters, settinge downe for them
 my multiplier 4 tymes, whiche thyng other I maye do at 4
 tymes seuerally, or elles I may gather that hole summe in my
 mynde fyrste, and then set it downe: as to saye 4 tymes 300 is
 1200: 4 tymes 60 are 240: and 4 tymes 5 make 20: y^t is in all
 1460, y^t shall I set downe also: as here you se.

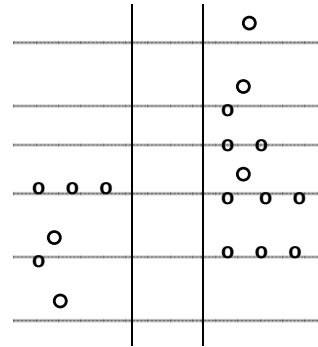
124 *a*. *whiche yf I
 ioyne in one summe
 with the formar
 numbers, it wyll
 appeare thus.
 Then to ende this
 multiplicacion, I
 remoue the fynger to



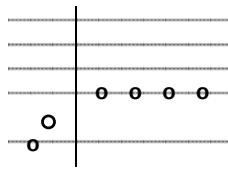
the lowest lyne, where are onely 2,
 them do I take vp, and in theyr stede
 do I set downe twyse 365, that is 730,
 for which I set 124 *b*. *one in the space
 aboue the thyrd lyne for 500, and 2
 more in the thyrd lyne with that one
 that is there all redye, and the reste in

theyr order, *and* so haue I ended the hole summe thus.

Wherby you se, that 1542 (which is the number of yeares syth Ch[r]ystes incarnation) beyng multiplyed by 365 (which is the number of dayes in one yeare) dothe amounte vnto 562830, which declareth y^e number of daies sith Chrystes incarnation vnto the ende of 1542¹ yeares. (besyde 385 dayes and 12 houres for lepe yeares).

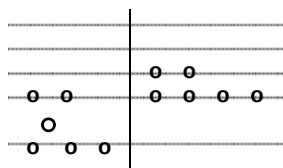
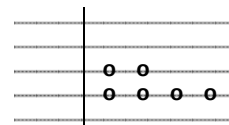


S. Now wyll I proue by an other example, as this: 40 labourers (after 6 d. y^e day for eche man) haue wrought 28 dayes, I wold 125 a. *know what theyr wages doth amount vnto: In this case muste I worke doublely: fyrst I must multiplye the number of the labourers by y^e wages of a man for one day, so wyll y^e charge of one daye amount: then secondarely shall I multiply that charge of one daye, by the hole number of dayes, *and* so wyll the hole summe appeare: fyrst therefore I shall set the summes thus.



Where in the fyrste space is the multiplyer (y^t is one dayes wages for one man) *and* in the second space is set the number of the worke men to be multiplyed: then saye I, 6 tymes 4 (rekenynge that second lyne as the lyne of vnites) maketh 24, for whiche summe I shulde set 2 counters in the thyrde lyne, and 4 in the seconde, therefore do I set 2 in the thyrde lyne, and let the 4 stand styll in the seconde lyne, thus.* 125 b.

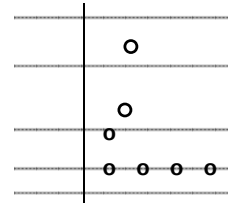
So apwereth the hole dayes wages to be 240d'. that is 20 s. Then do I multiply agayn the same summe by the number of dayes and fyrste I sette the nombers, thus.



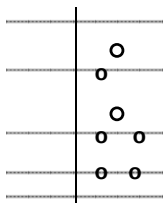
Then bycause there are counters in dyuers lynes, I shall begynne with the hyghest, and take them vp, settinge for them the multiplyer so many tymes, as I toke vp counters, y^t is twyse, then wyll y^e summe

stande thus.

Then come I to y^e seconde lyne, and take vp those 4 counters, settinge for them the multiplyer foure tymes, so wyll the hole summe appeare thus.* 126 a.



So is the hole wages of 40 workemen, for 28 dayes (after 6d'. eche daye for a man) 6720d'. that is 560 s. or 28 l'i.

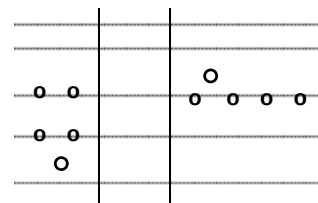


on on the Counting Board.

M. Now if you wold proue Multiplikation, the surest way is by Dyuision: therfore wyll I ouer passe it tyll I haue taught you y^e arte of Diuision, whiche you shall worke thus. Fyrste sette downe the Diuisor for feare of forgettynge, and then set the number that shalbe deuided, at y^e ryghte syde, so farre from the diuisor, that the quotient may be set betwene them: as for example: Yf 225 shepe cost 45 l'i. what dyd euery shepe cost? To knowe this, I shulde diuide the hole summe, that is 45 l'i. by 225, but that can not be, therfore must I fyrste reduce that 45 l'i. into a lesser denomination, as into shylynges: then I multiply 45 by 20, and it is 900, that summe shall I diuide by the number of 126 b. *shepe, whiche is 225, these two numbers therfore I sette thus.

Diuision.

Then begynne I at the hyghest lyne of the diuident, and seke how often I may haue the diuisor therin, and that maye I do 4 tymes, then say I, 4 tymes 2 are 8, whyche yf I take from 9, there resteth but 1, thus And bycause I founde the diuisor 4 tymes in the diuidente, I haue set (as you se) 4 in the myddle rounge, which 127 a. *is the place of the quotient: but now must I take the reste of the diuisor as often out of the remayner:

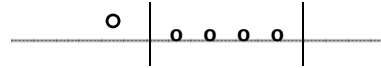


therfore come I to the seconde lyne of the diuisor, sayeng 2 foure tymes make 8, take 8 from 10, and there resteth 2, thus.

Then come I to the lowest number, which is 5, and multiply it 4 tymes, so

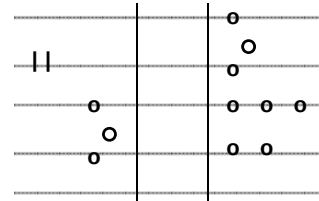


is it 20, that take I from 20, and there remayneth nothyng, so that I se my quotient to be 4, whiche are in valewe shylynges, for so was the diuident: and therby I knowe, that yf 225 shepe dyd coste 45 l'i. euery shepe coste 4 s.



S. This can I do, as you shall perceau by this example: Yf 160 sowldyars do spende euery moneth 68 l'i. what spendeth eche man? Fyrst 127 b. *bycause I can not diuide the 68 by 160, therefore I wyll turne the poundes into pennes by multiplicacion, so shall there be 16320 d'. Nowe muste I diuide this summe by the number of sowldyars, therefore I set them in order, thus.

Then begyn I at the hyghest place of the diuidente, sekyng my diuisor there, whiche I fynde ones, Therefore set I 1 in the nether lyne.

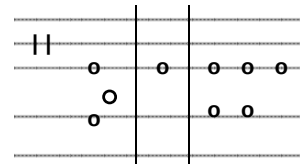


M. Not in the nether line of the hole summe, but in the nether lyne of that worke, whiche is the thyrd lyne.

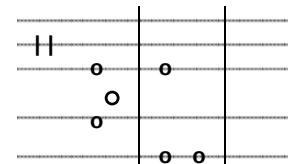
S. So standeth it with reason.

M. Then thus do they stande.* 128 a.

Then seke I agayne in the reste, how often I may fynde my diuisor, and I se that in the 300 I myghte fynde 100 thre tymes, but then the 60 wyll not be so often founde in 20, therefore I take 2 for my quotient: then take I 100 twyse from 300, and there resteth 100, out of whiche with the 20 (that maketh 120) I may take 60 also twyse, and then standeth the numbers thus,



128 b. *where I haue sette the quotient 2 in the lowest lyne: So is euery sowldyars portion 102 d'. that is 8 s. 6 d'.

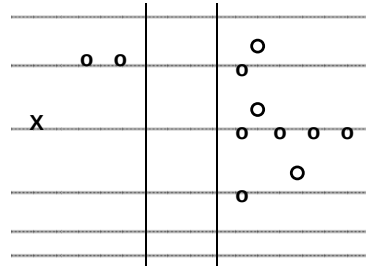


M. But yet bycause you shall perceau iustly the reason of Diuision, it shall be good that you do set your diuisor styll agaynst those nombres from whiche you do take it:

as by this example I wyll declare. Yf y^e purchase of 200 acres of ground dyd coste 290 l'i. what dyd one acre coste? Fyrst wyl I

turne the poundes into pennes, so wyll there be 69600 d'. Then in settinge downe these numbers I shall do thus.

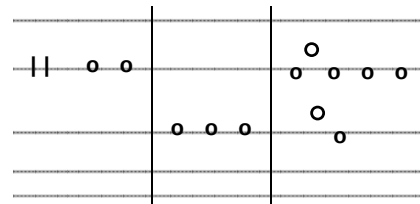
Fyrst set the diuident on the ryghte hande as it oughte, and then ^{129 a.} *the diuisor on the lefte hande agaynst those numbers, from which I entende to take hym fyrst as here you se, wher I haue set the diuisor two lynes hygher then is theyr owne place.



S. This is lyke the order of diuision by the penne.

M. Truth you say, and nowe must I set y^e quotient of this worke in the thyrde lyne, for that is the lyne of vnities in respecte to the diuisor in this worke. Then I seke howe often the diuisor maye be founde in the diuident, *and* that I fynde 3 tymes, then set I 3 in the thyrde lyne for the quotient, and take awaye that 60000 from the diuident, and farther I do set the diuisor one line lower, as yow se here.

^{129 b.} *And then seke I how often the diuisor wyll be taken from the nomber agaynste it, whiche wyll be 4 tymes and 1 remaynyng.

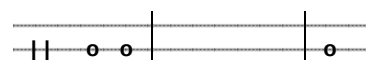


S. But what yf it chaunce that when the diuisor is so remoued, it can not be ones taken out of the diuident agaynste it?

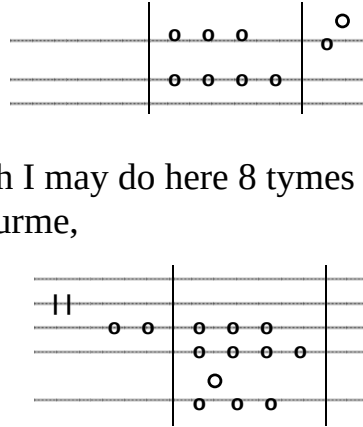
M. Then must the diuisor be set in an other line lower.

S. So was it in diuision by the penne, and therfore was there a cypher set in the quotient: but howe shall that be noted here?

M. Here nedeth no token, for the lynes do represente the places: onely loke that you set your quotient in that place which standeth for vnities in respecte of the diuisor: but now to returne to the example, I fynde the diuisor 4 tymes in the diuidente, and 1 remaynyng, for 4 tymes 2 make 8, which I take from 9, and there resteth 1, as this figure sheweth: and in the myddle space for the quotient I set 4 in the seconde lyne,



whiche is in this worke the place of vnities.* 130 a. Then remoue I y^e diuisor to the next lower line, and seke how often I may haue it in the dyuident, which I may do here 8 tymes iust, and nothyng remayne, as in this fourme, where you may se that the hole quotient is 348 d', that is 29 s. wherby I knowe that so moche coste the purchase of one aker.



S. Now resteth the profes of Multiplycation, and also of Diuision.

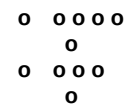
M. Ther best profes are eche 130 b. *one by the other, for Multiplycation is proued by Diuision, and Diuision by Multiplycation, as in the worke by the penne you learned.

S. Yf that be all, you shall not nede to repete agayne that, y^t was sufficyently taughte all redye: and excepte you wyll teache me any other feate, here maye you make an ende of this arte I suppose.

M. So wyll I do as touchynge hole number, and as for broken number, I wyll not trouble your wytte with it, tyll you haue practised this so well, y^t you be full perfecte, so that you nede not to doubte in any poynte that I haue taught you, and thenne maye I boldly enstructe you in y^e arte of fractions or broken number, wherein I wyll also shoue you the reasons of all that you haue nowe learned. But yet before I make an ende, I wyll shoue you the order of commen castyng, wher in are bothe pennes, shylynges, and poundes, procedynge by no grounded reason, but onely by a receaued 131 a. *fourme, and that dyuersly of dyuers men: for marchauntes vse one fourme, and auditors an other:

ants' Casting Counters.

But fyrste for marchauntes fourme marke this example here, in which I haue expressed this summe 198 l'i. ² 19 s. 11 d'. So that you maye se that the lowest lyne serueth for pennes, the next aboue for shylynges, the thyrd for poundes, and the fourth for scores of



Merchant casting.

poundes. And farther you maye se, that the space
 betwene pennes and shylynges may receaue but one
 counter (as all other spaces lyke wayes do) and that one standeth
 in that place for 6 d'. Lyke wayes betwene the shylynges *and* the
 poundes, one counter standeth for 10 s. And betwene the poundes
 and 20 l'i. one counter standeth for 10 poundes. But besyde those
 you maye see at the left syde of shylynges, that one counter
 standeth alone, *and* betokeneth 5 s. 131 b. *So agaynste the
 poundes, that one counter standeth for 5 l'i. And agaynst the 20
 poundes, the one counter standeth for 5 score poundes, that is
 100 l'i. so that euery syde counter is 5 tymes so moch as one of
 them agaynst whiche he standeth. Now for the accompt of
 auditors take this example.

o oooo
 o
 ooooo

Auditors'
 casting.

where I haue expressed y^e same summe
 198 l'i. 19 s. 11 d'. But here you se the

o o o o o o
 ooo ooo ooo ooo
 o o o

pennes stande toward y^e ryght hande, and
 the other encreasyng orderly toward the lefte hande. Agayne
 you maye se, that auditours wyll make 2 lynes (yea and more) for
 pennes, shylynges, *and* all other valewes, yf theyr summes
 extende therto. Also you se, that they set one counter at the ryght
 ende of eche rowe, whiche so set there standeth for 5 of that
 rouse: and on 132 a. *the lefte corner of the rowe it standeth for
 10, of y^e same row. But now yf you wold adde other subtracte
 after any of both those sortes, yf you marke y^e order of y^t other
 feate which I taught you, you may easely do the same here
 without moch teachynge: for in Addition you must fyrst set
 downe one summe and to the same set the other orderly, and lyke
 maner yf you haue many: but in Subtraction you must sette
 downe fyrst the greatest summe, and from it must you abate that
 other euery denomination from his dewe place.

S. I do not doubte but with a lytell practise I shall attayne these
 bothe: but how shall I multiply and diuide after these fourmes?

M. You can not duely do none of both by these sortes, therfore in
 suche case, you must resort to your other artes.

S. Syr, yet I se not by these sortes how to expresse hundreddes, yf they excede one hundred, nother yet thousandes.

M. They that vse such accomptes that it excede 200 ^{132 b.} *in one summe, they sette no 5 at the lefte hande of the scores of poundes, but they set all the hundredes in an other farther rowe and 500 at the lefte hand therof, and the thousandes they set in a farther rowe yet, *and* at the lefte syde therof they sette the 5000, and in the space ouer they sette the 10000, and in a hygher rowe 20000, whiche all I haue expressed in this example, which is 97869 l'i. 12 s. 9 d' ob. q. for I had not told you before where, nother how you shuld set downe farthynges, which (as you se here) must be set in a voyde space sydelynge beneth the pennes: for q one counter: for ob. 2 counters: for ob. q. 3 counters: *and* more there can not be, for 4 farthynges ^{133 a.} *do make 1 d'. which must be set in his dewe place.

```

      o o o o
      o
o  o o
o  o o o
  o o o
o  o o o o
  o
  o o
  o
  o o o
      o o
      o

```

rs' Casting Counters.

And yf you desyre y^e same summe after audytors maner, lo here it is.

```

      o o      o      o      o      o      o
o o o o o o o o o o o o o o o o o o o o o
o

```

But in this thyng, you shall take this for suffycient, and the reste you shall obserue as you maye se by the working of eche sorte: for the dyuers wittes of men haue inuented dyuers and sundry wayes almost vnnumerable. But one feate I shall teache you, whiche not only for the straungenes and secretnes is moche pleasaunt, but also for the good *commoditie* of it ryghte worthy to be well marked. This feate hath ben vused aboue 2000 yeares at the leaste, and yet was it neuer comenly knowen, especyally in Englysshe it was neuer taughte yet. This is the arte of nombrynge on the hand, with diuers gestures of the fynghers, expressynge any summe conceaued in the ^{133 b.} *mynde. And fyrst to begynne, yf you wyll expresse any summe vnder 100, you shall expresse it with your lefte hande: and from 100 vnto 10000, you shall

expresse it with your ryght hande, as here orderly by this table
folowyng you may perceave.

¶ Here foloweth the table
of the arte of the
hande

The arte of nombryng by the hande.

134

1	10	100	1000
2	20	200	2000
3	30	300	3000
4	40	400	4000
5	50	500	5000
6	60	600	6000
7	70	700	7000
8	80	800	8000
9	90	900	9000

134 b. *In which as you may se 1 is expressed by y^e lyttle fynger
of y^e lefte hande closely and harde croked. 1

³2 is declared by lyke bowynge of the weddynge fynger (whiche
is the nexte to the lyttell fynger) together with the lytell fynger. 2

3 is signified by the myddle fynger bowed in lyke maner, with
those other two. 3

4 is declared by the bowyng of the myddle fynger and the rynge
fynger, or weddynge fynger, with the other all stretched forth. 4

5 is represented by the myddle fynger onely bowed. 5

And 6 by the weddynge fynger only crooked: and this you may
marke in these a certayne order. But now 7, 8, and 9, are
expressed *with* the bowynge of the same fyngers as are 1, 2, and
3, but after an other fourme. 6

For 7 is declared by the bowynge of the lytell fynger, as is 1,
saue that for 1 the fynger is clasped in, harde *and* 135 a. *rounde,
but for to expresse 7, you shall bowe the myddle ioynte of the
lytell fynger only, and holde the other ioyntes streyght. 7

S. Yf you wyll geue me leue to expresse it after my rude maner,
thus I vnderstand your meanyng: that 1 is expressed by
crookyng in the lyttell fynger lyke the head of a bysshoppes
bagle: and 7 is declared by the same fynger bowed lyke a gybbet.

M. So I perceauē, you vnderstande it.

Then to expresse 8, you shall bowe after the same maner both the
lyttell fynger and the rynge fynger. 8

And yf you bowe lyke wayes with them the myddle fynger, then
doth it betoken 9. 9

Now to expresse 10, you shall bowe your fore fynger rounde, and
set the ende of it on the hyghest ioynte of the thombe. 10

And for to expresse 20, you must set your fyngers streyght, and
the ende of your thombe to the partition of the 135 b. *fore moste
and myddle fynger. 20

30 is represented by the ioynynge together of y ^e headdes of the foremost fynger and the thombe.	30
40 is declared by settynge of the thombe crossewayes on the foremost fynger.	40
50 is signified by ryght stretchyng forth of the fyngers ioyntly, and applyenge of the thombes ende to the partition of the myddle fynger <i>and</i> the rynge fynger, or weddynge fynger.	50
60 is formed by bendynge of the thombe croked and crossynge it with the fore fynger.	60
70 is expressed by the bowynge of the foremost fynger, and settynge the ende of the thombe between the 2 foremost or hyghest ioyntes of it.	70
80 is expressed by settynge of the foremost fynger crossewayes on the thombe, so that 80 dyffereth thus from 40, that for 80 the forefynger is set crosse on the thombe, and for 40 the thombe is set crosse ouer y ^e forefinger.	80
136 a. *90 is signified, by bendynge the fore fynger, and settyng the ende of it in the innermost ioynthe of y ^e thombe, that is euen at the foote of it. And thus are all the numbers ended vnder 100.	90
S. In dede these be all the numbers from 1 to 10, <i>and</i> then all the tenthes within 100, but this teacyed me not how to expresse 11, 12, 13, <i>etc.</i> 21, 22, 23, <i>etc.</i> and such lyke.	11, 12, 13 21, 22, 23
M. You can lytell vnderstande, yf you can not do that without teachynge: what is 11? is it not 10 and 1? then expresse 10 as you were taught, and 1 also, and that is 11: and for 12 expresse 10 and 2: for 23 set 20 and 3: and so for 68 you muste make 60 and there to 8: and so of all other sortes.	
But now yf you wolde represente 100 other any number aboue it, you muste do that with the ryghte hande, after this maner.	100
You must expresse 100 in the ryght hand, with the lytell fynger so bowed as you dyd expresse 1 in the left hand.	

136 b. *And as you expressed 2 in the lefte hande, the same fasshyon in the ryght hande doth declare 200.	200
The fourme of 3 in the ryght hand standeth for 300.	300
The fourme of 4, for 400.	400
Lykewayes the fourme of 5, for 500.	500
The fourme of 6, for 600. And to be shorte: loke how you did expresse single vnities and tenthes in the lefte hande, so must you expresse vnities <i>and</i> tenthes of hundredes, in the ryghte hande.	600
S. I vnderstande you thus: that yf I wold represent 900, I must so fourme the fyngers of my ryghte hande, as I shuld do in my left hand to expresse 9, And as in my lefte hand I expressed 10, so in my ryght hande must I expresse 1000.	900 1000
And so the fourme of euery tenthe in the lefte hande serueth to expresse lyke number of thousandes, so y ^e fourme of 40 standeth for 4000.	4000
The fourme of 80 for 8000.	8000
137 a.	9000

*And the fourme of 90 (whiche is
the greatest) for 9000, and aboue that
I can not expresse any number. *M.*
No not with one fynger: how be it,
with dyuers fyngers you maye expresse
9999, and all at one tyme, and that lac
keth but 1 of 10000. So that vnder
10000 you may by your fyngers ex-
presse any summe. And this shal suf-
fyce for Numeration on the fyngers.
And as for Addition, Subtraction,
Multiplication, and Diuision (which
yet were neuer taught by any man as
farre as I do knowe) I wyll enstruct
you after the treatyse of fractions.
And now for this tyme fare well,

and loke that you cease not to
practyse that you haue lear
ned. S. Syr, with moste
harty mynde I thanke
you, bothe for your
good learnyng, *and*
also your good
counsel, which
(god wyll yng) I truste to folow.
Finis.

- [1.](#) 1342 in original.
- [2.](#) 168 in original.
- [3.](#) Bracket ([]) denotes new paragraph in original.

For this e-text, the brackets have been omitted in favor of restoring the paragraph breaks. Numbers 200 and up were printed as separate paragraphs and are unchanged. Sidenote 4 was missing and has been supplied by the transcriber; the pairs 5, 6 and 9, 10 (originally on one line) have been separated.

APPENDIX I.

A Treatise on the Numeration of Algorism.

[From a MS. of the 14th Century.]

To alle suche even nombrys the most have cifrys as to ten.
 twenty. thirtty. an hundred. an thousand and suche other. but ye
 schal vnderstonde that a cifre tokeneth nothings but he maketh
 other the more significatyf that comith after hym. Also ye schal
 vnderstonde that in nombrys composyt and in alle other nombrys
 that ben of diverse figurys ye schal begynne in the riht syde and
 to rekene backwarde and so he schal be wryte as thus—1000. the
 sifre in the riht side was first wryte and yit he tokeneth nothings
 to the secunde no the thridde but thei maken that figure of 1 the
 more signifycatyf that comith after hem by as moche as he born
 oute of his first place where he schuld yf he stode ther tokene but
 one. And there he stondith nowe in the ferye place he tokeneth a
 thousand as by this rewle. In the first place he tokeneth but
 hymself. In the secunde place he tokeneth ten times hymself. In
 the thridde place he tokeneth an hundred tymes hymself. In the
 ferye he tokeneth a thousand tymes hymself. In the fyfthe place
 he tokeneth ten thousand tymes hymself. In the sexte place he
 tokeneth an hundred thousand tymes hymself. In the seveth place
 he tokeneth ten hundred thousand tymes hymself, &c. And ye
 schal vnderstond that this worde nombre is partyd into thre
 partyes. Somme is callyd nombre of digitys for alle ben digitys
 that ben withine ten as ix, viii, vii, vi, v, iv, iii, ii, i. Articles ben
 alle thei that mow be devyded into nombrys of ten as xx, xxx, xl,
 and suche other. Composittys be alle nombrys that ben
 componyd of a digyt and of an articule as fourtene fyftene
 thrittene and suche other. Fourtene is componyd of four that is a
 digyt and of ten that is an articule. Fyftene is componyd of fyve
 that is a digyt and of ten that is an articule and so of others
 But as to this rewle. In the firste place he tokeneth but
 himself that is to say he tokeneth but that and no more. If that he
 stonde in the secunde place he tokeneth ten tymes himself as this
 figure 2 here 21. this is oon and twenty. This figure 2 stondith in
 the secunde place and therfor he tokeneth ten tymes himself and
 ten tymes 2 is twenty and so forye of every figure and he stonde
 after another toward the lest syde he schal tokene ten tymes as
 moche more as he schuld token and he stode in that place ther
 that the figure afore him stondeth: lo an example as thus 9634.
 This figure of foure that hath this schape 4 tokeneth but himself

for he stondeth in the first place. The figure of thre that hath this schape 3 tokeneth ten tyme himself for he stondeth in the secunde place and that is thritti. The figure of sexe that hath this schape 6 tokeneth ten tyme more than he schuld and he stode in the place yer the figure of thre stondeth for ther he schuld tokene but sixty. And now he tokeneth ten tymes that is sexe hundrid. The figure of nyne that hath this schape 9 tokeneth ten tymes more than he schulde and he stode in the place ther the figure of 6 stondeth inne for thanne he schuld tokene but nyne hundryd. And in the place that he stondeth inne nowe he tokeneth nine thousand. Alle the hole nombre of these foure figurys. Nine thousand sexe hundrid and foure and thritti.

APPENDIX II.

Carmen de Algorismo.

[From a B.M. MS., 8 C. iv., with additions from 12 E. 1 & Eg. 2622.]

Hec algorismus ars presens dicitur¹; in qua
Talibus Indorum² fruimur his quinque figuris.

0. 9. 8. 7. 6. 5. 4. 3. 2. 1.

Prima significat unum: duo vero secunda:

4

Tercia significat tria: sic procede sinistre
Donec ad extremam venies, qua cifra vocatur;
³[Que nil significat; dat significare sequenti.]

Quelibet illarum si primo limite ponas,

8

Simpliciter se significat: si vero secundo,
Se decies: sursum procedas multiplicando.⁴

[Namque figura sequens quevis signat decies plus,

12

Ipsa locata loco quam significet pereunte:
Nam precedentes plus ultima significabit.]

⁵Post predicta scias quod tres breuiter numerorum
Distincte species sunt; nam quidam digiti sunt;

16

Articuli quidam; quidam quoque compositi sunt.

[Sunt digiti numeri qui citra denarium sunt;

Articuli decupli degitorum; compositi sunt

Illi qui constant ex articulis digitisque.]

20

Ergo, proposito numero tibi scribere, primo

Respicias quis sit numerus; quia si digitus sit,

⁵[Una figura satis sibi; sed si compositus sit,]

Primo scribe loco digitum post articulum fac

24

Articulus si sit, cifram post articulum sit,

[Articulum vero reliquenti in scribe figure.]

Quolibet in numero, si par sit prima figura,

Par erit et totum, quicquid sibi continetur;

28

Impar si fuerit, totum sibi fiet et impar.

Septem ⁶ sunt partes, non plures, istius artis;

Addere, subtrahere, duplare, dimidiare;

Sexta est diuidere, set quinta est multiplicare;

32

Radicem extrahere pars septima dicitur esse.

Subtrahis aut addis a dextris vel mediabis;

A leua dupla, diuide, multiplicaque;

Extrahe radicem semper sub parte sinistra.

36

Addere si numero numerum vis, ordine tali

Incipe; scribe duas primo series numerorum

Prima sub prima recte ponendo figuram,

Et sic de reliquis facias, si sint tibi plures.

40

Inde duas adde primas hac condicione;

Si digitus crescat ex addicione priorum,

Primo scribe loco digitum, quicumque sit ille;

Si sit compositus, in limite scribe sequenti

44

Addition.

Articulus, primo digitum; quia sic iubet ordo.
Articulus si sit, in primo limite cifram,
Articulus vero reliquis inscribe figuris;
Vel per se scribas si nulla figura sequatur.

48

Si tibi cifra superueniens occurrerit, illam
Deme suppositam; post illic scribe figuram:
Postea procedas reliquas addendo figuras.

A numero numerum si sit tibi demere cura,

Subtractioni

52

Scribe figurarum series, vt in addicione;
Maiori numero numerum suppone minorem,
Siue pari numero supponatur numerus par.
Postea si possis a prima subtrahe primam,

56

Scribens quod remanet, cifram si nil remanebit.
Set si non possis a prima demere primam;
Procedens, vnum de limite deme sequenti;
Et demptum pro denario reputabis ab illo,

60

Subtrahe totaliter numerum quem proposuisti.
Quo facto, scribe supra quicquid remanebit,
Facque novenarios de cifris, cum remanebis,
Occurrant si forte cifre, dum demseris vnum;

64

Postea procedas reliquas demendo figuras.

⁷[Si subtractio sit bene facta probare valebis,
Quas subtraxisti primas addendo figuras.
Nam, subtractio si bene sit, primas retinebis,

Proof.

68

Et subtractio facta tibi probat additionem.]

Si vis duplare numerum, sic incipe; solam
Scribe figurarum seriem, quamcumque voles que
Postea procedas primam duplando figuram;

Duplation.

72

Inde quod excrescet, scribens, vbi iusserit ordo,
Juxta precepta que dantur in addicione.

Nam si sit digitus, in primo limite scribe;
Articulus si sit, in primo limite cifram,

76

Articulum vero reliquis inscribe figuris;
Vel per se scribas, si nulla figura sequatur:
Compositus si sit, in limite scribe sequenti
Articulum primo, digitum; quia sic jubet ordo:

80

Et sic de reliquis facias, si sint tibi plures.
⁸[Si super extremam nota sit, monadem dat eidem,
Quod tibi contingit, si primo dimidiabis.]

Incipe sic, si vis aliquem numerum mediare:

Mediation.

84

Scribe figurarum seriem solam, velud ante;
Postea procedens medias, et prima figura
Si par aut impar videas; quia si fuerit par,
Dimidiabis eam, scribens quicquit remanebit;

88

Impar si fuerit, vnum demas, mediare,
Nonne presumas, sed quod superest mediabis;
Inde super tractum, fac demptum quod notat unum;
Si monos, dele; sit ibi cifra post nota supra.

92

Postea procedas hac condicione secunda: ⁹
Impar ¹⁰ si fuerit hic vnum deme priori,
Inscribens quinque, nam denos significabit
Monos prædictam: si vero secunda dat vnam,

96

Illa deleta, scribatur cifra; priori
Tradendo quinque pro denario mediato;
Nec cifra scribatur, nisi inde figura sequatur:
Postea procedas reliquas mediando figuras,

100

Quin supra docui, si sint tibi mille figure.
¹¹[Si mediatio sit bene facta probare valebis,
Duplando numerum quem primo dimidiasti.]

Si tu per numerum numerum vis multiplicare,

Multiplicat
n.

104

Scribe duas, quascunque volis, series numerorum;
Ordo tamen seruetur vt vltima multiplicandi
Ponatur super anteriorem multiplicantis;

¹²[A leua relique sint scripte multiplicantes.]

108

In digitum cures digitum si ducere, major
Per quantes distat a denis respice, debes
Namque suo decuplo tociens delere minorem;
Sicque tibi numerus veniens exinde patebit.

112

Postea procedas postremam multiplicando,
Iuste multiplicans per cunctas inferiores,
Condicione tamen tali; quod multiplicantis
Scribas in capite, quicquid processerit inde;

116

Set postquam fuerit hec multiplicata, figure
Anterorentur seriei multiplicantis;
Et sic multiplica, velut istam multiplicasti,
Qui sequitur numerum scriptum quicunque figuris.

120

Set cum multiplicas, primo sic est operandum,
Si dabit articulum tibi multiplicacio solum;
Proposita cifra, summam transferre memento.
Sin autem digitus excrescerit articulusque,

124

Articulus supraposito digito salit ultra;
Si digitus tamen, ponas illum super ipsam,
Subdita multiplicans hanc que super incidit illi
Delet eam penitus, scribens quod provenit inde;

128

Sed si multiplices illam posite super ipsam,
Adiungens numerum quem prebet ductus earum;
Si suprainpositam cifra debet multiplicare,
Prorsus eam delet, scribi que loco cifra debet,

132

¹²[Si cifra multiplicat aliam positam super ipsam,
Sitque locus supra vacuus super hanc cifra fiet;]
Si supra fuerit cifra semper pretereunda est;

Si dubites, an sit bene multiplicando secunda,

136

Diuide totalem numerum per multiplicantem,

Et reddet numerus emergens inde priorem.

¹³[Per numerum si vis numerum quoque multiplicare

Tantum per normas subtiles absque figuris

140

Has normas poteris per versus scire sequentes.

Si tu per digitum digitum quilibet multiplicabis

Regula precedens dat qualiter est operandum

Articulum si per reliquum vis multiplicare

144

In proprium digitum debebit uterque resolvi

Articulus digitos post per se multiplicantes

Ex digitis quociens teneret multiplicatum

Articuli faciunt tot centum multiplicati.

148

Articulum digito si multiplicamus oportet

Articulum digitum sumi quo multiplicare

Debemus reliquum quod multiplicaris ab illis

Per reliquo decuplum sic omne latere nequibit

152

In numerum mixtum digitum si ducere cures

Articulus mixti sumatur deinde resolvas

In digitum post hec fac ita de digitis nec

Articulusque docet excrescens in detinendo

156

In digitum mixti post ducas multiplicantem

De digitis ut norma docet sit juncta secundo

Multiplica summam et postea summa patebit

Junctus in articulum purum articulumque

160

¹⁴[Articulum purum comittes articulum que]

Mixti pro digitis post fiat et articulus vt

Norma jubet retinendo quod egreditur ab illis

Articuli digitum post in digitum mixti duc

164

Regula de digitis ut percipit articulusque

Mental
Multiplica

Ex quibus excrescens summe tu junge priori
Sic manifesta cito fiet tibi summa petita.
Compositum numerum mixto sic multiplicabis

168

Vndecies tredecem sic est ex hiis operandum
In reliquum primum demum duc post in eundem
Unum post deinde duc in tertia deinde per unum
Multiplices tertia demum tunc omnia multiplicata

172

In summa duces quam que fuerit te dices
Hic ut hic mixtus intentus est operandum
Multiplicandorum de normis sufficiunt hec.]
Si vis dividere numerum, sic incipe primo;

Division.

176

Scribe duas, quascunque voles, series numerorum;
Majori numero numerum suppone minorem,
¹⁵[Nam docet ut major teneat bis terve minorem;]
Et sub supprima supprimam pone figuram,

180

Sic reliquis reliquas a dextra parte locabis;
Postea de prima primam sub parte sinistra
Subtrahe, si possis, quociens potes adminus istud,
Scribens quod remanet sub tali conditione;

184

Ut totiens demas demendas a remanente,
Que serie recte ponentur in anteriori,
Unica si, tantum sit ibi decet operari;
Set si non possis a prima demere primam,

188

Procedas, et eam numero suppone sequenti;
Hanc uno retrahendo gradu quo comites retrahantur,
Et, quotiens poteris, ab eadem deme priorem,
Ut totiens demas demendas a remanenti,

192

Nec plus quam novies quicquam tibi demere debes,
Nascitur hinc numerus quociens supraque sequentem
Hunc primo scribas, retrahas exinde figuras,
Dum fuerit major supra positus inferiori,

196

Et rursum fiat divisio more priori;
Et numerum quotiens supra scribas pereunti,
Si fiat saliens retrahendo, cifra locetur,
Et pereat numero quotiens, proponas eidem

200

Cifram, ne numerum pereat vis, dum locus illic
Restat, et expletis divisio non valet ultra:
Dum fuerit numerus numerorum inferiore seorsum
Illum servabis; hinc multiplicando probabis,

204

Proof.

Si bene fecisti, divisor multiplicetur
Per numerum quotiens; cum multiplicaveris, adde
Totali summæ, quod servatum fuit ante,
Reddeturque tibi numerus quem proposuisti;

208

Et si nil remanet, hunc multiplicando reddet,
Cum ducis numerum per se, qui provenit inde
Sit tibi quadratus, ductus radix erit hujus,
Nec numeros omnes quadratos dicere debes,

212

Square
Numbers.

Est autem omnis numerus radix alicujus.
Quando voles numeri radicem querere, scribe
Debet; inde notes si sit locus ulterius impar,
Estque figura loco talis scribenda sub illo,

216

Que, per se dicta, numerum tibi destruat illum,
Vel quantum poterit ex inde delebis eandem;
Vel retrahendo duplex retrahens duplando sub ista
Que primo sequitur, duplicatur per duplicationem,

220

Post per se minuens pro posse quod est minuendum.

¹⁶Post his propones digitum, qui, more priori

Per precedentes, post per se multiplicatus,
Destruat in quantum poterit numerum remanentem,

224

Et sic procedens retrahens duplando figuram,
Preponendo novam donec totum peragatur,

Subdupla propriis servare docetque duplatis;
Si det compositum numerum duplacio, debet

228

Inscribi digitus a parte dextra parte propinqua,
Articulusque loco quo non duplicata resessit;
Si dabit articulum, sit cifra loco pereunte
Articulusque locum tenet unum, de duplicata resessit;

232

Si donet digitum, sub prima pone sequente,
Si supraposita fuerit duplicata figura
Major proponi debet tantummodo cifra,
Has retrahens solito propones more figuram,

236

Usque sub extrema ita fac retrahendo figuras,
Si totum deles numerum quem proposuisti,
Quadratus fuerit, de dupla quod duplicasti,
Sicque tibi radix illius certa patebit,

240

Si de duplatis fit juncta supprima figura;
Radicem per se multiplices habeasque
Primo propositum, bene te fecisse probasti;
Non est quadratus, si quis restat, sed habentur

244

Radix quadrati qui stat major sub eadem;
Vel quicquid remanet tabula servare memento;
Hoc casu radix per se quoque multiplicetur,
Vel sic quadratus sub primo major habetur,

248

Hinc addas remanens, et prius debes haberi;
Si locus extremus fuerit par, scribe figuram
Sub pereunte loco per quam debes operari,
Que quantum poterit supprimas destruat ambas,

252

Vel penitus legem teneas operando priorem,
Si suppositum digitus suo fine repertus,
Omnino delet illic scribi cifra debet,
A leva si qua sit ei sociata figura;

256

Si cifre remanent in fine pares decet harum
Radices, numero mediam proponere partem,
Tali quesita radix patet arte reperta.
Per numerum recte si nosti multiplicare

260

Ejus quadratum, numerus qui pervenit inde
Dicetur cubicus; primus radix erit ejus;
Nec numeros omnes cubicatos dicere debes,
Est autem omnis numerus radix alicujus;

264

Cube Roo

Si curas cubici radicem quærere, primo
Inscriptum numerum distinguere per loca debes;
Que tibi mille notant a mille notante suprema
Initiam, summa operandi parte sinistra,

268

Illic sub scribas digitum, qui multiplicatus
In semet cubice suprapositum sibi perdat,
Et si quid fuerit adjunctum parte sinistra
Si non omnino, quantum poteris minuendo,

272

Hinc triplans retrahe saltum, faciendo sub illa
Que manet a digito deleta terna, figuram
Illi propones quo sub triplo asocietur,
Ut cum subtriplo per eam tripla multiplicatur;

276

Hinc per eam solam productum multiplicabis,
Postea totalem numerum, qui provenit inde
A suprapositis respectu tolle triplate
Addita supprimo cubice tunc multiplicetur,

280

Respectu cujus, numerus qui progredietur
Ex cubito ductu, supra omnes adimetur;
Tunc ipsam delens triples saltum faciendo,
Semper sub ternas, retrahens alias triplicatas

284

Ex hinc triplatis aliam propone figuram,
Que per triplatas ducatur more priori;
Primo sub triplis sibi junctis, postea per se,

In numerum ducta, productum de triplicatis:

288

Utque prius dixi numerus qui provenit inde
A suprapositis has respiciendo trahatur,
Huic cubice ductum sub primo multiplicabis,
Respectumque sui, removebis de remanenti,

292

Et sic procedas retrahendo triplando figuram.
Et proponendo nonam, donec totum peragatur,
Subtripla sub propriis servare decet triplicatis;
Si nil in fine remanet, numerus datus ante

296

Est cubicus; cubicam radicem sub tripla prebent,
Cum digito juncto quem supprimo posuisti,
Hec cubice ducta, numerum reddant tibi primum.
Si quid erit remanens non est cubicus, sed habetur

300

Major sub primo qui stat radix cubicam,
Servari debet quicquid radice remansit,
Extracto numero, decet hec addi cubicato.
Quo facto, numerus reddi debet tibi primus.

304

Nam debes per se radicem multiplicare
Ex hinc in numerum duces, qui provenit inde
Sub primo cubicus major sic invenietur;
Illi jungatur remanens, et primus habetur,

308

Si per triplatum numerum nequeas operari;
Cifram propones, nil vero per hanc operare
Set retrahens illam cum saltu deinde triplata,
Propones illi digitum sub lege priori,

312

Cumque cifram retrahas saliendo, non triplicabis,
Namque nihil cifre triplacio dicitur esse;
At tu cum cifram protraxeris aut triplicata,
Hanc cum subtriplo semper servare memento:

316

Si det compositum, digiti triplacio debet

Illius scribi, digitus saliendo sub ipsam;
Digito delete, que terna dicitur esse;
Jungitur articulus cum triplata pereunte,

320

Set facit hunc scribi per se triplacio prima,
Que si det digitum per se scribi facit illum;
Consumpto numero, si sole fuit tibi cifre
Triplato, propone cifram saltum faciendo,

324

Cumque cifram retrahe triplam, scribendo figuram,
Preponas cifre, sic procedens operare,
Si tres vel duo serie in sint, pone sub yma,
A dextris digitum servando prius documentum.

328

Si sit continua progressio terminus nuper
Per majus medium totalem multiplicato;
Si par, per medium tunc multiplicato sequentem.
Set si continua non sit progressio finis:

332

Impar, tunc majus medium si multiplicabis,

333

Si par per medium sibi multiplicato propinquum.

[1.](#) “Hec præsens ars dicitur algorismus ab Algore rege ejus inventore, vel dicitur ab *algos* quod est ars, et *rodos* quod est numerus; quæ est ars numerorum vel numerandi, ad quam artem bene sciendum inveniebantur apud Indos bis quinque (id est decem) figuræ.” —*Comment. Thomæ de Novo-Mercatu*. MS. Bib. Reg. Mus. Brit. 12 E. 1.

[2.](#) “Hæ necessariæ figuræ sunt Indorum characteros.” *MS. de numeratione*. Bib. Sloan. Mus. Brit. 513, fol. 58. “Cum vidissem Yndos constituisse ix literas in universo numero suo propter dispositionem suam quam posuerunt, volui patefacere de opere quod sit per eas aliquidque esset levius discentibus, si Deus voluerit. Si autem Indi hoc voluerunt et intentio illorum nihil novem literis fuit, causa que mihi potuit. Deus direxit me ad hoc. Si vero alia dicam preter

eam quam ego exposui, hoc fecerunt per hoc quod ego exposui, eadem tam certissime et absque ulla dubitatione poterit inveniri. Levitasque patebit aspicientibus et discentibus.” MS. U.L.C., Ii. vi. 5, f. 102.

[3.](#) From Eg. 2622.

[4.](#) 8 C. iv. inserts Nullum cipa significat: dat significare sequenti.

[5.](#) From 12 E. 1.

[6.](#)

En argorisme devon prendre
Vii especes
Adision subtracion
Doubloison mediacion
Monteploie et division
Et de radix eustracion
A chez vii especes savoir
Doit chascun en memoire avoir
Letres qui figures sont dites
Et qui excellens sont ecrites.—MS. *Seld. Arch.* B. 26.

[7.](#) From 12 E. 1.

[8.](#) From 12 E. 1.

[9.](#) 8 C. iv. inserts Atque figura prior nuper fuerit mediando.

[10.](#) *I.e.* figura secundo loco posita.

[11.](#) So 12 E. 1; 8 C. iv. inserts—

Si super extremam nota sit monades dat eidem
Quod contingat cum primo dimiabis
Atque figura prior nuper fuerit mediando.

[12.](#) 12 E. 1 inserts.

[13.](#) 12 E. 1 inserts to l. 174.

[14.](#) 12 E. 1 omits, Eg. 2622 inserts.

[15.](#) 12 E. 1 inserts.

[16.](#) 8 C. iv. inserts—

Hinc illam dele duplans sub ei psalliendo
Que sequitur retrahens quicquid fuerit duplicatum.

INDEX OF TECHNICAL TERMS ¹

algorisme, [33/12](#); **algorym**, **augrym**, [3/3](#); the art of computing, using the so-called Arabic numerals.

The word in its various forms is derived from the Arabic *al-Khowarazmi* (i.e. the native of Khwarazm (Khiva)). This was the surname of Ja'far Mohammad ben Musa, who wrote a treatise early in the 9th century (see p. xiv).

The form *algorithm* is also found, being suggested by a supposed derivation from the Greek ἀριθμός (number).

antery, [24/11](#); to move figures to the right of the position in which they are first written. This operation is performed repeatedly upon the multiplier in multiplication, and upon certain figures which arise in the process of root extraction.

anterioracioun, [50/5](#); the operation of moving figures to the right.

article, [34/23](#); **articul**, [5/31](#); **articals**, [9/36](#), [29/7,8](#); a number divisible by ten without remainder.

cast, [8/12](#); to add one number to another.

'Addition is a *casting* together of two numbers into one number,' [8/10](#).

cifre, [4/1](#); the name of the figure 0. The word is derived from the Arabic *sifr* = empty, nothing. Hence *zero*.

A cipher is the symbol of the absence of number or of zero quantity. It may be used alone or in conjunction with digits or other ciphers, and in the latter case, according to the position which it occupies relative to the other figures, indicates the absence of units, or tens, or hundreds, etc. The great superiority of the Arabic to all other systems of notation resides in the employment of this symbol. When the cipher is not used, the place value of digits has to be indicated by writing them in assigned rows or columns. Ciphers, however, may be interpolated amongst the significant figures used, and as they sufficiently indicate the positions of the empty rows or columns, the latter need not be indicated in any other way. The practical performance of calculations is thus enormously facilitated (see [p. xvi](#)).

compone, [33/24](#); **composyt**, [5/35](#); with reference to numbers, one compounded of a multiple of ten and a digit.

conuertide = conversely, [46/29](#), [47/9](#).

cubicede, [50/13](#); **to be c.**, to have its cube root found.

cubike nombre, [47/8](#); a number formed by multiplying a given number twice by itself, *e.g.* $27 = 3 \times 3 \times 3$. Now called simply a cube.

decuple, [22/12](#); the product of a number by ten. Tenfold.

departys = divides, [5/29](#).

digit, [5/30](#); **digitalle**, [33/24](#); a number less than ten, represented by one of the nine Arabic numerals.

dimydicion, [7/23](#); the operation of dividing a number by two. Halving.

duccioun, multiplication, [43/9](#).

duplacion, [7/23](#), [14/15](#); the operation of multiplying a number by two. Doubling.

i-mediet = halved, [19/23](#).

intercise = broken, [46/2](#); intercise Progression is the name given to either of the Progressions 1, 3, 5, 7, etc.; 2, 4, 6, 8, etc., in which the common difference is 2.

lede into, multiply by, [47/18](#).

lyneal nombre, [46/14](#); a number such as that which expresses the measure of the length of a line, and therefore is not *necessarily* the product of two or more numbers (*vide* Superficial, Solid). This appears to be the meaning of the phrase as used in *The Art of Nombryng*. It is possible that the numbers so designated are the prime numbers, that is, numbers not divisible by any other number except themselves and unity, but it is not clear that this limitation is intended.

mediacioun, [16/36](#), [38/16](#); dividing by two (see also **dimydicion**).

medlede nombre, [34/1](#); a number formed of a multiple of ten and a digit (*vide* componede, composyt).

medye, [17/8](#), to halve; **mediete**, halved, [17/30](#); **ymedit**, [20/9](#).

naturelle progressioun, [45/22](#); the series of numbers 1, 2, 3, etc.

produccioun, multiplication, [50/11](#).

quadrat nombre, [46/12](#); a number formed by multiplying a given number by itself, *e.g.* $9 = 3 \times 3$, a square.

rote, [7/25](#); **roote**, [47/11](#); root. The roots of squares and cubes are the numbers from which the squares and cubes are derived by multiplication into themselves.

significatyf, significant, [5/14](#); The significant figures of a number are, strictly speaking, those other than zero, *e.g.* in 3 6 5 0 4 0 0, the significant figures are 3, 6, 5, 4. Modern usage, however, regards all figures between the two extreme significant figures as significant, even when some are zero. Thus, in the above example, 3 6 5 0 4 are considered significant.

solide nombre, [46/37](#); a number which is the product of three other numbers, *e.g.* $66 = 11 \times 2 \times 3$.

superficial nombre, [46/18](#); a number which is the product of two other numbers, *e.g.* $6 = 2 \times 3$.

ternary, consisting of three digits, [51/7](#).

vnder double, a digit which has been doubled, [48/3](#).

vnder-trebille, a digit which has been trebled, [49/28](#); **vnder-triplat**, [49/39](#).

w, a symbol used to denote half a unit, [17/33](#).

1. This Index has been kindly prepared by Professor J. B. Dale, of King's College, University of London, and the best thanks of the Society are due to him for his valuable contribution.

GLOSSARY

Words whose first appearance is earlier than the page cited in the Glossary are identified in supplementary notes, and both occurrences are marked in the

main text.

ablacioun, taking away, [36/21](#)

addyst, haddest, [10/37](#)

agregacioun, addition, [45/22](#). (First example in N.E.D., 1547.)

a-3enenes, against, [23/10](#)

allgate, always, [8/39](#)

als, as, [22/24](#)

and, if, [29/8](#);

&, [4/27](#);

& yf, [20/7](#)

a-nendes, towards, [23/15](#)

aproprede, appropriated, [34/27](#)

apwereth, appears, [61/8](#)

a-risy3t, arises, [14/24](#)

a-rowe, in a row, [29/10](#)

arsemetrike, arithmetic, [33/1](#)

ayene, again, [45/15](#)

bagle, crozier, [67/12](#)

bordure = ordure, row, [43/30](#)

borro, *inf.* borrow, [11/38](#);

imp. s. **borowe**, [12/20](#);

pp. **borwed**, [12/15](#);

borred, [12/19](#)

boue, above, [42/34](#)

caputule, chapter, [7/26](#)

certayn, assuredly, [18/34](#)

clepede, called, [47/7](#)

competently, conveniently, [35/8](#)

compt, count, [47/29](#)

contynes, contains, [21/12](#);

pp. **contenythe**, [38/39](#)

craft, art, [3/4](#)

distingue, divide, [51/5](#)

egalle, equal, [45/21](#)

excep, except, [5/16](#)

exclusede, excluded, [34/37](#)

excrement, resulting, [35/16](#)

exeant, resulting, [43/26](#)

expone, expound, [3/23](#)

ferye = ferþe, fourth, [70/12](#)

figure = figures, [5/1](#)

for-by, past, [12/11](#)

fors; no f., no matter, [22/24](#)

forseth, matters, [53/30](#)

forye = forþe, forth, [71/8](#)

fyfitye = fyftþe, fifth, [70/16](#)

grewe, Greek, [33/13](#)

haluendel, half, [16/16](#);

haldel, [19/4](#);

pl. **haluedels**, [16/16](#)

hayst, hast, [17/3](#), 32

hast, haste, [22/25](#)

heer, higher, [9/35](#)

here, their, [7/26](#)

here-a-fore, heretofore, [13/7](#)

heyth, was called, [3/5](#)

hole, whole, [4/39](#);

holle, [17/1](#);

hoole, of three dimensions, [46/15](#)

holdyþe, holds good, [30/5](#)

how be it that, although, [44/4](#)

lede = lete, let, [8/37](#)

lene, lend, [12/39](#)

lest, least, [43/27](#)

lest = left, [71/9](#)

leue, leave, [6/5](#);

pr. 3 s. **leues**, remains, [11/19](#);

leus, [11/28](#);

pp. **laft**, left, [19/24](#)

lewder, more ignorant, [3/3](#)

lust, desirest to, [45/13](#)

ly3t, easy, [15/31](#)

lymytes, limits, [34/18](#);

lynes, [34/12](#);

lynees, [34/17](#);

Lat. limes, *pl.* limites.

maystery, achievement;

First used in [10/40](#)

no m., no achievement, i.e. easy, [19/10](#)

me, *indef. pron.* one, [42/1](#) First used in [34/16](#)

mo, more, [9/16](#)

moder = more (Lat. majorem), [43/22](#)

most, must, [30/3](#) First used in [3/12](#)

multipliede, to be m. = multiplying, [40/9](#)

mynvtes, the sixty parts into which a unit is divided, [38/25](#)

myse-wroȝt, mis-wrought, [14/11](#)

nether, nor, [34/25](#)

nex, next, [19/9](#)

noȝt, nought, [5/7](#) First used in [4/8](#)

note, not, [30/5](#)

oo, one, [42/20](#); **o**, [42/21](#) First used in [34/27](#) (oo); [33/22](#) (o)

omest, uppermost, higher, [35/26](#);

omyst, [35/28](#)

omwhile, sometimes, [45/31](#) First used in [39/17](#)

on, one, [8/29](#)

opyne, plain, [47/8](#)

or, before, [13/25](#)

or = þe oþer, the other, [28/34](#)

ordure, order, [34/9](#);

row, [43/1](#) Word form is "order"

other, or, [33/13](#), [43/26](#); Note also "one other other" in [35/24](#)

other . . . or, either . . . or, [38/37](#) First used in [37/5](#)

ouerer, upper, [42/15](#)

ouer-hippede, passed over, [43/19](#)

recte, directly, [27/20](#) First used in [26/31](#)

remayner, remainder, [56/28](#)

representithe, represented, [39/14](#)

resteth, remains, [63/29](#) First used in [57/29](#)

rewarde, regard, [48/6](#)

rew, row, [4/8](#)

rewle, row, [4/20](#), [7/12](#);

rewele, [4/18](#);

rewles, rules, [5/33](#)

s. = scilicet, [3/8](#)

sentens, meaning, [14/29](#)

signifye(tyf), [5/13](#). The last three letters are added above the line, evidently because of the word 'significatyf' in l. 14. But the 'Solucio,' which contained the word, has been omitted.

sithen, since, [33/8](#)

some, sum, result, [40/17](#), 32

First used in [36/21](#)

sowne, pronounce, [6/29](#)

singillatim, singly, [7/25](#)

spices, species, kinds, [34/4](#)

First used in [5/34](#)

spyl, waste, [14/26](#)

styde, stead, [18/20](#)

subtrahe, subtract, [48/12](#);

pp. subtrayd, [13/21](#)

sythes, times, [21/16](#)

taȝt, taught, [16/36](#)

take, *pp.* taken;

t. fro, starting from, [45/22](#)

taward, toward, [23/34](#)

thouȝt, though, [5/20](#)

trebille, multiply by three, [49/26](#)

twene, two, [8/11](#)

First used in [4/23](#)

þow, though, [25/15](#)

þowȝt, thought;

be þ., mentally, [28/4](#)

þus = þis, this, [20/33](#)

vny, unite, [45/10](#)

wel, wilt, [14/31](#)

wete, wit, [15/16](#);

wyte, know, [8/38](#);

pr. 2 s. **wost**, [12/38](#)

wex, become, [50/18](#)

where, whether, [29/12](#)

wher-thurghe, whence, [49/15](#)

worth, work, [8/19](#);

First used in [7/35](#)

wrich, [8/35](#);

wyrch, [6/19](#);

imp. s. **worth**, [15/9](#);

First used in [9/6](#)

pp. **y-wroth**, [13/24](#)

write, written, [29/19](#);

First used in [4/5](#)

y-write, [16/1](#)

wryrchynge = wyrchyng, working, [30/4](#)

t

w, with, [55/8](#)

y-broth, brought, [21/18](#)

ychon, each one, [29/10](#)

y~~do~~, done, added, [9/6](#)

First used in [8/37](#)

y~~lke~~, same, [5/12](#)

y-~~lyech~~, alike, [22/23](#)

y-~~myzt~~, been able, [12/2](#)

y-~~nowzt~~, enough, [15/31](#);

ynovzt, [18/34](#)

yove, given, [45/33](#)

t
y_, that, [52/8](#)

y-write, v. [write](#).

y-wroth, v. [worch](#).

MARGINAL NOTES

Headnotes have been moved to the beginning of the appropriate paragraph. Headnotes were omitted from the two Appendixes, as sidenotes give the same information.

Line Numbers are cited in the Index and Glossary. They have been omitted from the e-text except in the one verse selection (App. II, *Carmen de Algorismo*). Instead, the Index and Glossary are linked directly to each word.

Numbered Notes:

Numbered sidenotes show page or leaf numbers from the original MSS. In the e-text, sidenote numbers have been replaced with simple asterisks.

Footnotes give textual information such as variant readings. They have been numbered sequentially within each title.

Sidenotes giving a running synopsis of the text have been kept as close as possible to their original format and location.

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